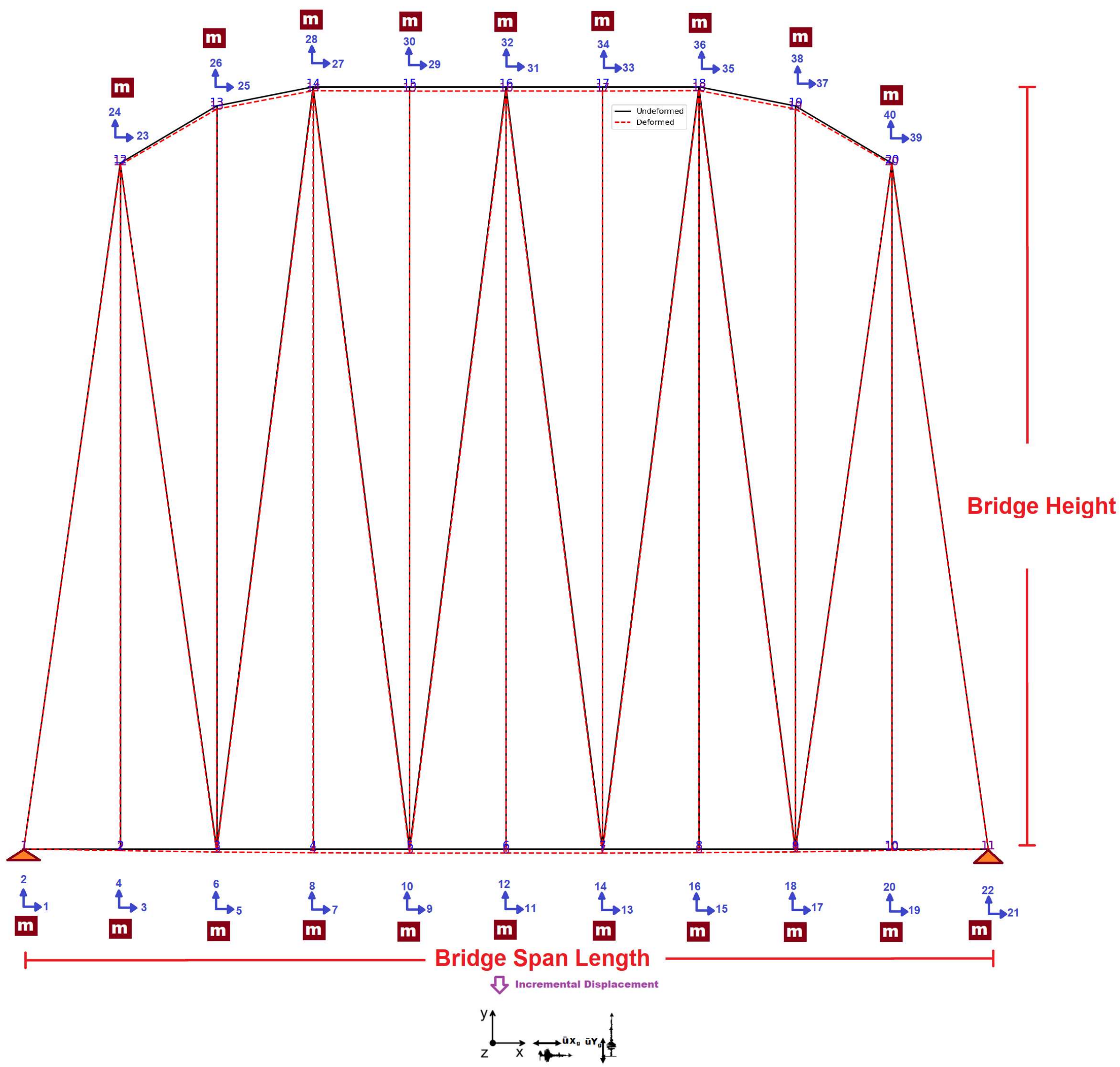
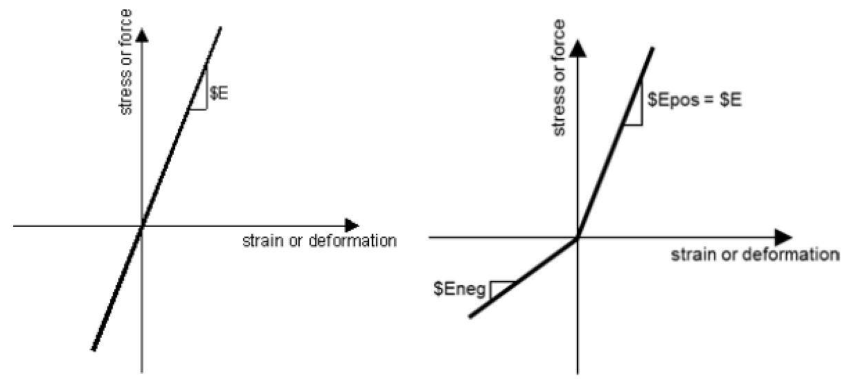
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

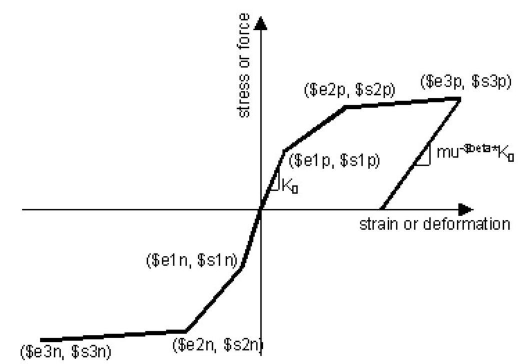
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL TRUSS BRIDGES: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

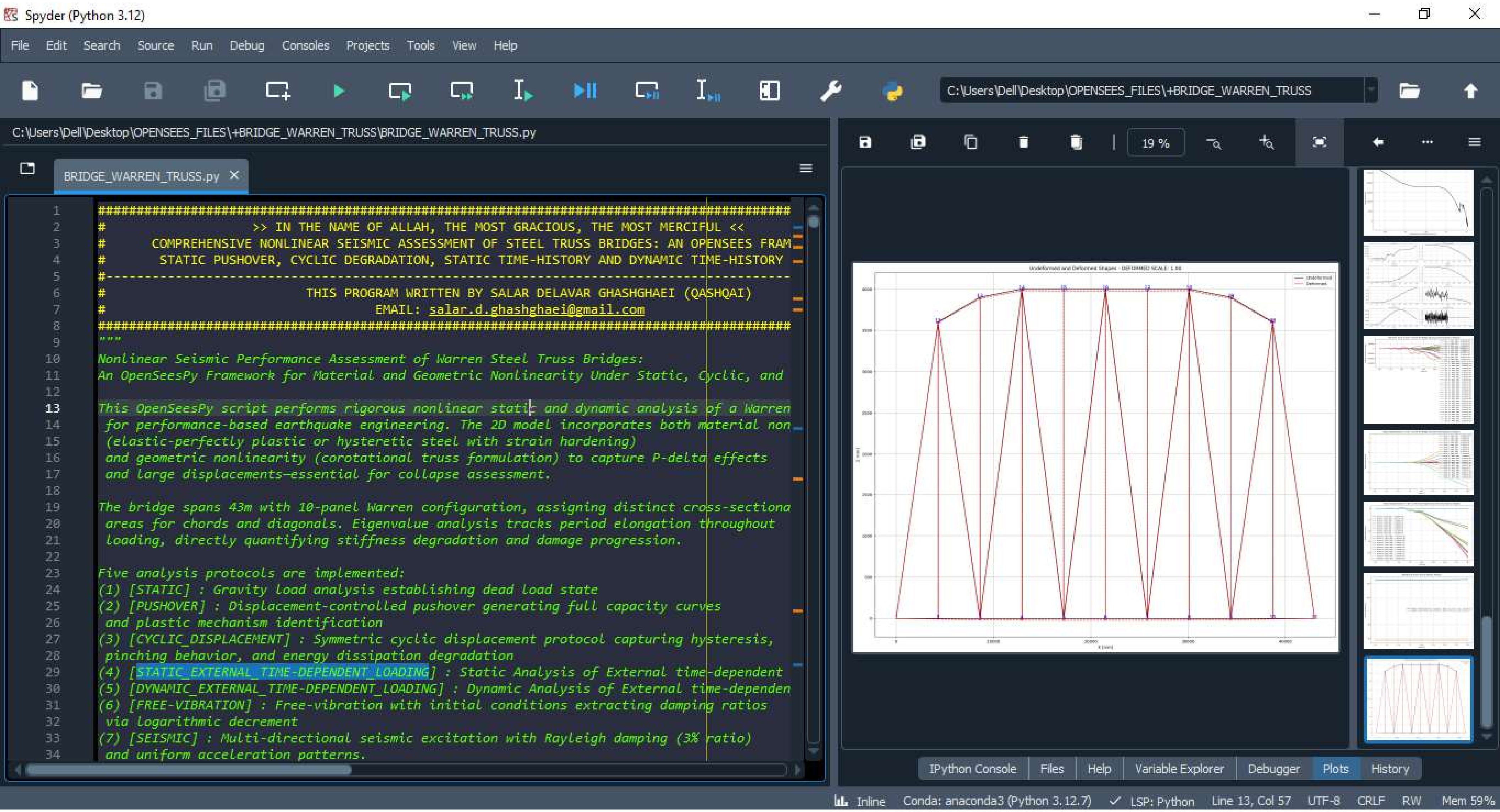




TRUSS ELEMENT ELASTIC MATERIAL



TRUSS ELEMENT INELASTIC MATERIAL

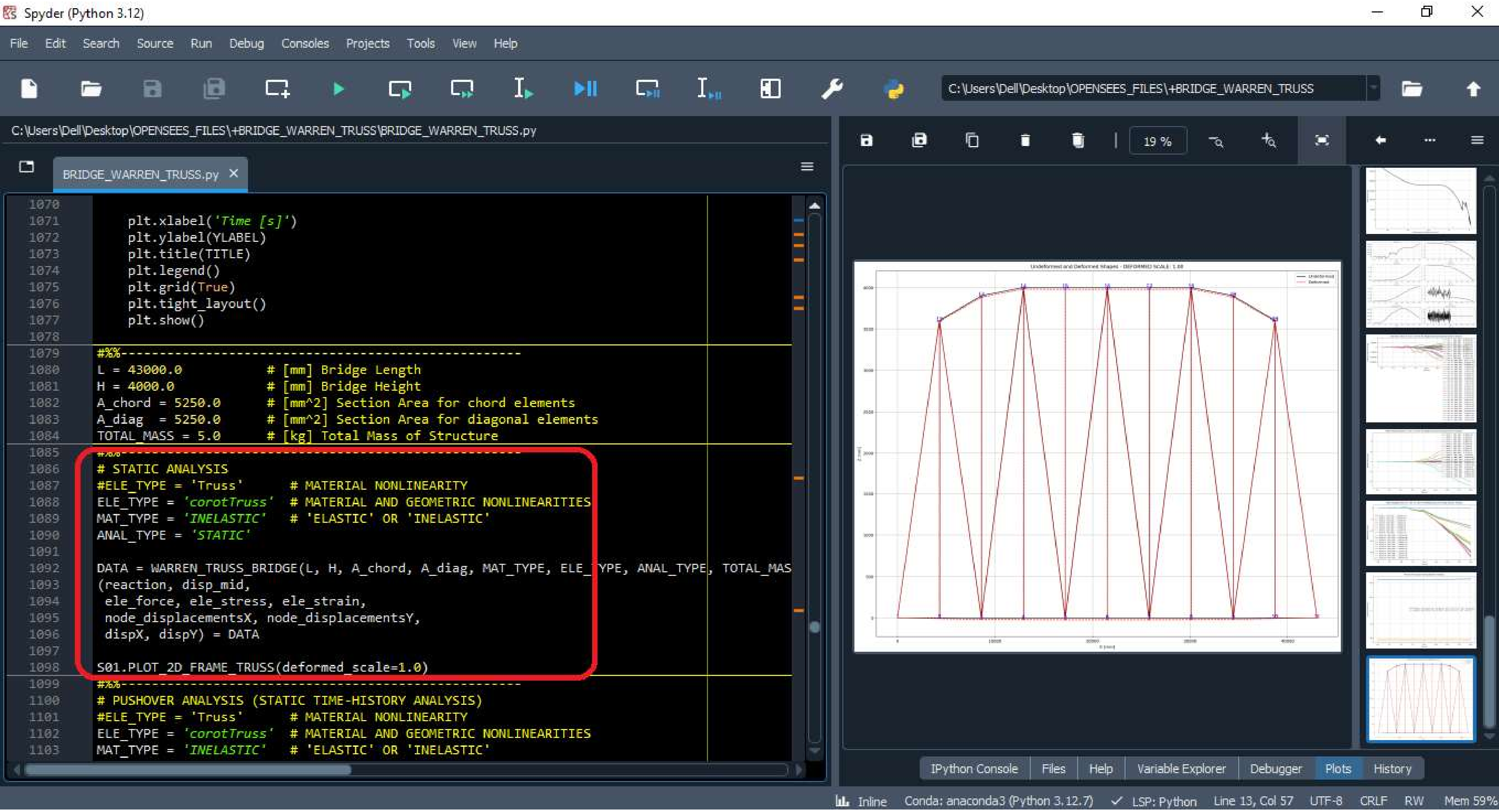

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_WARREN_TRUSS
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_WARREN_TRUSS\BRIDGE_WARREN_TRUSS.py
BRIDGE_WARREN_TRUSS.py
##
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL TRUSS BRIDGES: AN OPENSEES FRAM
STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY
#--
THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
EMAIL: salar.d.ghashghaei@gmail.com
##
"""
Nonlinear Seismic Performance Assessment of Warren Steel Truss Bridges:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and

This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a Warren
 for performance-based earthquake engineering. The 2D model incorporates both material non
 (elastic-perfectly plastic or hysteretic steel with strain hardening)
 and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
 and large displacements—essential for collapse assessment.

The bridge spans 43m with 10-panel Warren configuration, assigning distinct cross-sectiona
 areas for chords and diagonals. Eigenvalue analysis tracks period elongation throughout
 loading, directly quantifying stiffness degradation and damage progression.

Five analysis protocols are implemented:
(1) [STATIC] : Gravity load analysis establishing dead load state
(2) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
 and plastic mechanism identification
(3) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
 pinching behavior, and energy dissipation degradation
(4) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External time-dependent
(5) [DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Dynamic Analysis of External time-dependen
(6) [FREE-VIBRATION] : Free-vibration with initial conditions extracting damping ratios
 via logarithmic decrement
(7) [SEISMIC] : Multi-directional seismic excitation with Rayleigh damping (3% ratio)
 and uniform acceleration patterns.
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 13, Col 57 UTF-8 CRLF RW Mem 59%

STATIC ANALYSIS


Spyder (Python 3.12)
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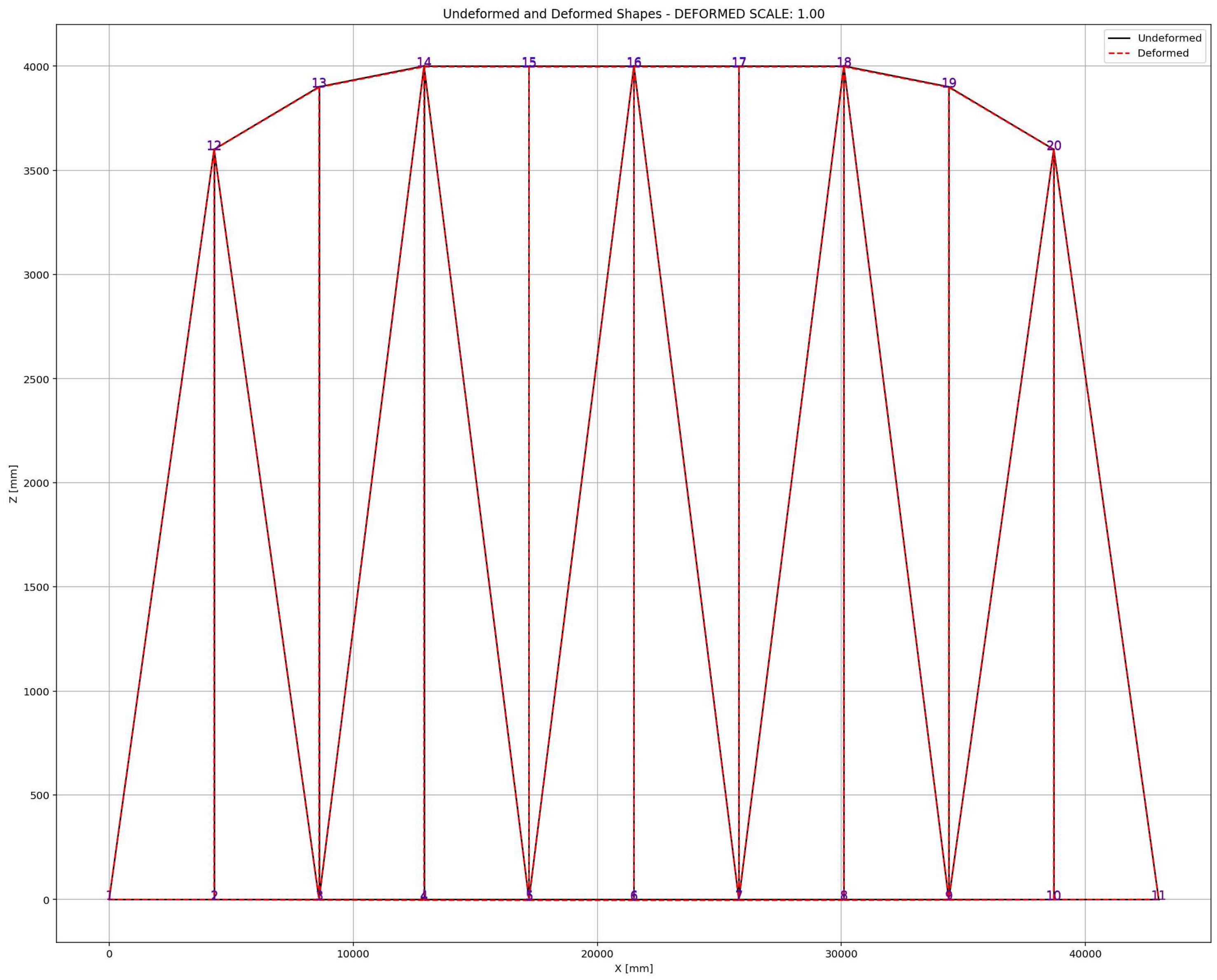
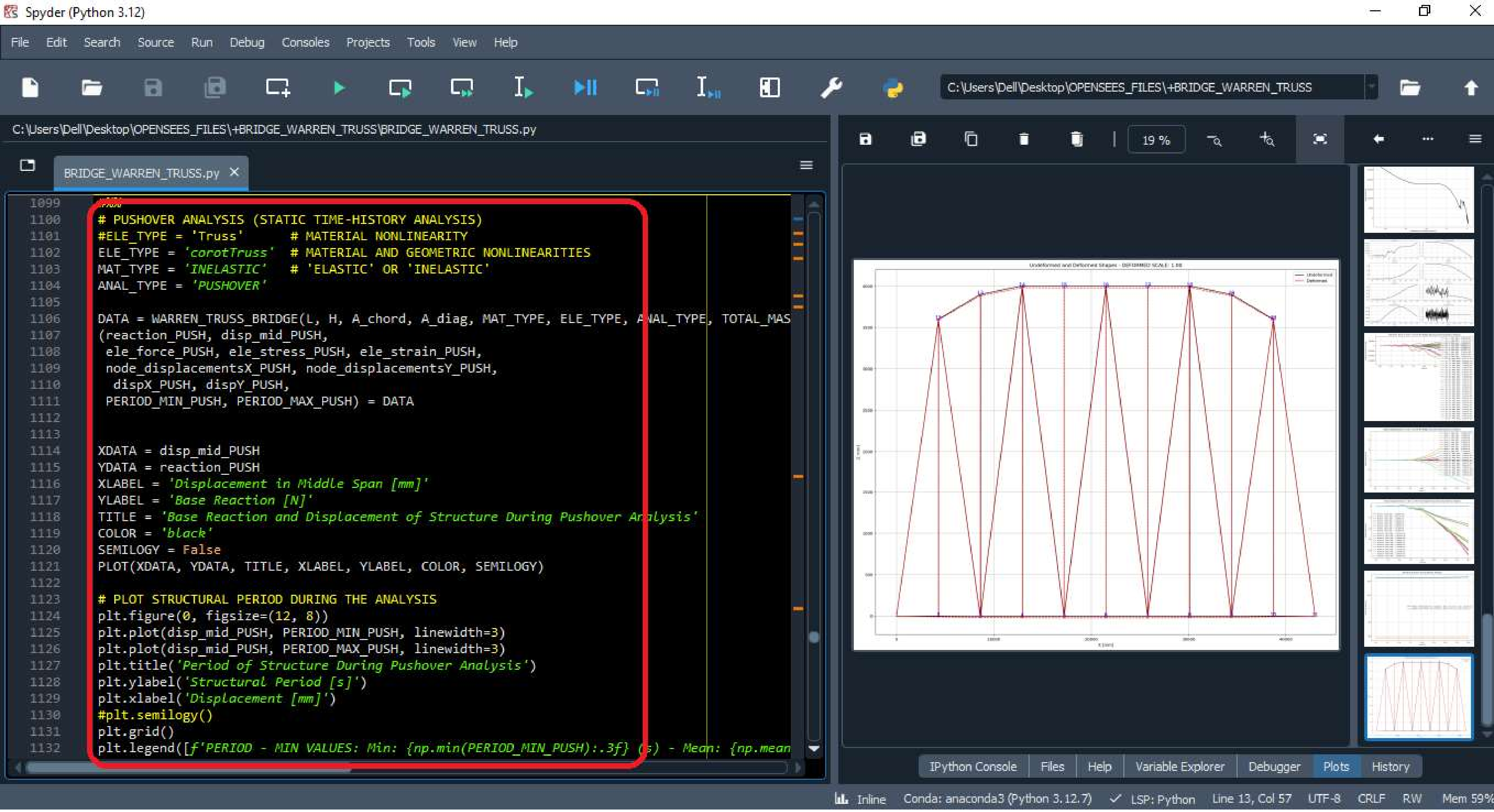
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_WARREN_TRUSS\BRIDGE_WARREN_TRUSS.py
BRIDGE_WARREN_TRUSS.py
    plt.xlabel('Time [s]')
    plt.ylabel(YLABEL)
    plt.title(TITLE)
    plt.legend()
    plt.grid(True)
    plt.tight_layout()
    plt.show()
#%%---
L = 43000.0   # [mm] Bridge Length
H = 4000.0   # [mm] Bridge Height
A_chord = 5250.0   # [mm^2] Section Area for chord elements
A_diag = 5250.0   # [mm^2] Section Area for diagonal elements
TOTAL_MASS = 5.0   # [kg] Total Mass of Structure
#%%---
STATIC ANALYSIS
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(reaction, disp_mid,
 ele_force, ele_stress, ele_strain,
 node_displacementsX, node_displacementsY,
 dispX, dispY) = DATA
S01.PLOT_2D_FRAME_TRUSS(deformed_scale=1.0)
#%%---
PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 13, Col 57 UTF-8 CRLF RW Mem 59%

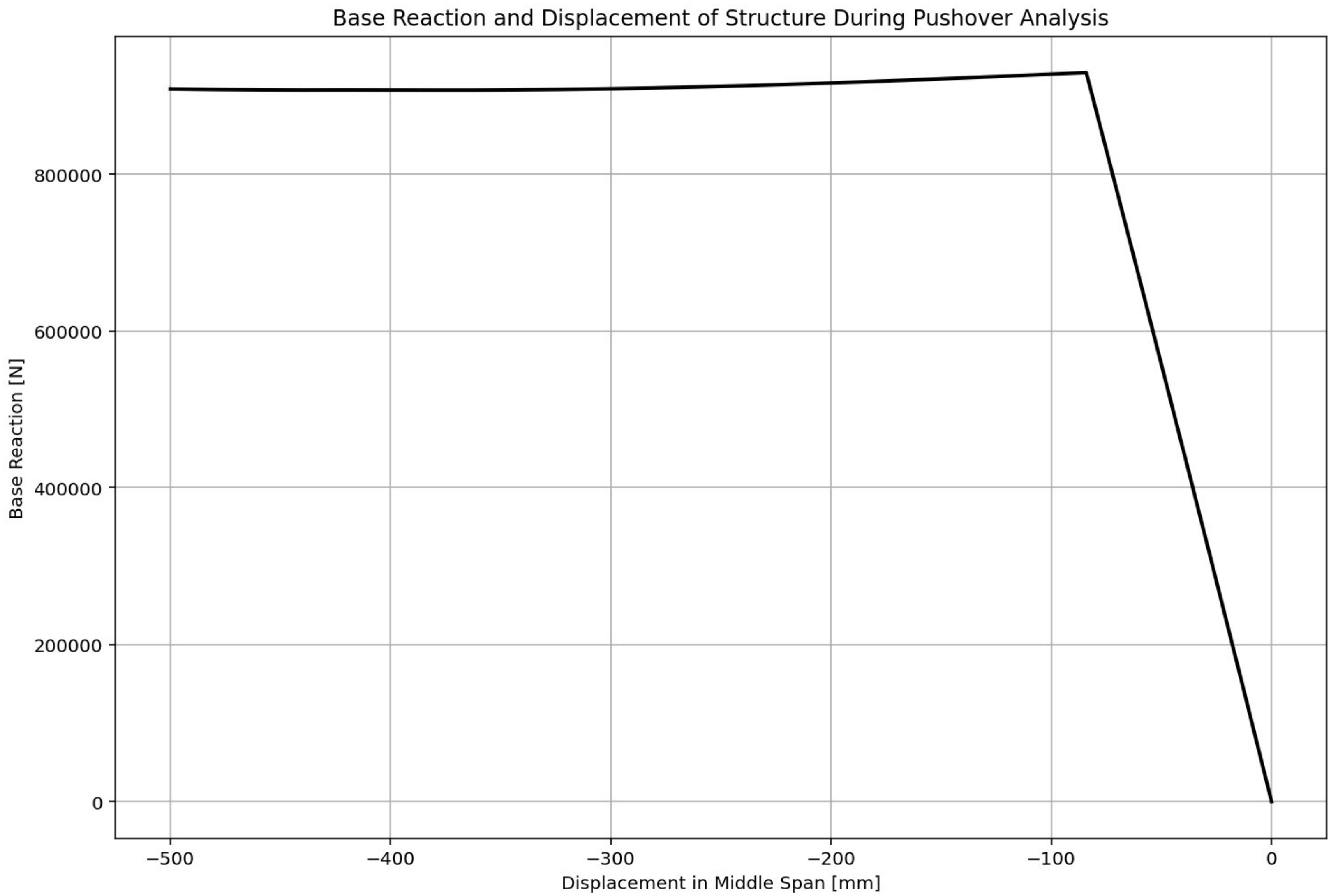


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
3500
4000
0
10000
20000
30000
40000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

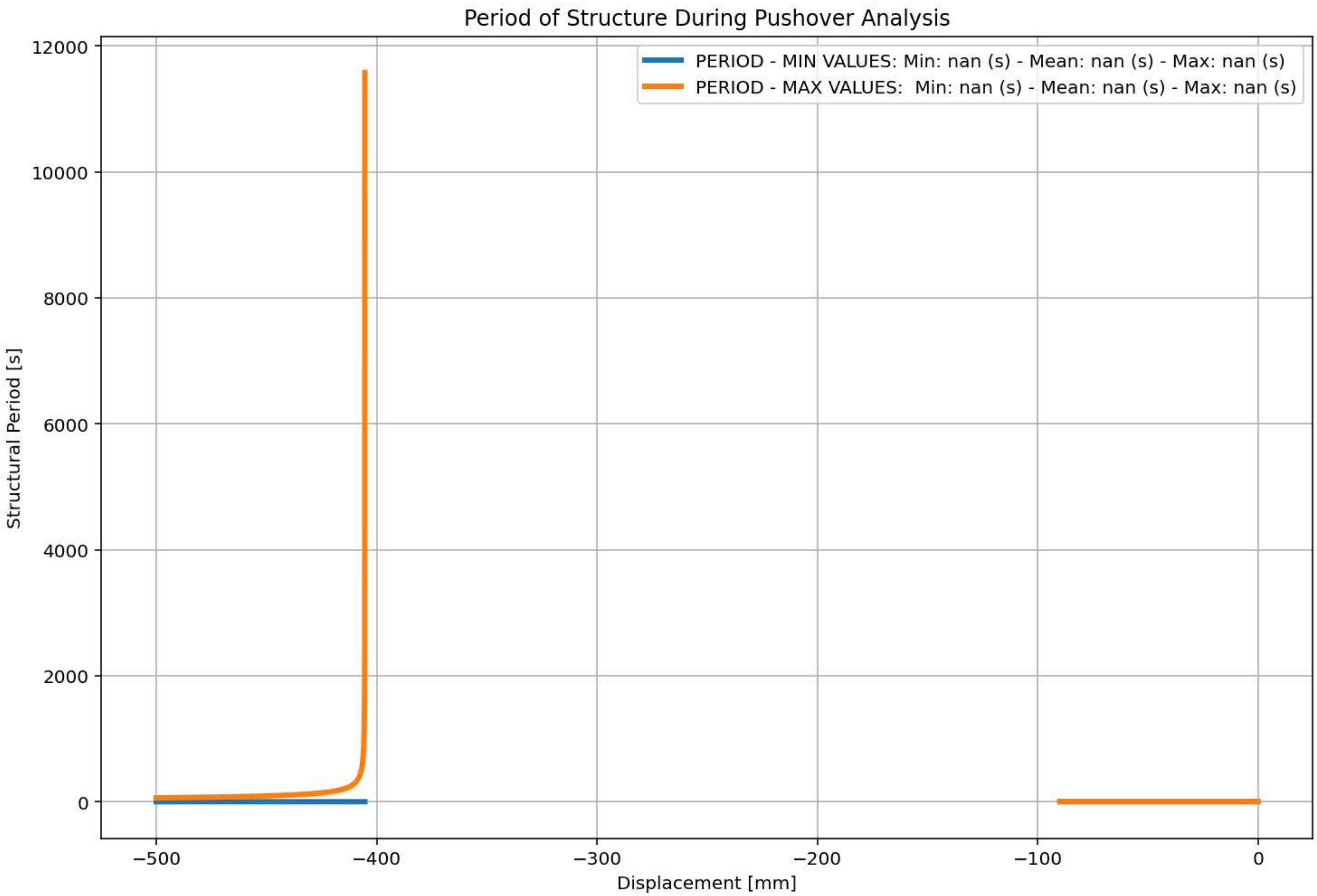
PUSHOVER ANALYSIS

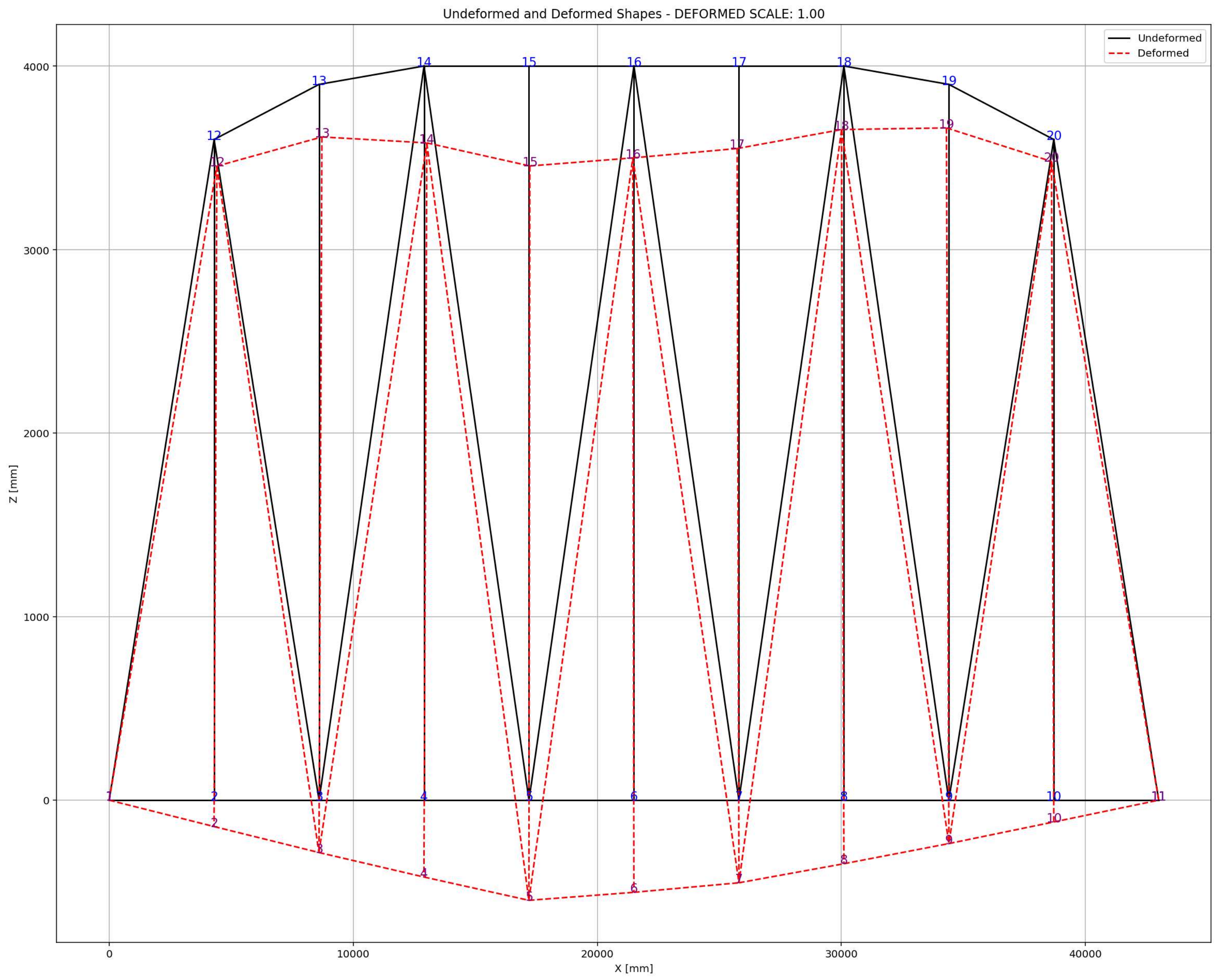


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_WARREN_TRUSS
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_WARREN_TRUSS\BRIDGE_WARREN_TRUSS.py
BRIDGE_WARREN_TRUSS.py
PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'PUSHOVER'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(reaction_PUSH, disp_mid_PUSH,
ele_force_PUSH, ele_stress_PUSH, ele_strain_PUSH,
node_displacementsX_PUSH, node_displacementsY_PUSH,
dispX_PUSH, dispY_PUSH,
PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
XDATA = disp_mid_PUSH
YDATA = reaction_PUSH
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_PUSH, PERIOD_MIN_PUSH, linewidth=3)
plt.plot(disp_mid_PUSH, PERIOD_MAX_PUSH, linewidth=3)
plt.title('Period of Structure During Pushover Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [mm]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_PUSH):.3f} (s) - Mean: {np.mean
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 13, Col 57 UTF-8 CRLF RW Mem 59%

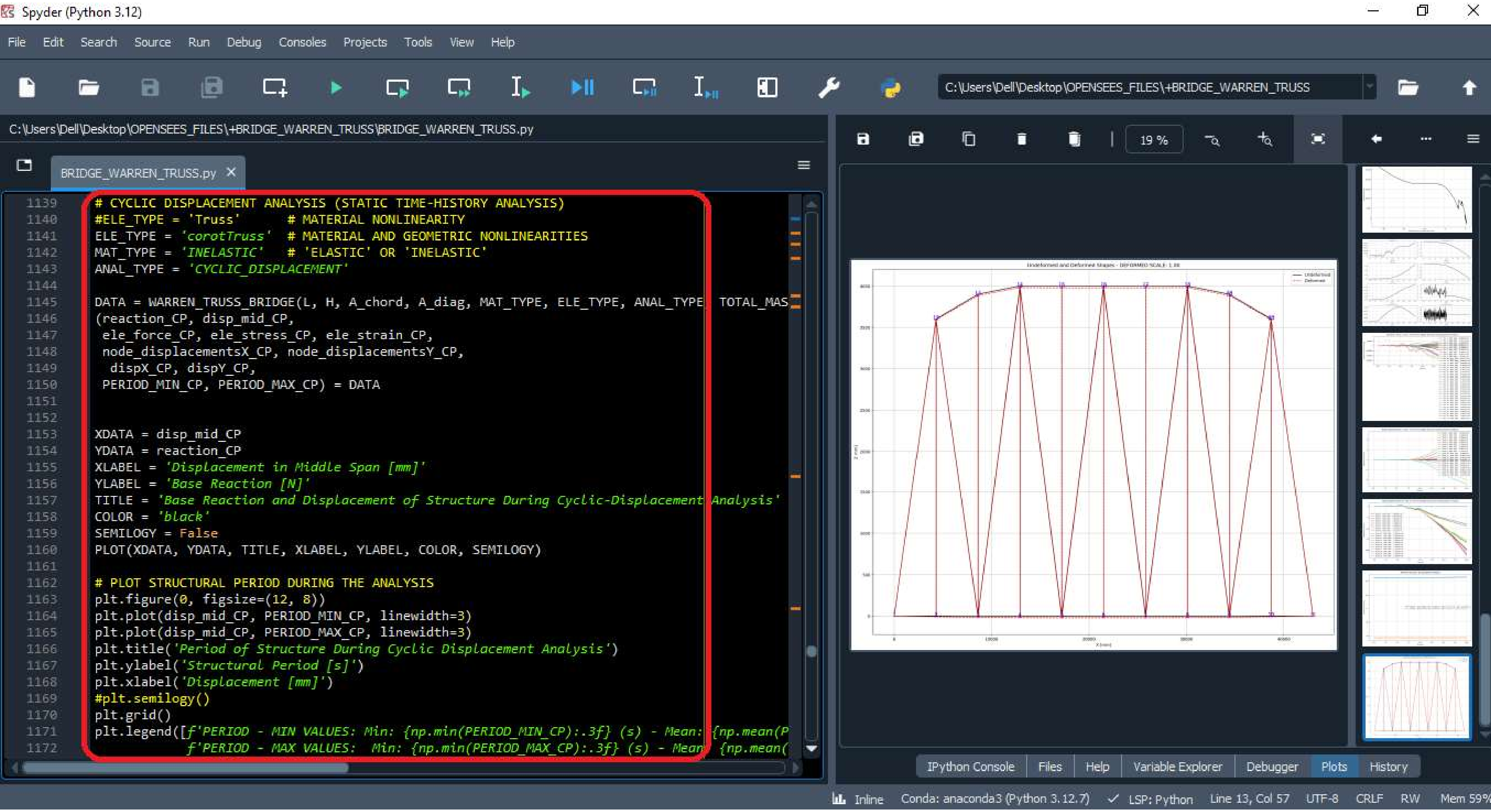


Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
800000
600000
400000
200000
0
−500
−400
−300
−200
−100
0
Displacement in Middle Span [mm]


Period of Structure During Pushover Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
Displacement [mm]
12000
10000
8000
6000
4000
2000
0
−500
−400
−300
−200
−100
0


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
1000
2000
3000
4000
0
10000
20000
30000
40000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

CYCLIC DISPLACEMENT ANALYSIS


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
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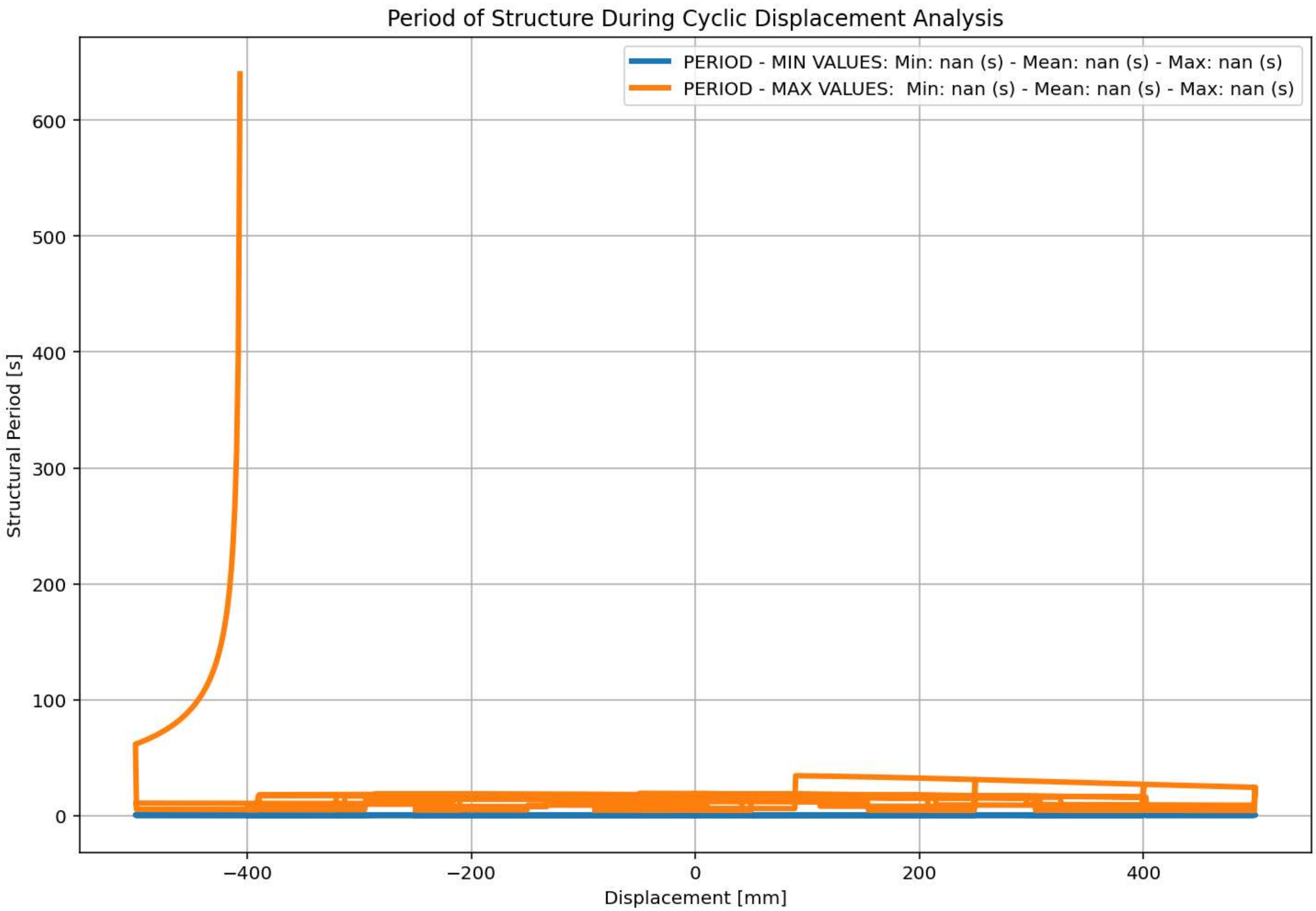
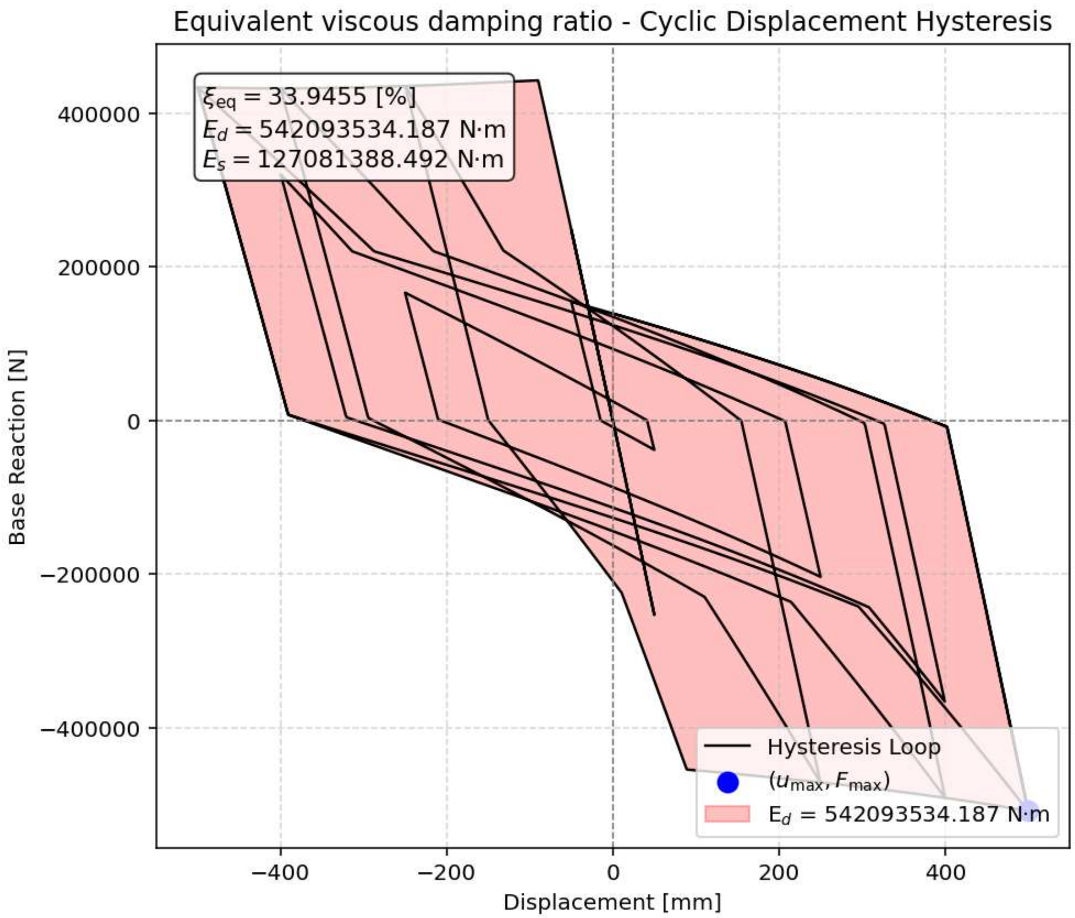
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_WARREN_TRUSS\BRIDGE_WARREN_TRUSS.py
BRIDGE_WARREN_TRUSS.py
CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(reaction_CP, disp_mid_CP,
 ele_force_CP, ele_stress_CP, ele_strain_CP,
 node_displacementsX_CP, node_displacementsY_CP,
  dispX_CP, dispY_CP,
 PERIOD_MIN_CP, PERIOD_MAX_CP) = DATA
XDATA = disp_mid_CP
YDATA = reaction_CP
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_CP, PERIOD_MIN_CP, linewidth=3)
plt.plot(disp_mid_CP, PERIOD_MAX_CP, linewidth=3)
plt.title('Period of Structure During Cyclic Displacement Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [mm]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_CP):.3f} (s) - Mean: {np.mean(P
            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_CP):.3f} (s) - Mean: {np.mean(
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 13, Col 57 UTF-8 CRLF RW Mem 59%

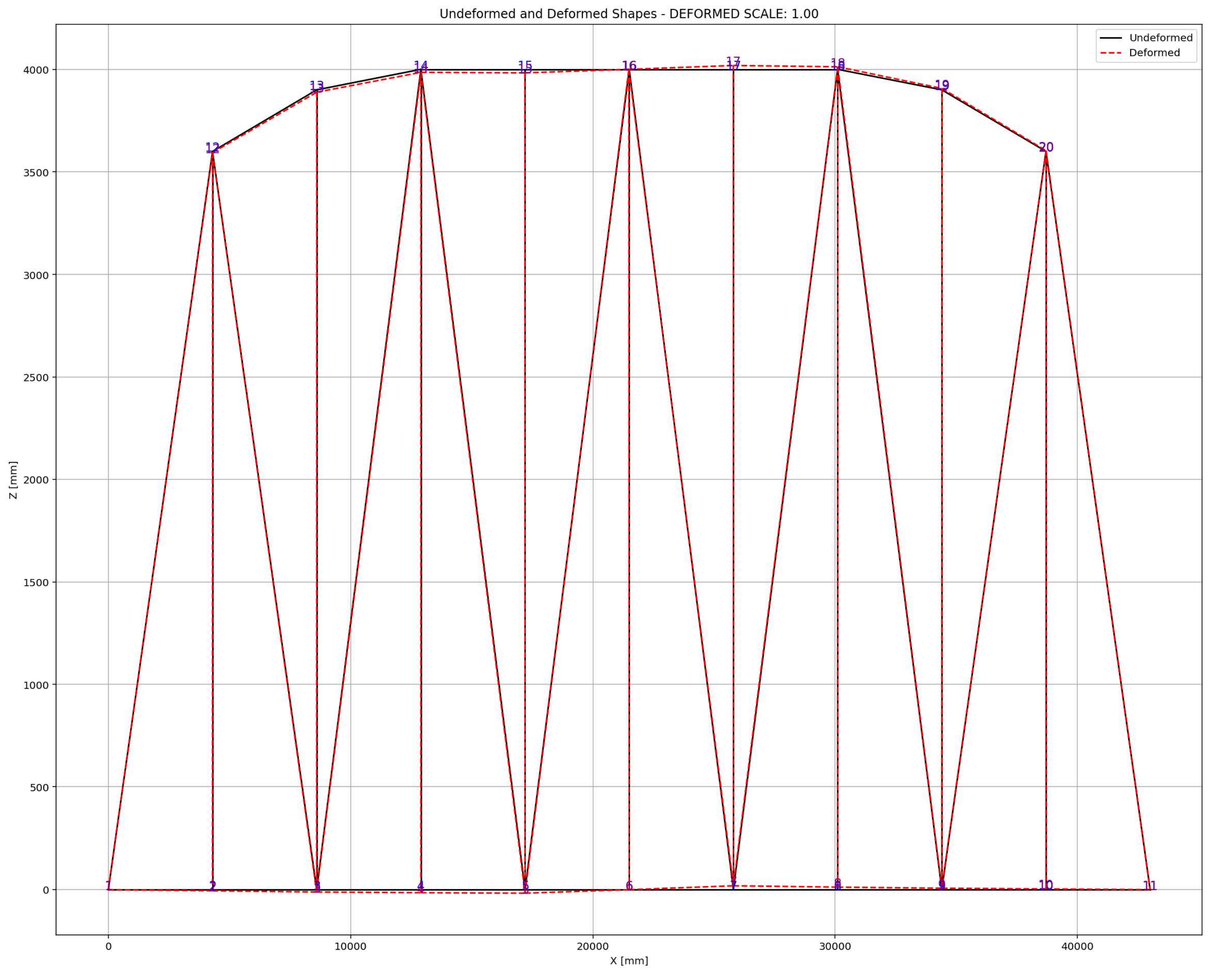

Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
1e6
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Base Reaction [N]
−400
−200
0
200
400
Displacement in Middle Span [mm]



Period of Structure During Cyclic Displacement Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
Displacement [mm]
600
500
400
300
200
100
0
−400
−200
0
200
400

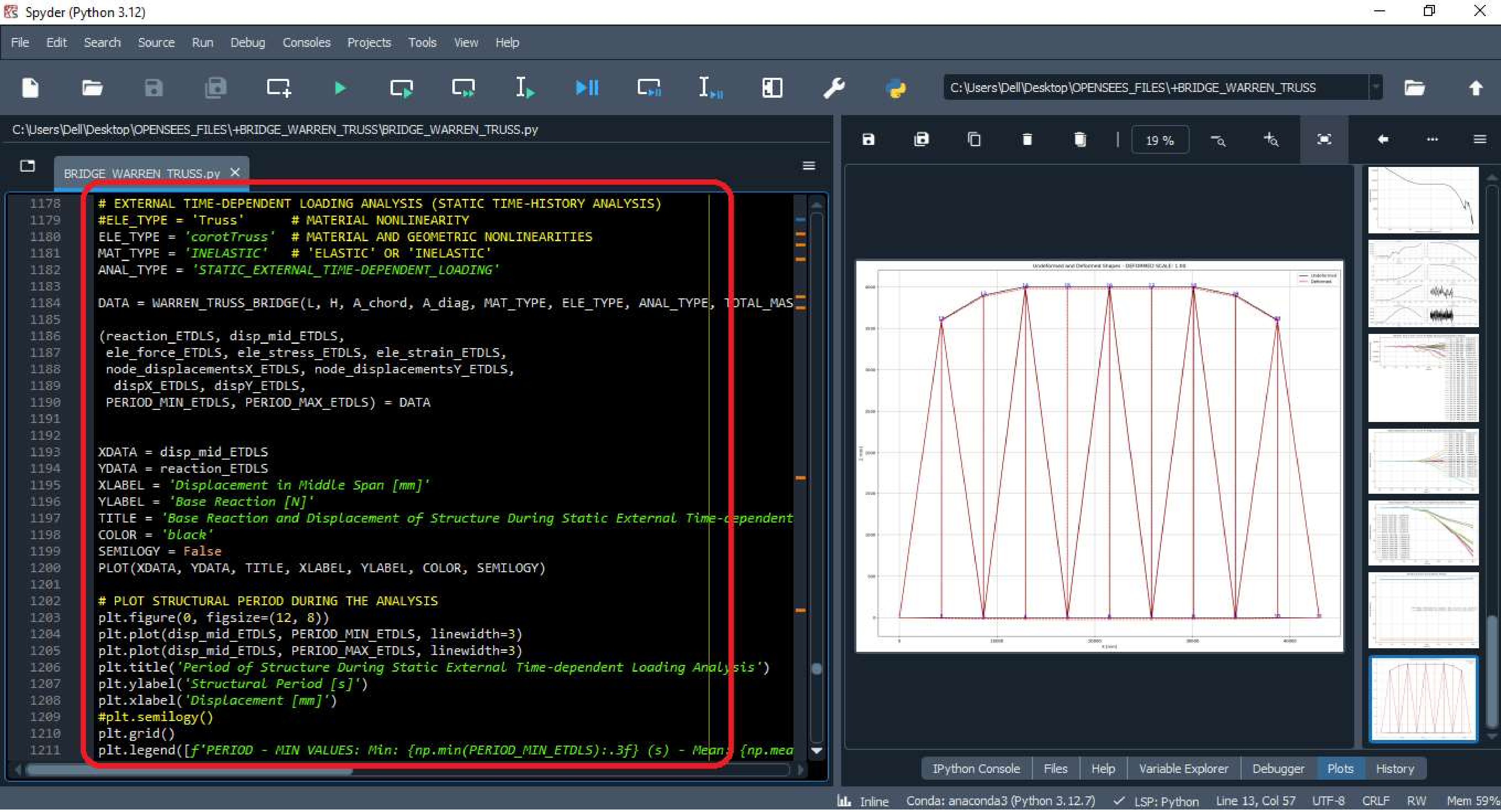


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
$\xi_{eq} = 33.9455$ [%]
$E_d = 542093534.187$ N·m
$E_s = 127081388.492$ N·m
400000
200000
0
−200000
−400000
Base Reaction [N]
−400
−200
0
200
400
Displacement [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 542093534.187 N·m



Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
10000
20000
30000
40000
500
1000
1500
2000
2500
3000
3500
4000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS


Spyder (Python 3.12)
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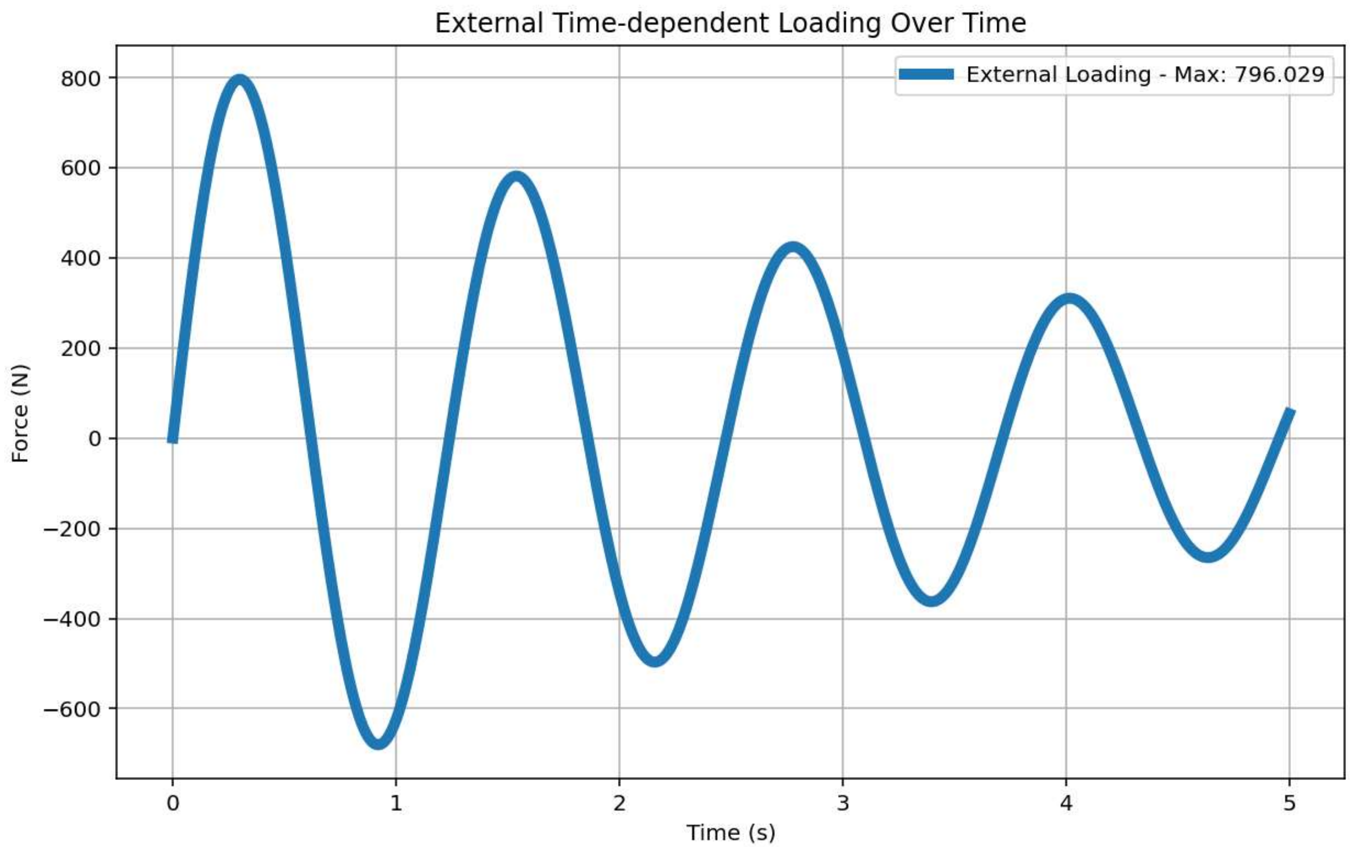
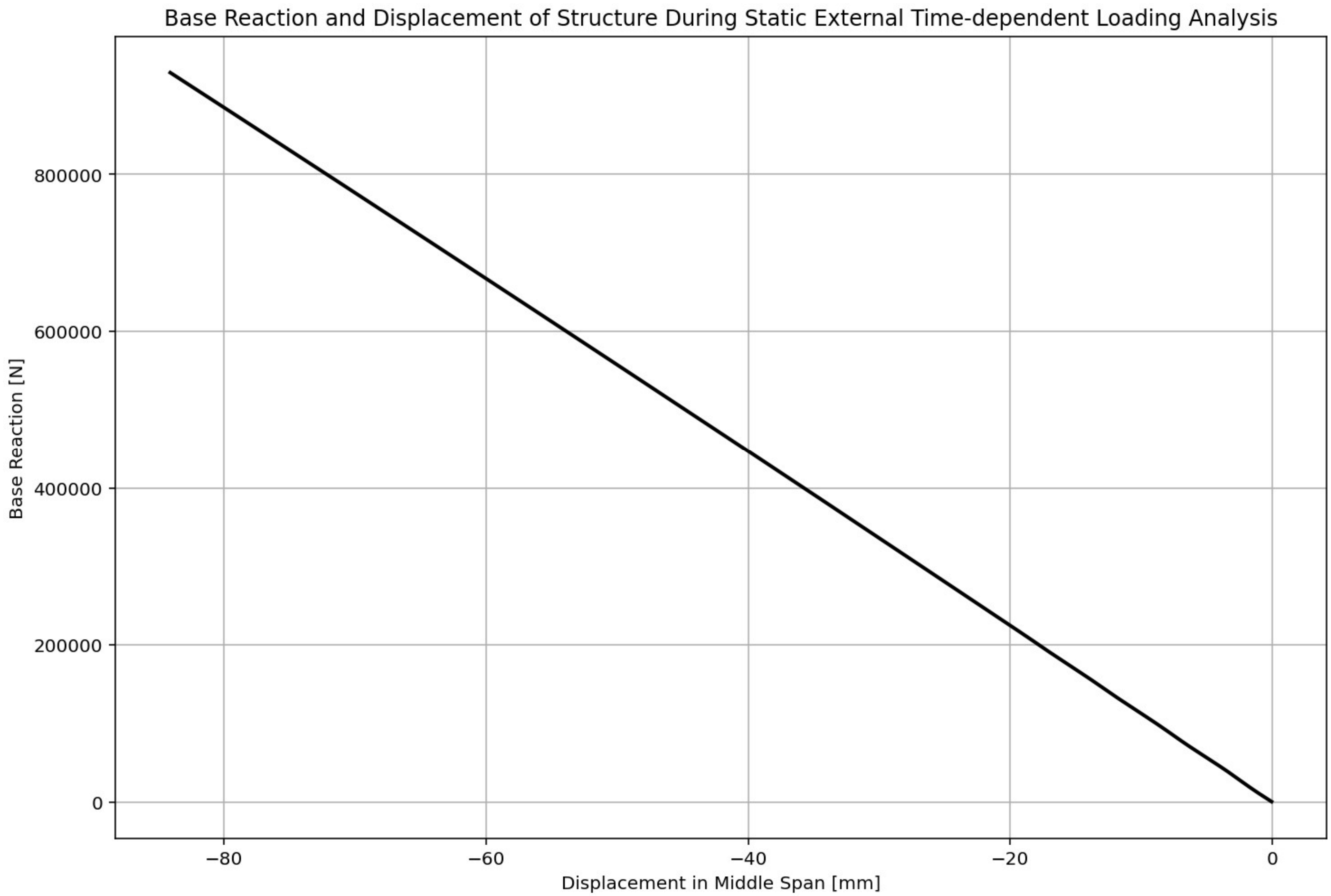
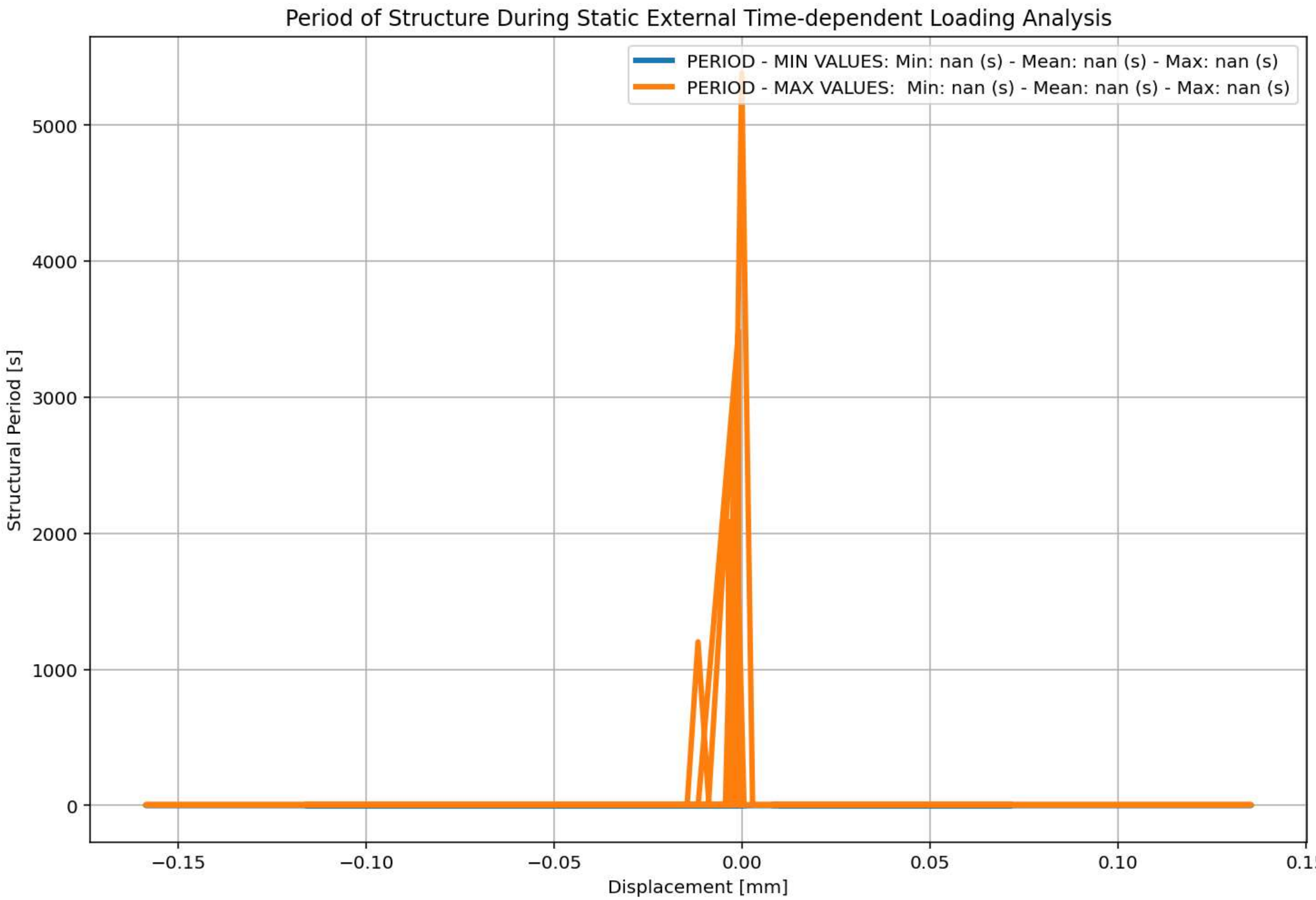
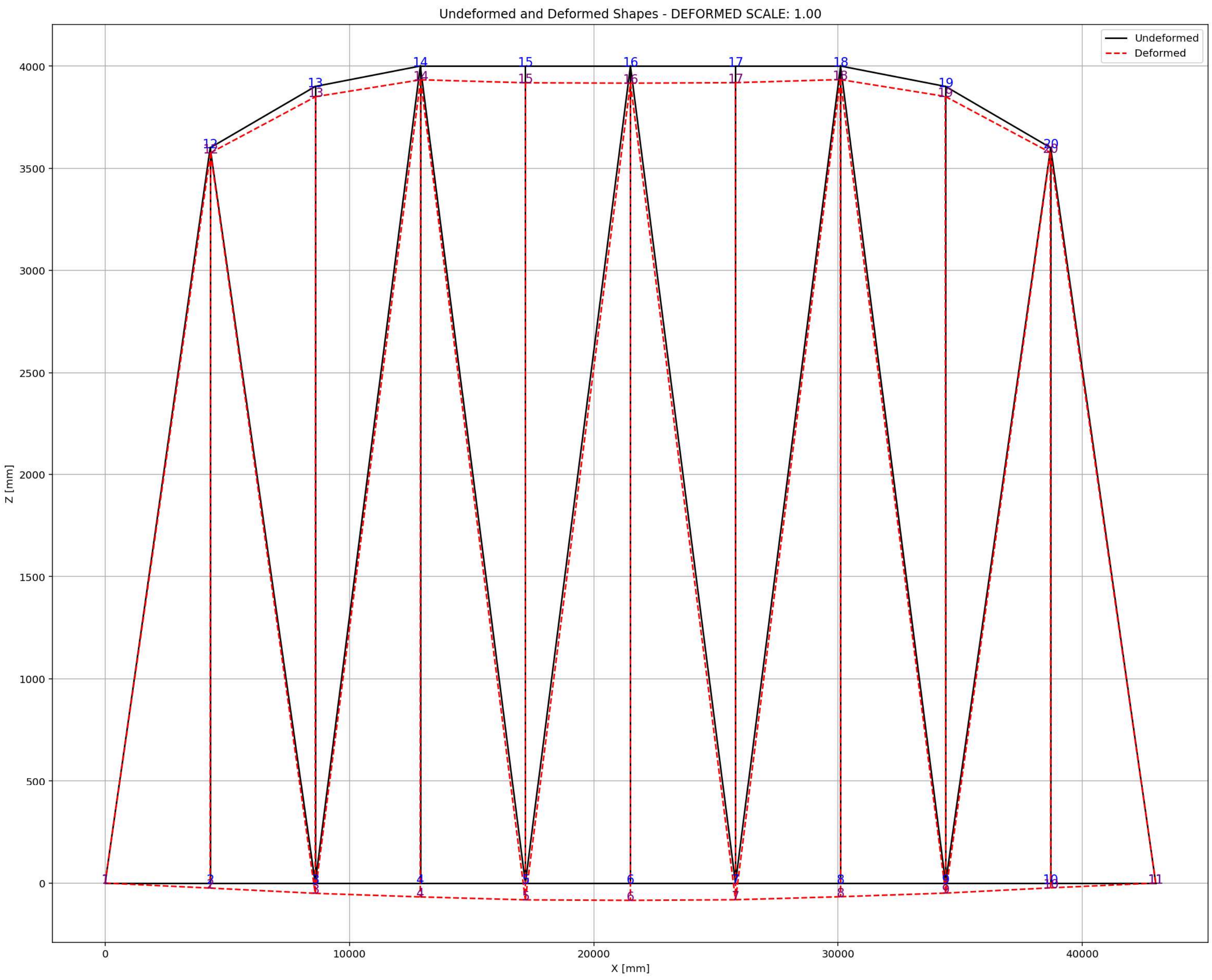
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_WARREN_TRUSS\BRIDGE_WARREN_TRUSS.py
BRIDGE_WARREN_TRUSS.py
EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(reaction_ETDLS, disp_mid_ETDLS,
 ele_force_ETDLS, ele_stress_ETDLS, ele_strain_ETDLS,
 node_displacementsX_ETDLS, node_displacementsY_ETDLS,
  dispX_ETDLS, dispY_ETDLS,
 PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS) = DATA
XDATA = disp_mid_ETDLS
YDATA = reaction_ETDLS
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [mm]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLS):.3f} (s) - Mean: {np.mea
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 13, Col 57 UTF-8 CRLF RW Mem 59%


External Time-dependent Loading Over Time
External Loading - Max: 796.029
800
600
400
200
0
−200
−400
−600
Force (N)
0
1
2
3
4
5
Time (s)

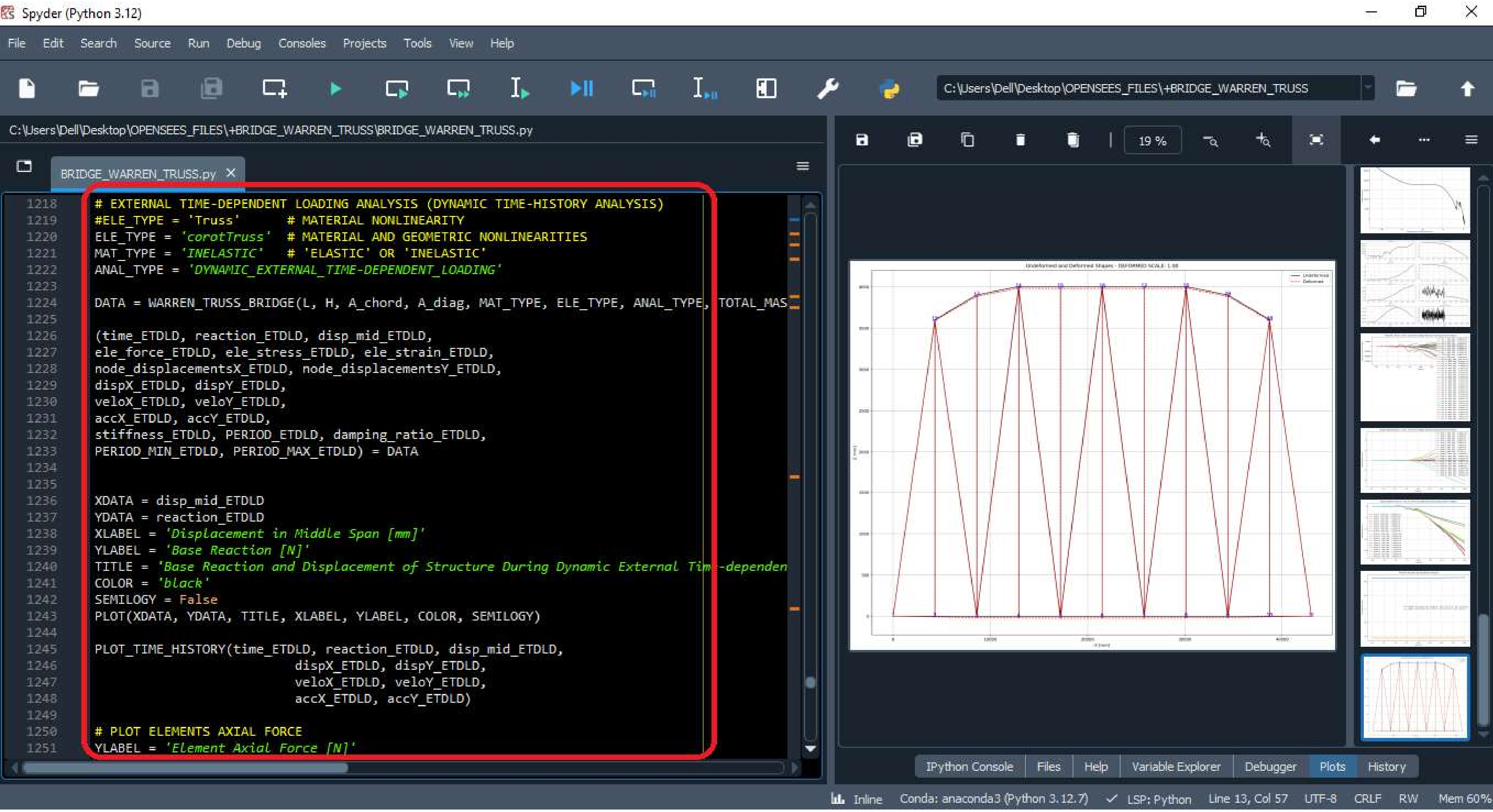

Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
800000
600000
400000
200000
0
−80
−60
−40
−20
0
Displacement in Middle Span [mm]


Period of Structure During Static External Time-dependent Loading Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
5000
4000
3000
2000
1000
0
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
Displacement [mm]



Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
3500
4000
10000
20000
30000
40000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS


Spyder (Python 3.12)
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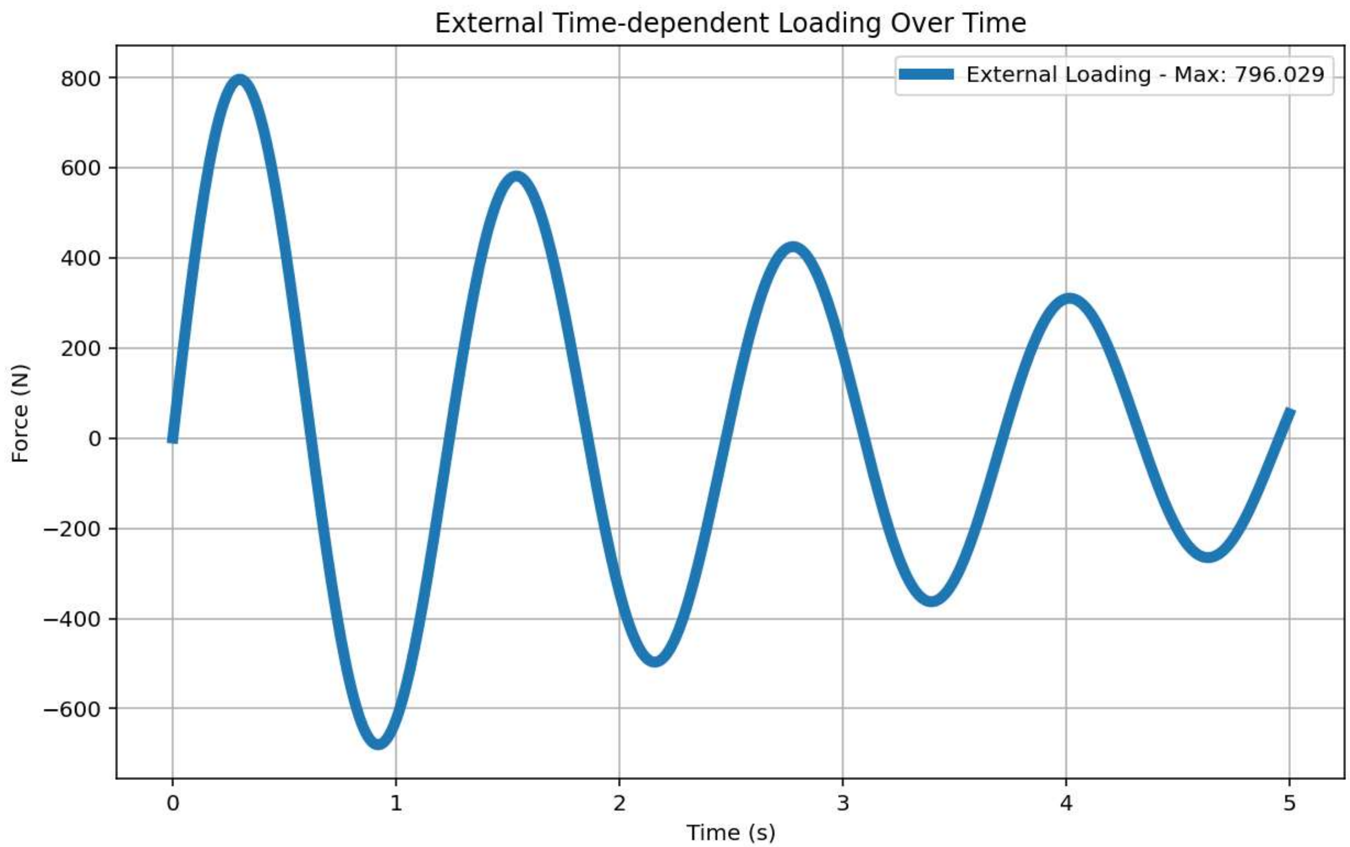
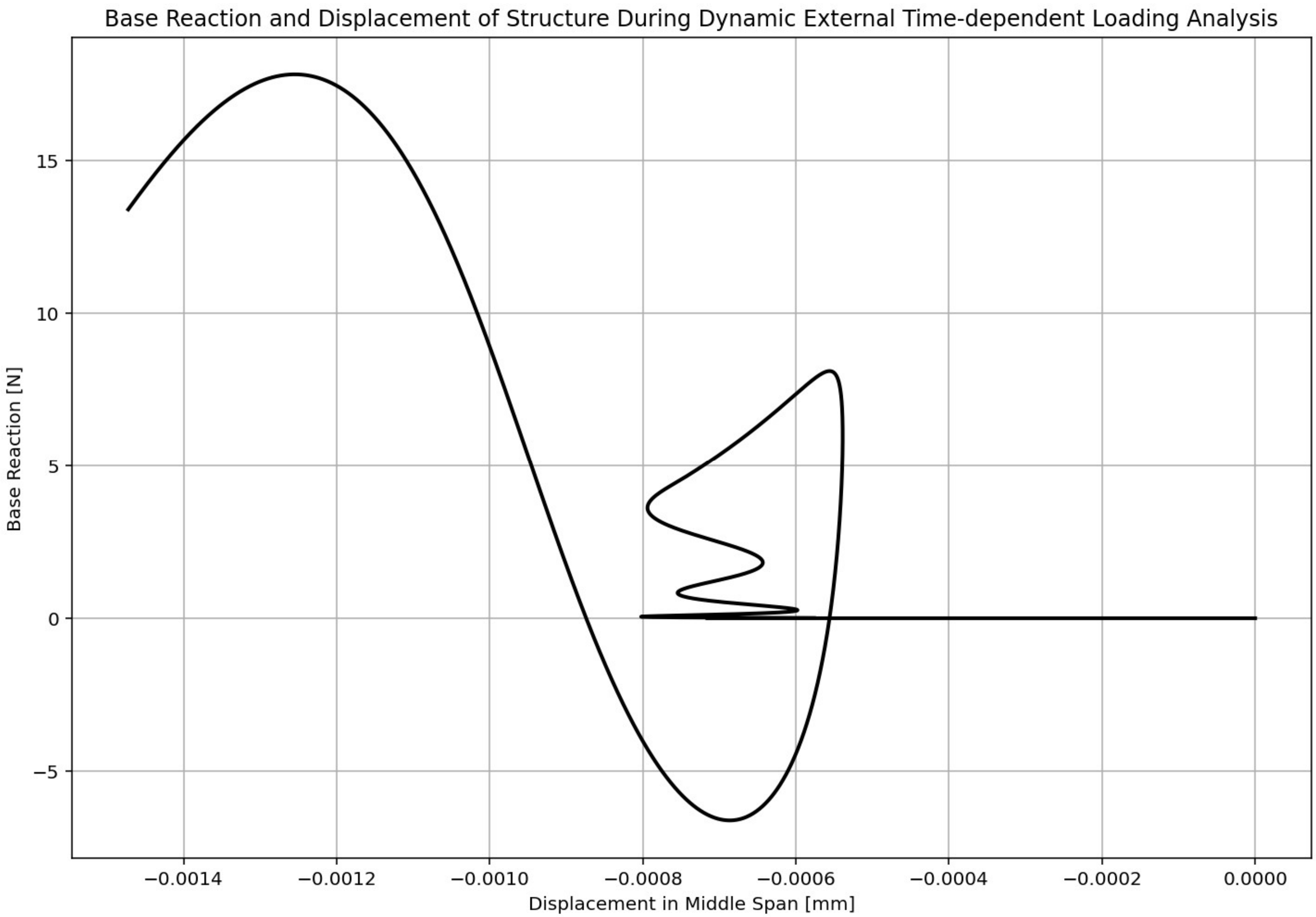
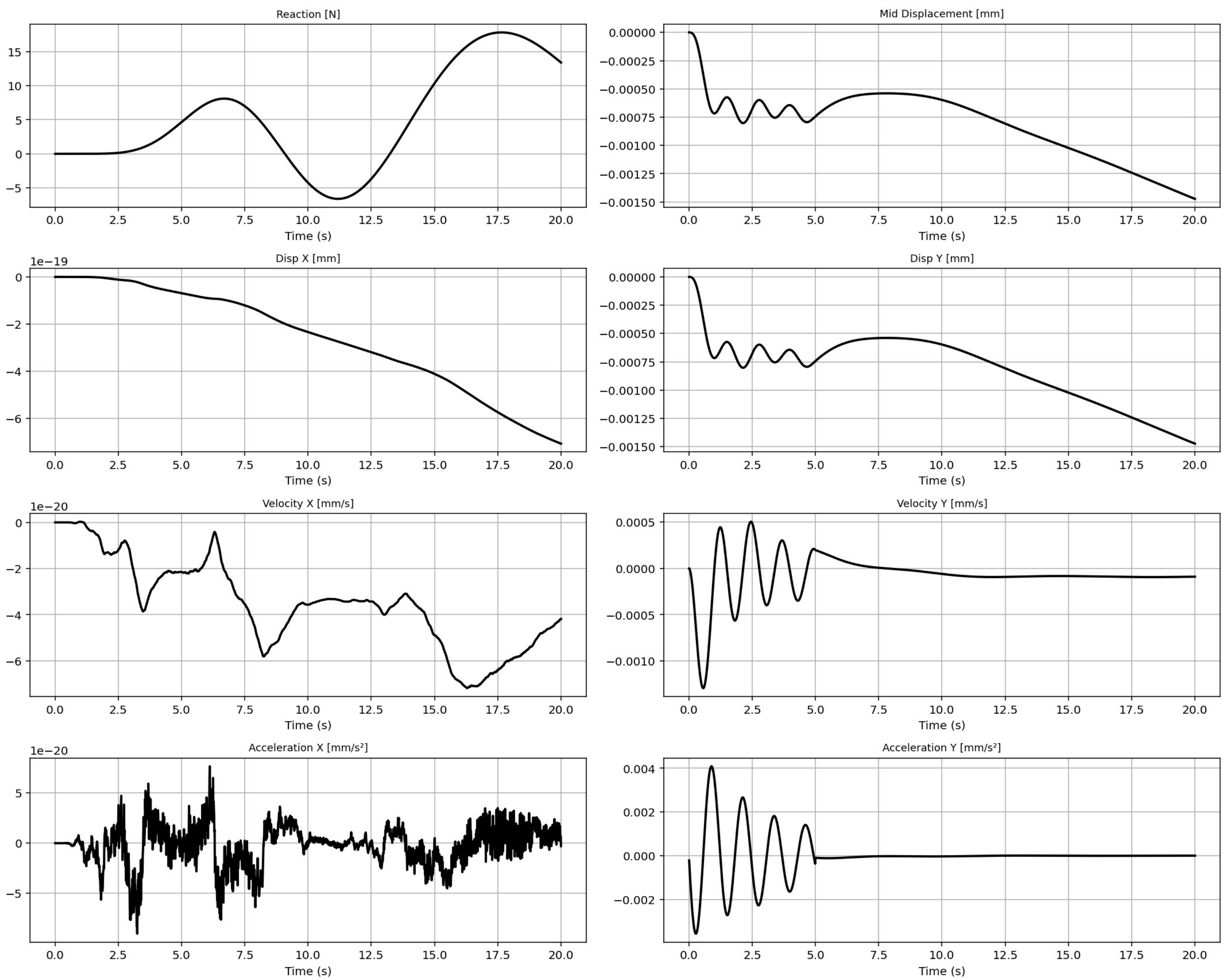
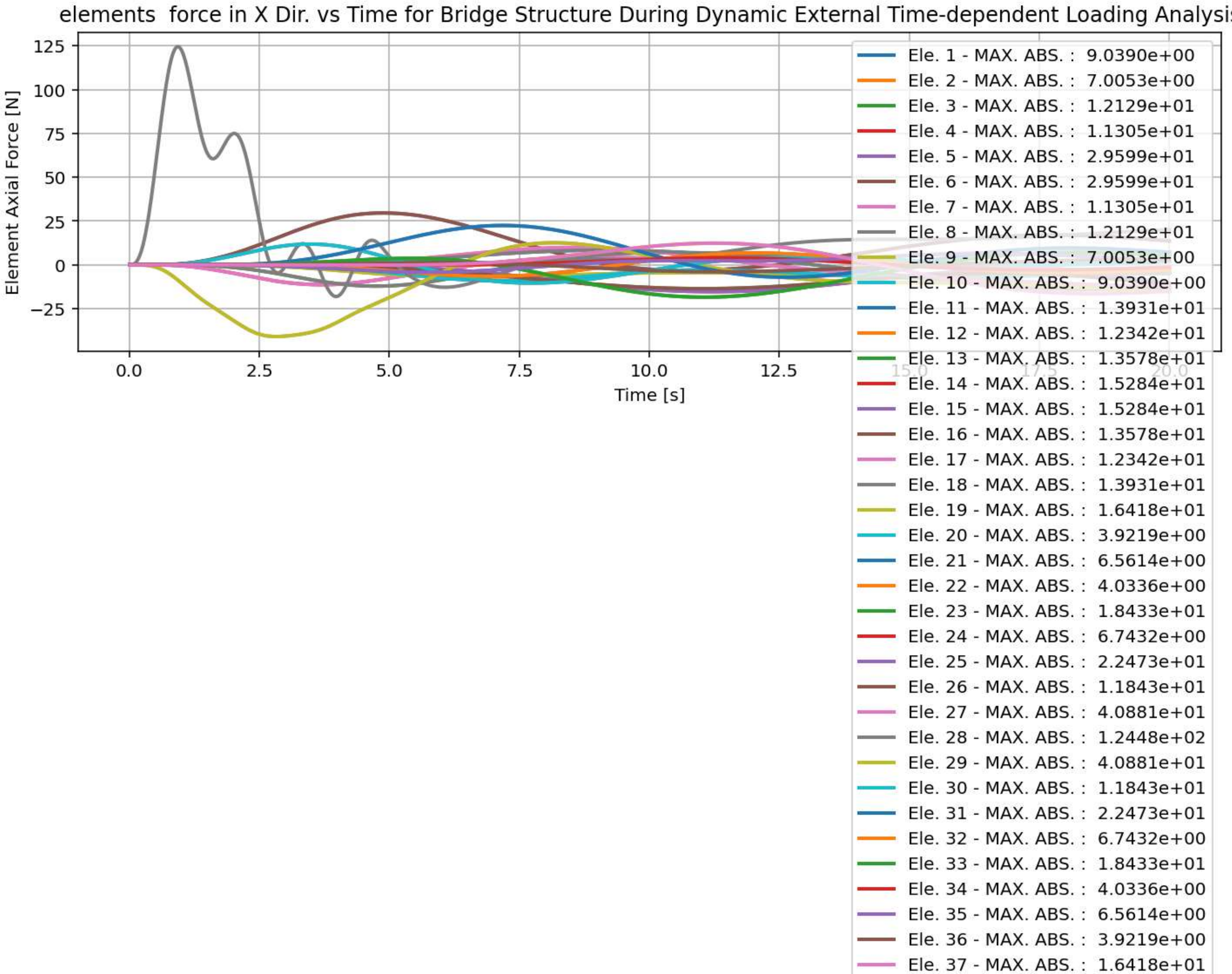
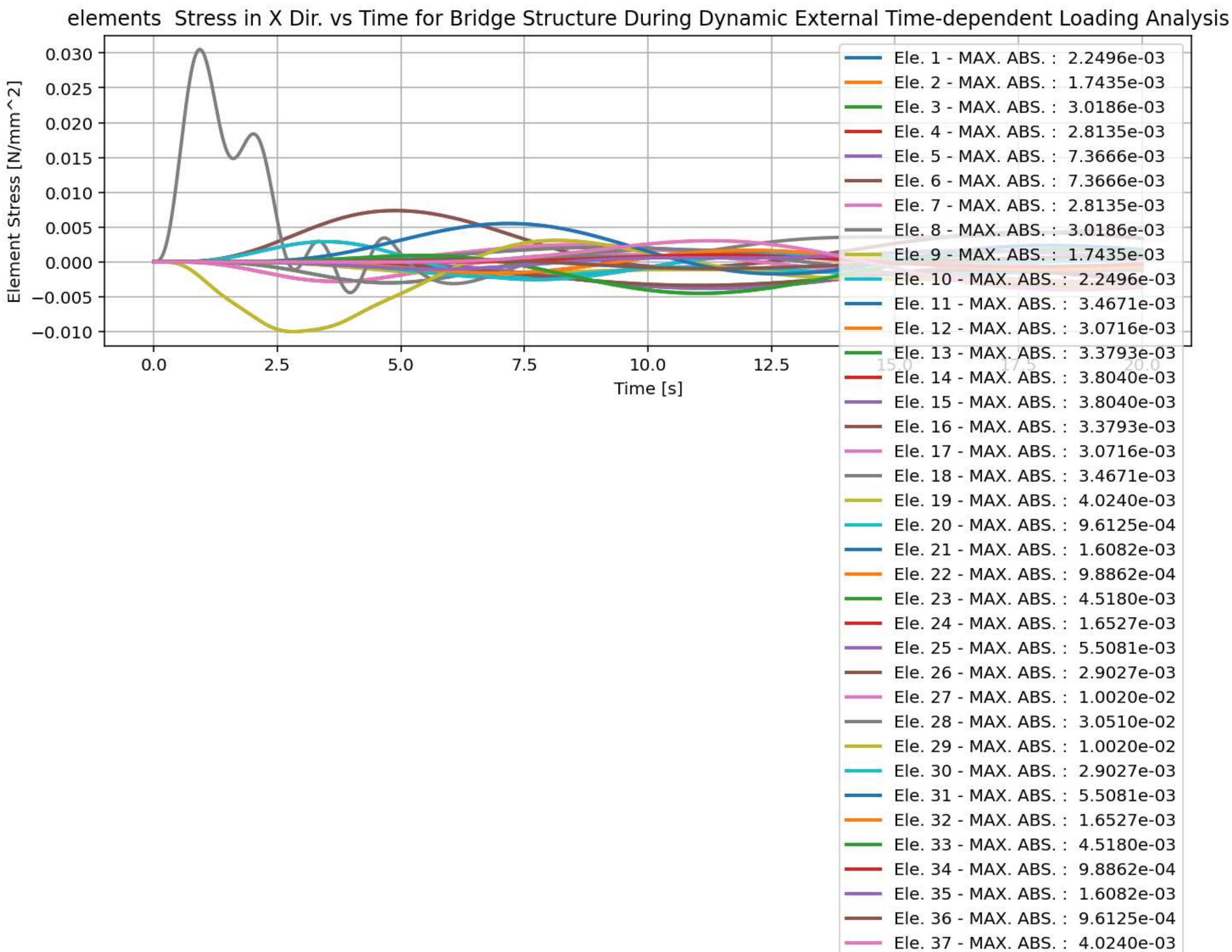
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_WARREN_TRUSS\BRIDGE_WARREN_TRUSS.py
BRIDGE_WARREN_TRUSS.py
EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
ele_force_ETDLD, ele_stress_ETDLD, ele_strain_ETDLD,
node_displacementsX_ETDLD, node_displacementsY_ETDLD,
dispX_ETDLD, dispY_ETDLD,
veloX_ETDLD, veloY_ETDLD,
accX_ETDLD, accY_ETDLD,
stiffness_ETDLD, PERIOD_ETDLD, damping_ratio_ETDLD,
PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD) = DATA
XDATA = disp_mid_ETDLD
YDATA = reaction_ETDLD
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependen
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
dispX_ETDLD, dispY_ETDLD,
veloX_ETDLD, veloY_ETDLD,
accX_ETDLD, accY_ETDLD)
# PLOT ELEMENTS AXIAL FORCE
YLABEL = 'Element Axial Force [N]'
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 13, Col 57 UTF-8 CRLF RW Mem 60%

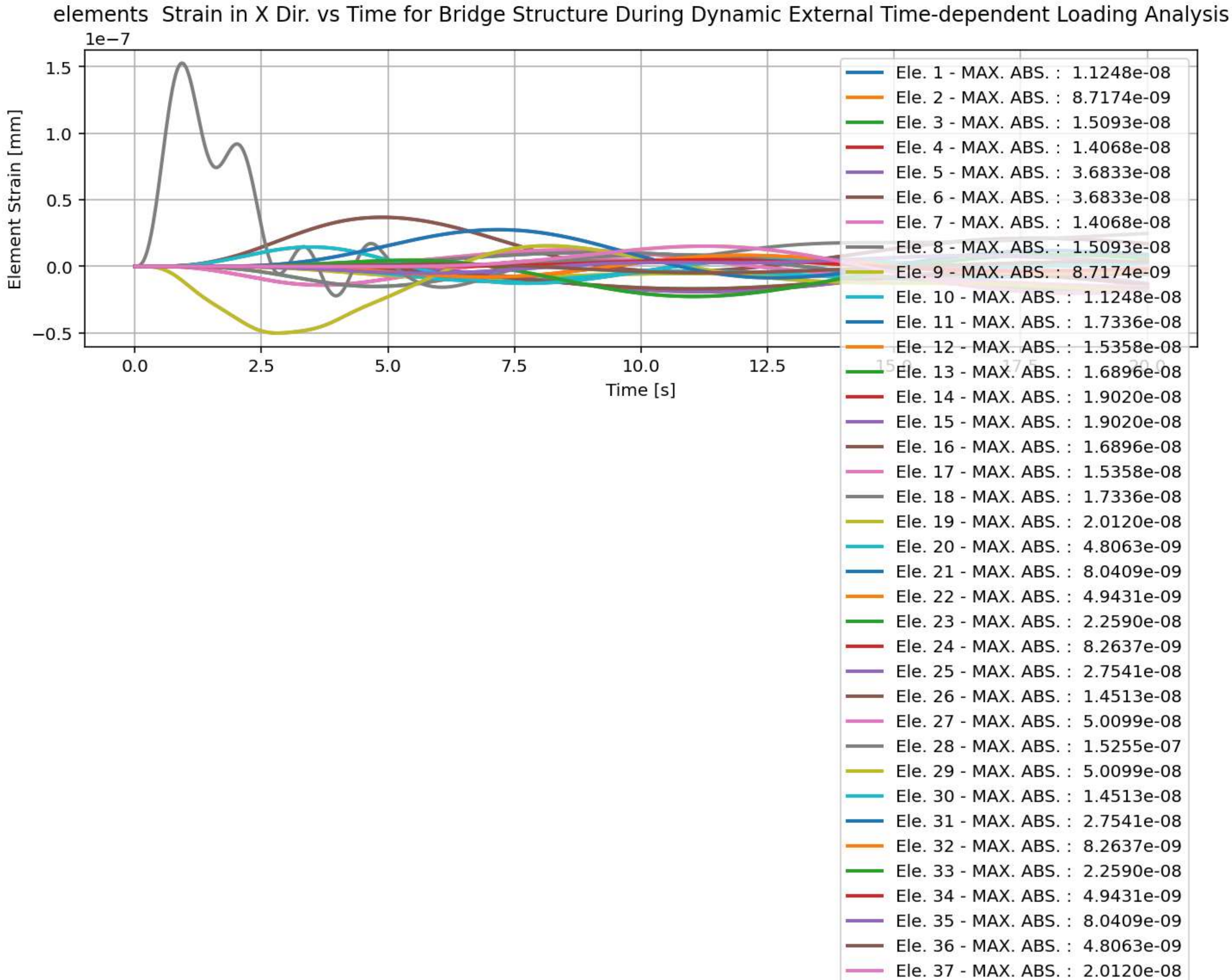

External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
Time (s)
800
600
400
200
0
−200
−400
−600
0
1
2
3
4
5

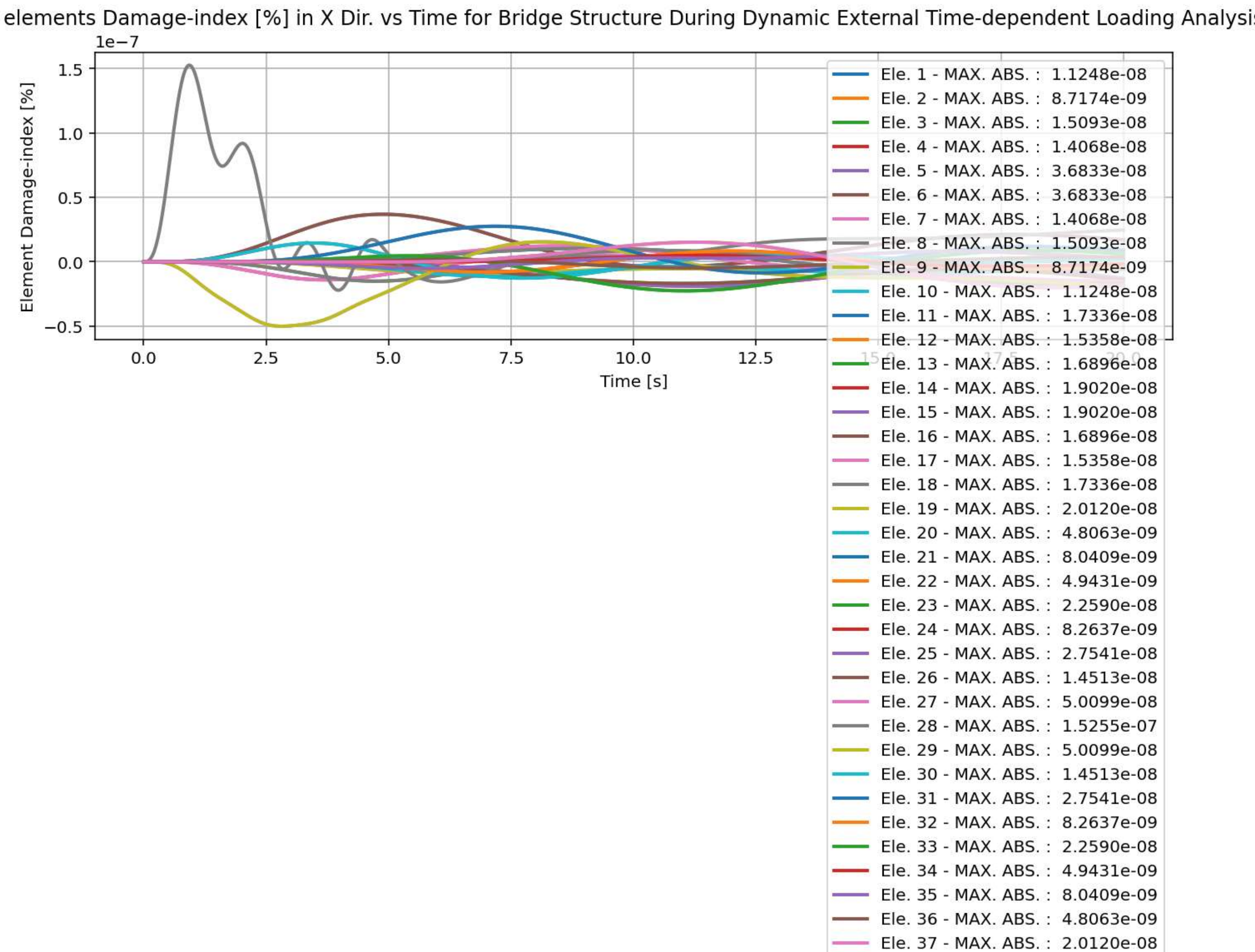

Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
15
10
5
0
−5
−0.0014
−0.0012
−0.0010
−0.0008
−0.0006
−0.0004
−0.0002
0.0000


Reaction [N]
Mid Displacement [mm]
Disp X [mm]
1e−19
Disp Y [mm]
Velocity X [mm/s]
1e−20
Velocity Y [mm/s]
Acceleration X [mm/s²]
1e−20
Acceleration Y [mm/s²]
Time (s)

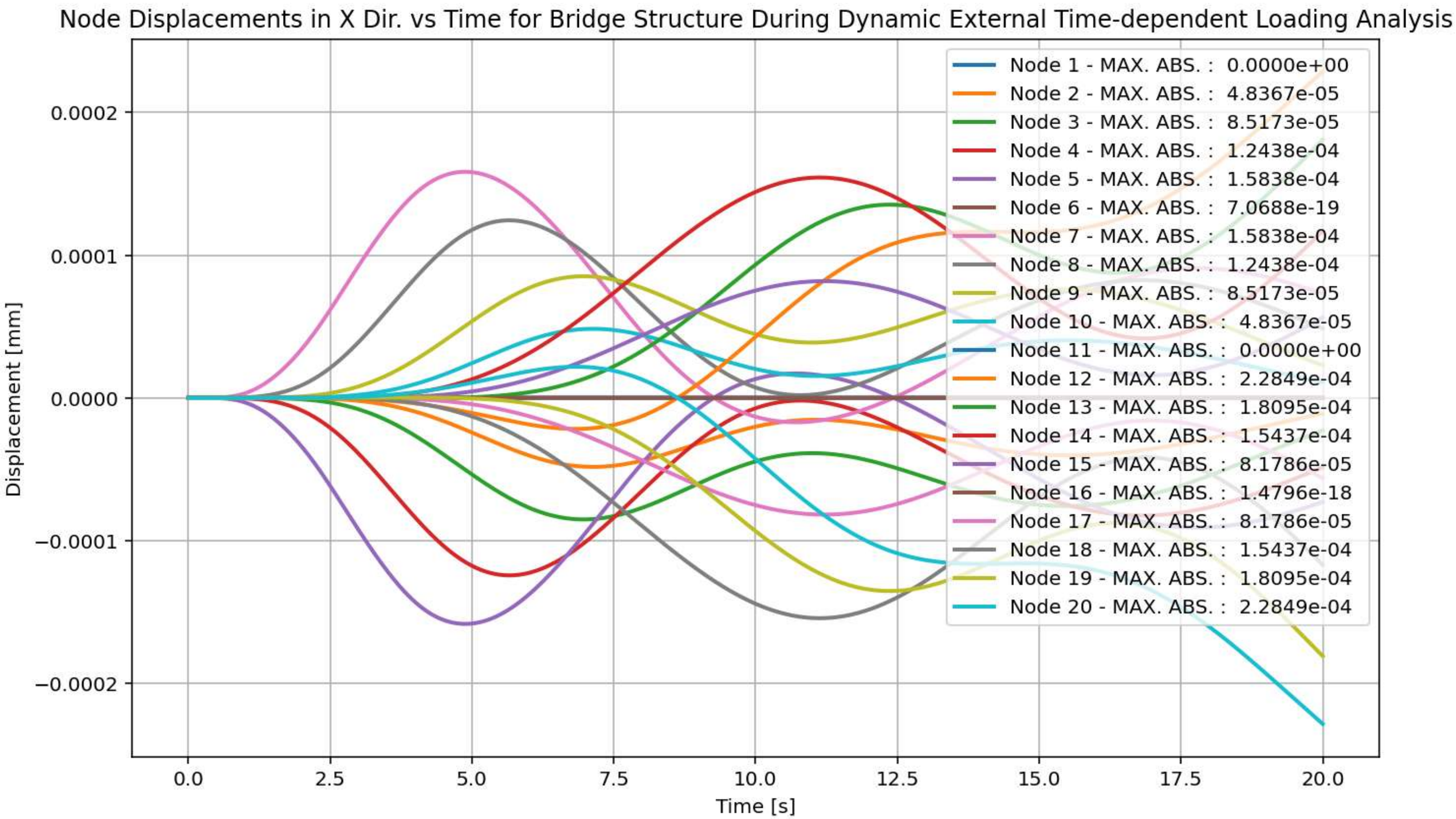

elements force in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Element Axial Force [N]
Time [s]
125
100
75
50
25
0
−25
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 9.0390e+00
Ele. 2 - MAX. ABS. : 7.0053e+00
Ele. 3 - MAX. ABS. : 1.2129e+01
Ele. 4 - MAX. ABS. : 1.1305e+01
Ele. 5 - MAX. ABS. : 2.9599e+01
Ele. 6 - MAX. ABS. : 2.9599e+01
Ele. 7 - MAX. ABS. : 1.1305e+01
Ele. 8 - MAX. ABS. : 1.2129e+01
Ele. 9 - MAX. ABS. : 7.0053e+00
Ele. 10 - MAX. ABS. : 9.0390e+00
Ele. 11 - MAX. ABS. : 1.3931e+01
Ele. 12 - MAX. ABS. : 1.2342e+01
Ele. 13 - MAX. ABS. : 1.3578e+01
Ele. 14 - MAX. ABS. : 1.5284e+01
Ele. 15 - MAX. ABS. : 1.5284e+01
Ele. 16 - MAX. ABS. : 1.3578e+01
Ele. 17 - MAX. ABS. : 1.2342e+01
Ele. 18 - MAX. ABS. : 1.3931e+01
Ele. 19 - MAX. ABS. : 1.6418e+01
Ele. 20 - MAX. ABS. : 3.9219e+00
Ele. 21 - MAX. ABS. : 6.5614e+00
Ele. 22 - MAX. ABS. : 4.0336e+00
Ele. 23 - MAX. ABS. : 1.8433e+01
Ele. 24 - MAX. ABS. : 6.7432e+00
Ele. 25 - MAX. ABS. : 2.2473e+01
Ele. 26 - MAX. ABS. : 1.1843e+01
Ele. 27 - MAX. ABS. : 4.0881e+01
Ele. 28 - MAX. ABS. : 1.2448e+02
Ele. 29 - MAX. ABS. : 4.0881e+01
Ele. 30 - MAX. ABS. : 1.1843e+01
Ele. 31 - MAX. ABS. : 2.2473e+01
Ele. 32 - MAX. ABS. : 6.7432e+00
Ele. 33 - MAX. ABS. : 1.8433e+01
Ele. 34 - MAX. ABS. : 4.0336e+00
Ele. 35 - MAX. ABS. : 6.5614e+00
Ele. 36 - MAX. ABS. : 3.9219e+00
Ele. 37 - MAX. ABS. : 1.6418e+01

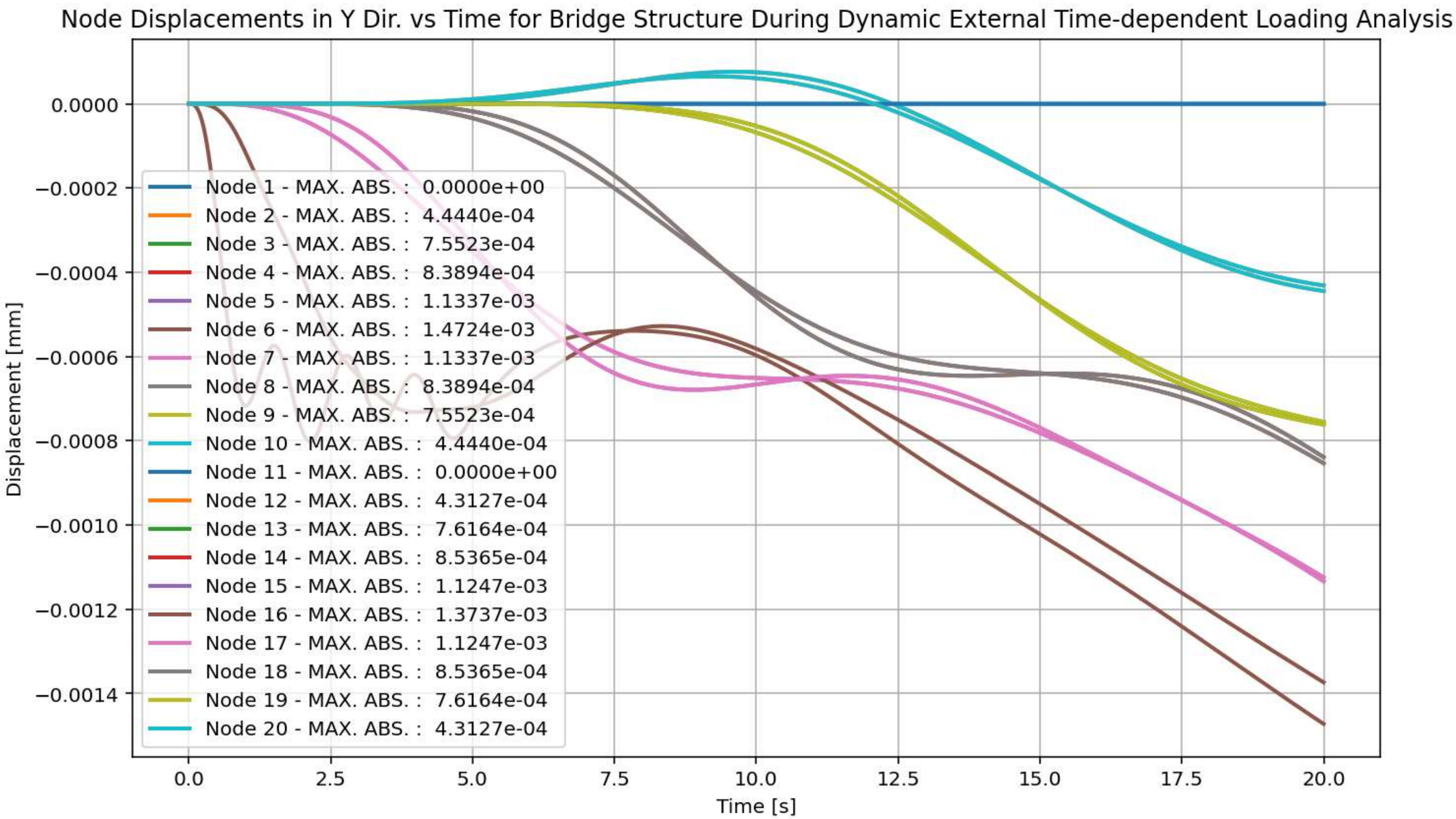

elements Stress in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Element Stress [N/mm^2]
Time [s]
0.030
0.025
0.020
0.015
0.010
0.005
0.000
−0.005
−0.010
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 2.2496e-03
Ele. 2 - MAX. ABS. : 1.7435e-03
Ele. 3 - MAX. ABS. : 3.0186e-03
Ele. 4 - MAX. ABS. : 2.8135e-03
Ele. 5 - MAX. ABS. : 7.3666e-03
Ele. 6 - MAX. ABS. : 7.3666e-03
Ele. 7 - MAX. ABS. : 2.8135e-03
Ele. 8 - MAX. ABS. : 3.0186e-03
Ele. 9 - MAX. ABS. : 1.7435e-03
Ele. 10 - MAX. ABS. : 2.2496e-03
Ele. 11 - MAX. ABS. : 3.4671e-03
Ele. 12 - MAX. ABS. : 3.0716e-03
Ele. 13 - MAX. ABS. : 3.3793e-03
Ele. 14 - MAX. ABS. : 3.8040e-03
Ele. 15 - MAX. ABS. : 3.8040e-03
Ele. 16 - MAX. ABS. : 3.3793e-03
Ele. 17 - MAX. ABS. : 3.0716e-03
Ele. 18 - MAX. ABS. : 3.4671e-03
Ele. 19 - MAX. ABS. : 4.0240e-03
Ele. 20 - MAX. ABS. : 9.6125e-04
Ele. 21 - MAX. ABS. : 1.6082e-03
Ele. 22 - MAX. ABS. : 9.8862e-04
Ele. 23 - MAX. ABS. : 4.5180e-03
Ele. 24 - MAX. ABS. : 1.6527e-03
Ele. 25 - MAX. ABS. : 5.5081e-03
Ele. 26 - MAX. ABS. : 2.9027e-03
Ele. 27 - MAX. ABS. : 1.0020e-02
Ele. 28 - MAX. ABS. : 3.0510e-02
Ele. 29 - MAX. ABS. : 1.0020e-02
Ele. 30 - MAX. ABS. : 2.9027e-03
Ele. 31 - MAX. ABS. : 5.5081e-03
Ele. 32 - MAX. ABS. : 1.6527e-03
Ele. 33 - MAX. ABS. : 4.5180e-03
Ele. 34 - MAX. ABS. : 9.8862e-04
Ele. 35 - MAX. ABS. : 1.6082e-03
Ele. 36 - MAX. ABS. : 9.6125e-04
Ele. 37 - MAX. ABS. : 4.0240e-03


elements Strain in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
1e−7
Element Strain [mm]
Time [s]
Ele. 1 - MAX. ABS. : 1.1248e-08
Ele. 2 - MAX. ABS. : 8.7174e-09
Ele. 3 - MAX. ABS. : 1.5093e-08
Ele. 4 - MAX. ABS. : 1.4068e-08
Ele. 5 - MAX. ABS. : 3.6833e-08
Ele. 6 - MAX. ABS. : 3.6833e-08
Ele. 7 - MAX. ABS. : 1.4068e-08
Ele. 8 - MAX. ABS. : 1.5093e-08
Ele. 9 - MAX. ABS. : 8.7174e-09
Ele. 10 - MAX. ABS. : 1.1248e-08
Ele. 11 - MAX. ABS. : 1.7336e-08
Ele. 12 - MAX. ABS. : 1.5358e-08
Ele. 13 - MAX. ABS. : 1.6896e-08
Ele. 14 - MAX. ABS. : 1.9020e-08
Ele. 15 - MAX. ABS. : 1.9020e-08
Ele. 16 - MAX. ABS. : 1.6896e-08
Ele. 17 - MAX. ABS. : 1.5358e-08
Ele. 18 - MAX. ABS. : 1.7336e-08
Ele. 19 - MAX. ABS. : 2.0120e-08
Ele. 20 - MAX. ABS. : 4.8063e-09
Ele. 21 - MAX. ABS. : 8.0409e-09
Ele. 22 - MAX. ABS. : 4.9431e-09
Ele. 23 - MAX. ABS. : 2.2590e-08
Ele. 24 - MAX. ABS. : 8.2637e-09
Ele. 25 - MAX. ABS. : 2.7541e-08
Ele. 26 - MAX. ABS. : 1.4513e-08
Ele. 27 - MAX. ABS. : 5.0099e-08
Ele. 28 - MAX. ABS. : 1.5255e-07
Ele. 29 - MAX. ABS. : 5.0099e-08
Ele. 30 - MAX. ABS. : 1.4513e-08
Ele. 31 - MAX. ABS. : 2.7541e-08
Ele. 32 - MAX. ABS. : 8.2637e-09
Ele. 33 - MAX. ABS. : 2.2590e-08
Ele. 34 - MAX. ABS. : 4.9431e-09
Ele. 35 - MAX. ABS. : 8.0409e-09
Ele. 36 - MAX. ABS. : 4.8063e-09
Ele. 37 - MAX. ABS. : 2.0120e-08

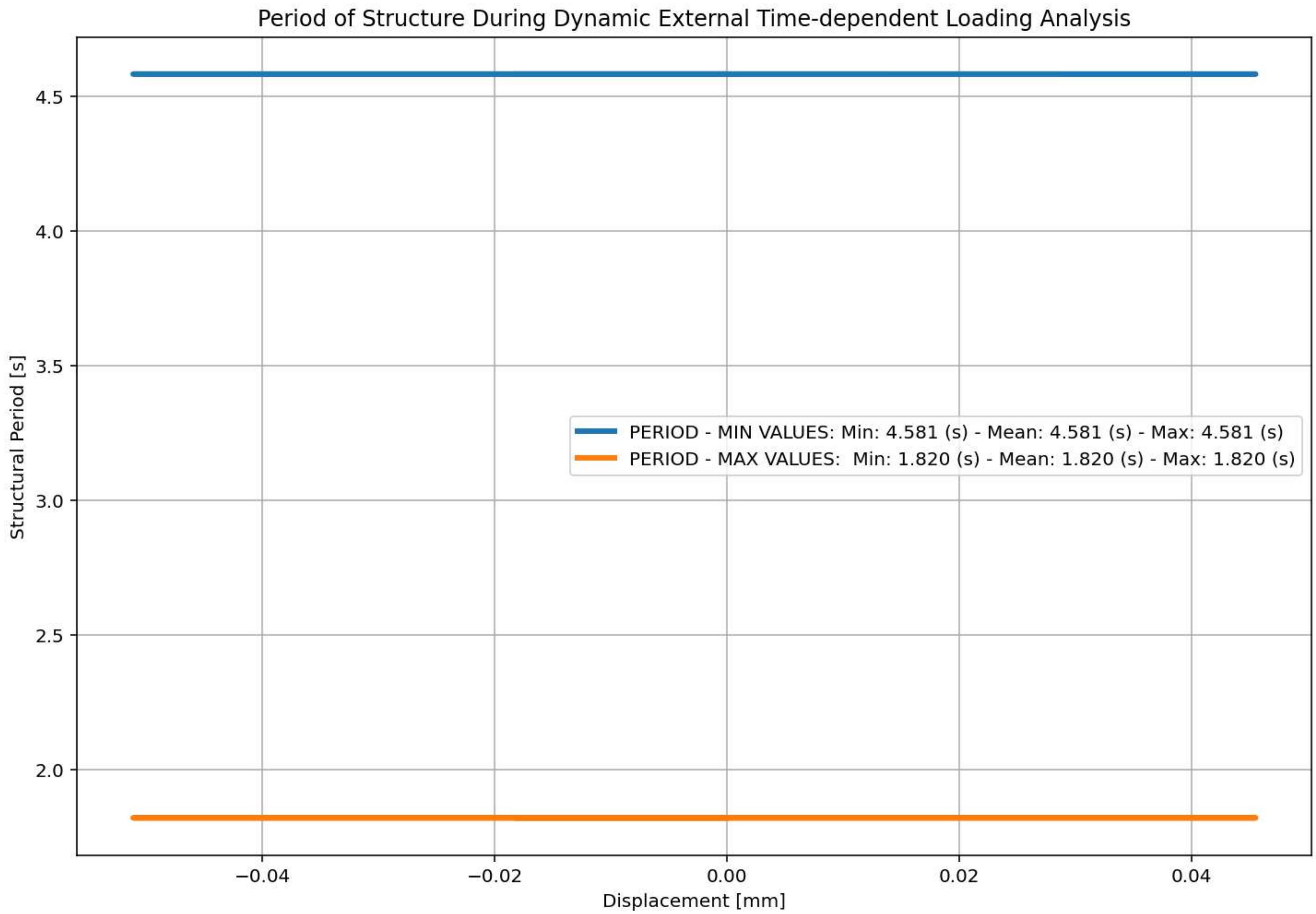


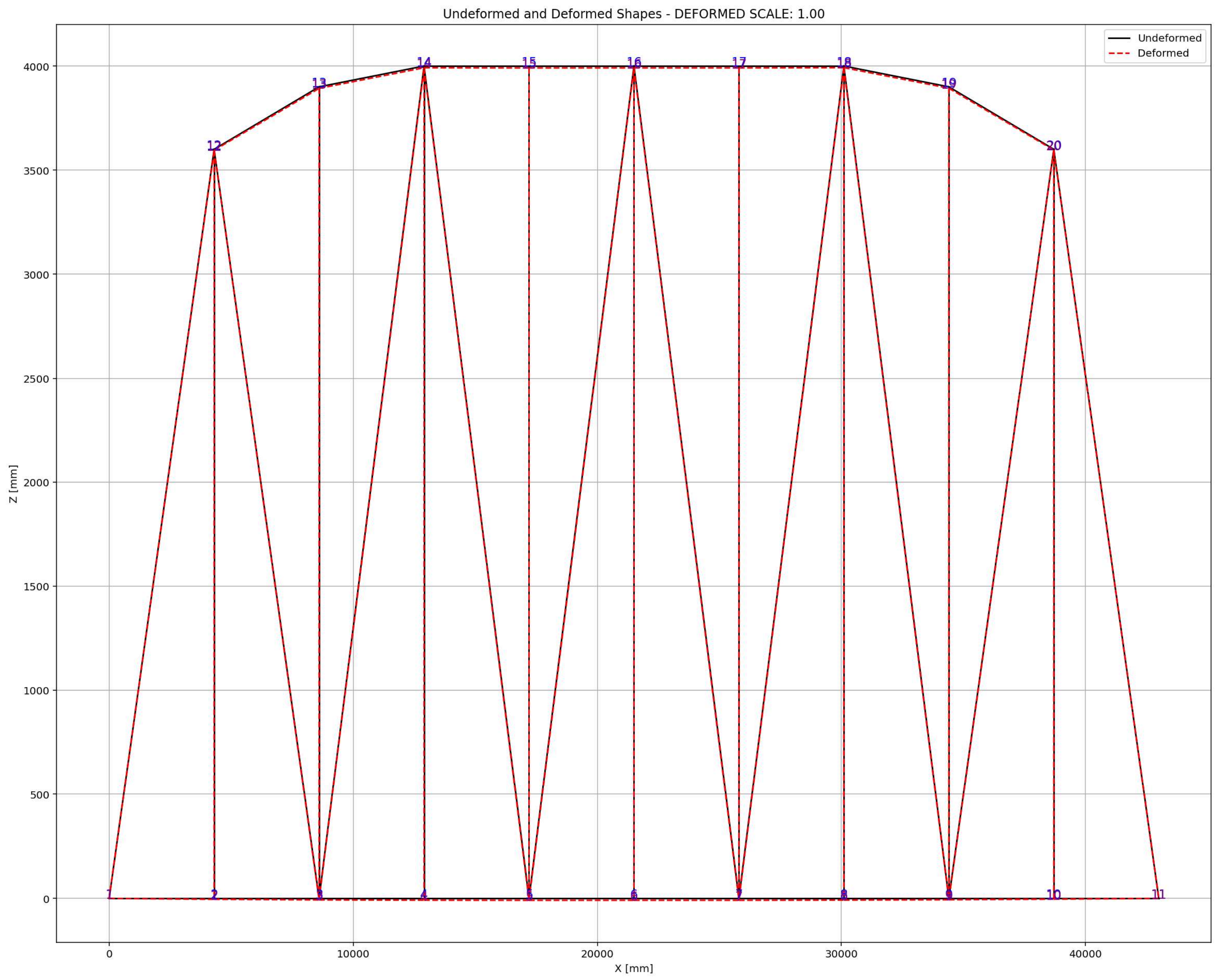
elements Damage-index [%] in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
1e−7
1.5
1.0
0.5
0.0
−0.5
Element Damage-index [%]
0.0
2.5
5.0
7.5
10.0
12.5
Time [s]
Ele. 1 - MAX. ABS. : 1.1248e-08
Ele. 2 - MAX. ABS. : 8.7174e-09
Ele. 3 - MAX. ABS. : 1.5093e-08
Ele. 4 - MAX. ABS. : 1.4068e-08
Ele. 5 - MAX. ABS. : 3.6833e-08
Ele. 6 - MAX. ABS. : 3.6833e-08
Ele. 7 - MAX. ABS. : 1.4068e-08
Ele. 8 - MAX. ABS. : 1.5093e-08
Ele. 9 - MAX. ABS. : 8.7174e-09
Ele. 10 - MAX. ABS. : 1.1248e-08
Ele. 11 - MAX. ABS. : 1.7336e-08
Ele. 12 - MAX. ABS. : 1.5358e-08
Ele. 13 - MAX. ABS. : 1.6896e-08
Ele. 14 - MAX. ABS. : 1.9020e-08
Ele. 15 - MAX. ABS. : 1.9020e-08
Ele. 16 - MAX. ABS. : 1.6896e-08
Ele. 17 - MAX. ABS. : 1.5358e-08
Ele. 18 - MAX. ABS. : 1.7336e-08
Ele. 19 - MAX. ABS. : 2.0120e-08
Ele. 20 - MAX. ABS. : 4.8063e-09
Ele. 21 - MAX. ABS. : 8.0409e-09
Ele. 22 - MAX. ABS. : 4.9431e-09
Ele. 23 - MAX. ABS. : 2.2590e-08
Ele. 24 - MAX. ABS. : 8.2637e-09
Ele. 25 - MAX. ABS. : 2.7541e-08
Ele. 26 - MAX. ABS. : 1.4513e-08
Ele. 27 - MAX. ABS. : 5.0099e-08
Ele. 28 - MAX. ABS. : 1.5255e-07
Ele. 29 - MAX. ABS. : 5.0099e-08
Ele. 30 - MAX. ABS. : 1.4513e-08
Ele. 31 - MAX. ABS. : 2.7541e-08
Ele. 32 - MAX. ABS. : 8.2637e-09
Ele. 33 - MAX. ABS. : 2.2590e-08
Ele. 34 - MAX. ABS. : 4.9431e-09
Ele. 35 - MAX. ABS. : 8.0409e-09
Ele. 36 - MAX. ABS. : 4.8063e-09
Ele. 37 - MAX. ABS. : 2.0120e-08


Node Displacements in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Displacement [mm]
Time [s]
0.0002
0.0001
0.0000
−0.0001
−0.0002
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 4.8367e-05
Node 3 - MAX. ABS. : 8.5173e-05
Node 4 - MAX. ABS. : 1.2438e-04
Node 5 - MAX. ABS. : 1.5838e-04
Node 6 - MAX. ABS. : 7.0688e-19
Node 7 - MAX. ABS. : 1.5838e-04
Node 8 - MAX. ABS. : 1.2438e-04
Node 9 - MAX. ABS. : 8.5173e-05
Node 10 - MAX. ABS. : 4.8367e-05
Node 11 - MAX. ABS. : 0.0000e+00
Node 12 - MAX. ABS. : 2.2849e-04
Node 13 - MAX. ABS. : 1.8095e-04
Node 14 - MAX. ABS. : 1.5437e-04
Node 15 - MAX. ABS. : 8.1786e-05
Node 16 - MAX. ABS. : 1.4796e-18
Node 17 - MAX. ABS. : 8.1786e-05
Node 18 - MAX. ABS. : 1.5437e-04
Node 19 - MAX. ABS. : 1.8095e-04
Node 20 - MAX. ABS. : 2.2849e-04

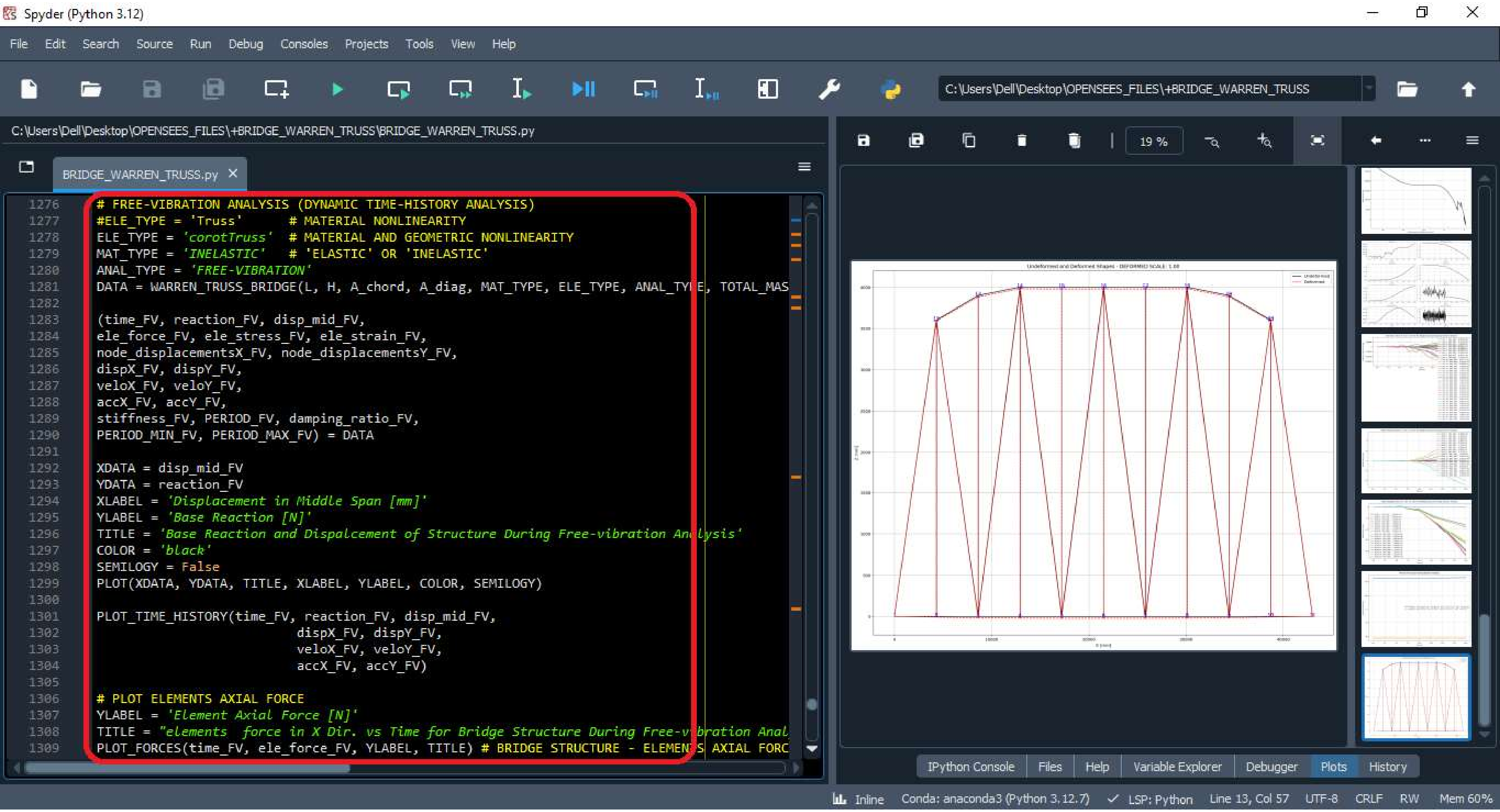


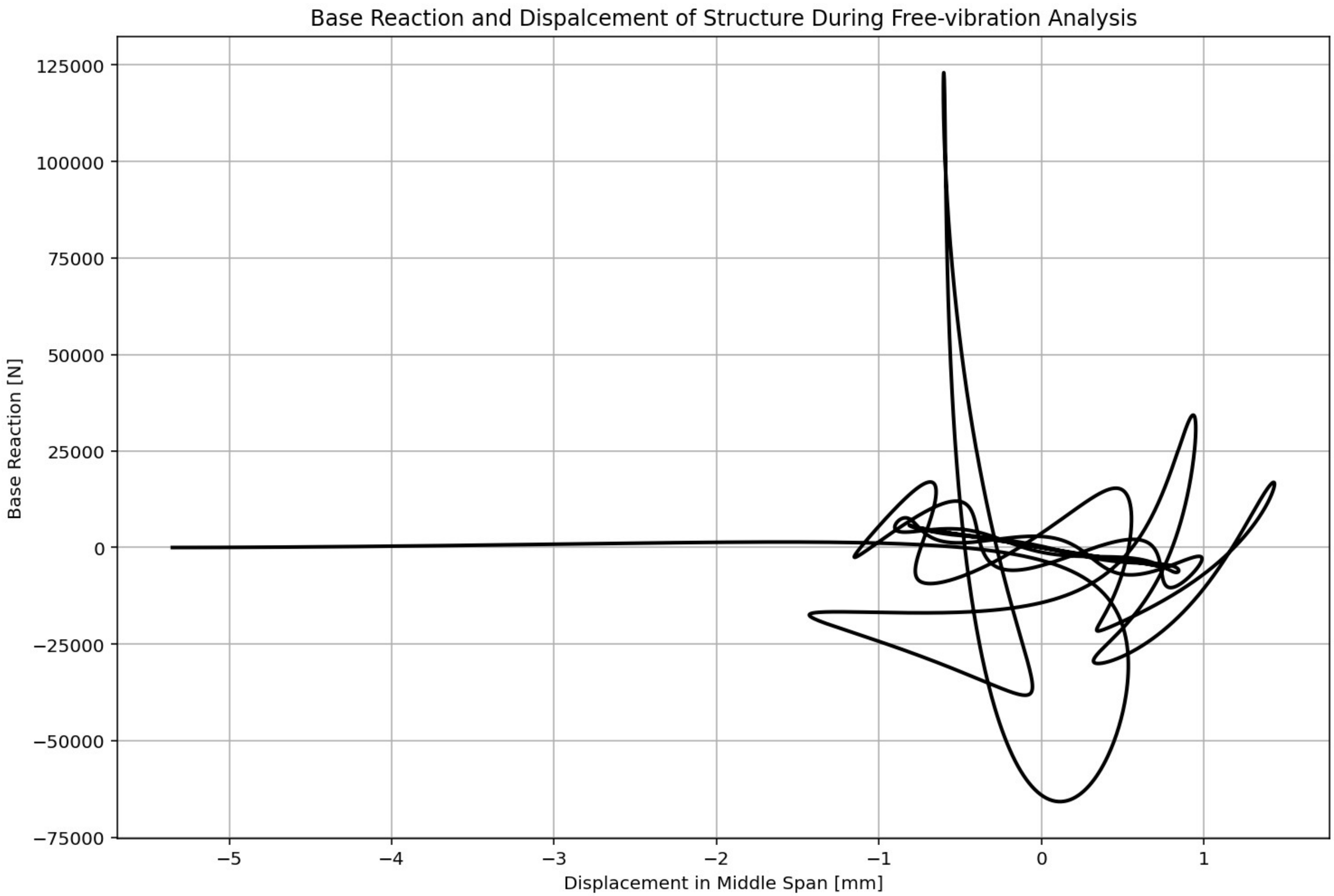
Node Displacements in Y Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Displacement [mm]
Time [s]
0.0000
−0.0002
−0.0004
−0.0006
−0.0008
−0.0010
−0.0012
−0.0014
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 4.4440e-04
Node 3 - MAX. ABS. : 7.5523e-04
Node 4 - MAX. ABS. : 8.3894e-04
Node 5 - MAX. ABS. : 1.1337e-03
Node 6 - MAX. ABS. : 1.4724e-03
Node 7 - MAX. ABS. : 1.1337e-03
Node 8 - MAX. ABS. : 8.3894e-04
Node 9 - MAX. ABS. : 7.5523e-04
Node 10 - MAX. ABS. : 4.4440e-04
Node 11 - MAX. ABS. : 0.0000e+00
Node 12 - MAX. ABS. : 4.3127e-04
Node 13 - MAX. ABS. : 7.6164e-04
Node 14 - MAX. ABS. : 8.5365e-04
Node 15 - MAX. ABS. : 1.1247e-03
Node 16 - MAX. ABS. : 1.3737e-03
Node 17 - MAX. ABS. : 1.1247e-03
Node 18 - MAX. ABS. : 8.5365e-04
Node 19 - MAX. ABS. : 7.6164e-04
Node 20 - MAX. ABS. : 4.3127e-04

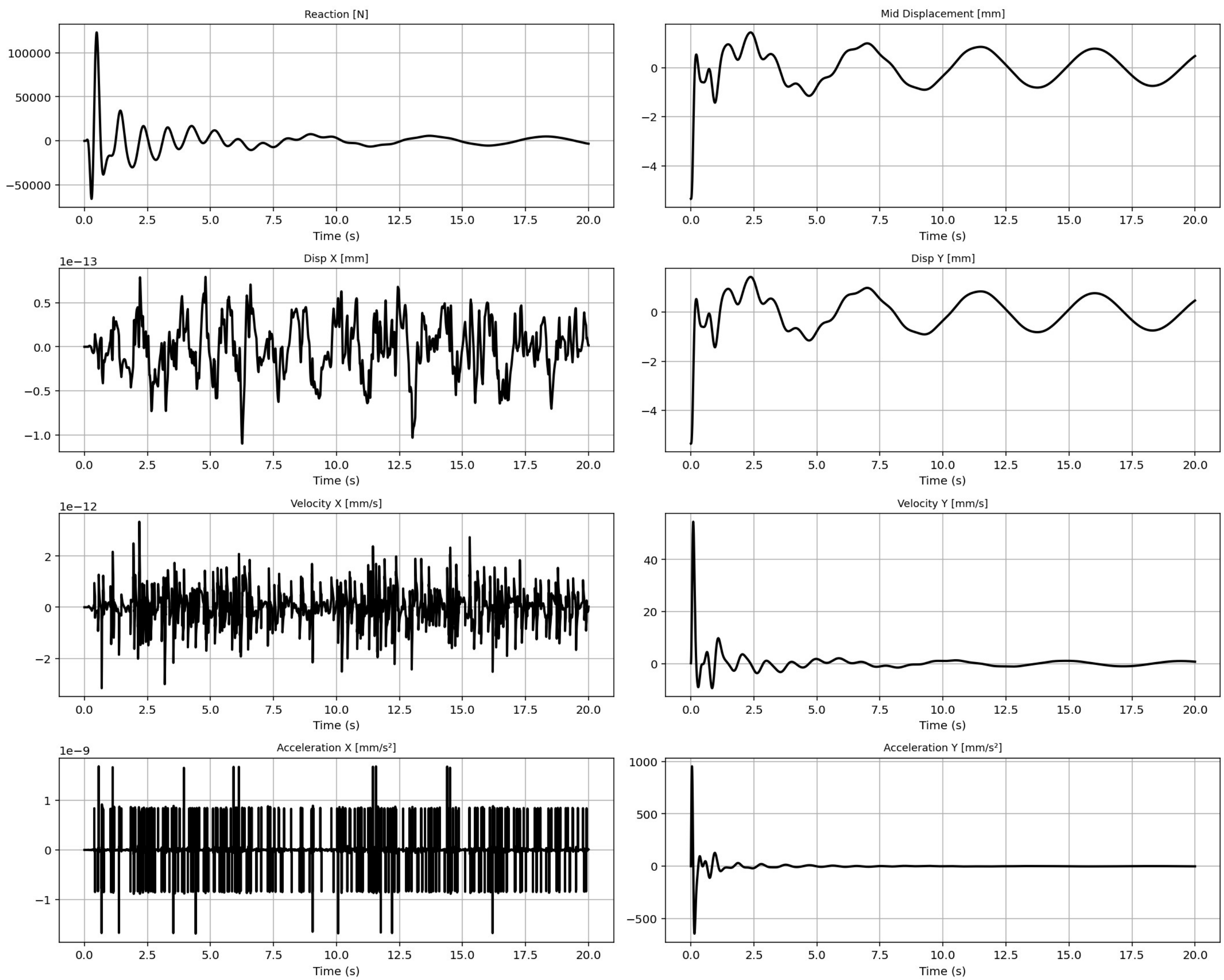

Period of Structure During Dynamic External Time-dependent Loading Analysis
Structural Period [s]
4.5
4.0
3.5
3.0
2.5
2.0
PERIOD - MIN VALUES: Min: 4.581 (s) - Mean: 4.581 (s) - Max: 4.581 (s)
PERIOD - MAX VALUES: Min: 1.820 (s) - Mean: 1.820 (s) - Max: 1.820 (s)
−0.04
−0.02
0.00
0.02
0.04
Displacement [mm]

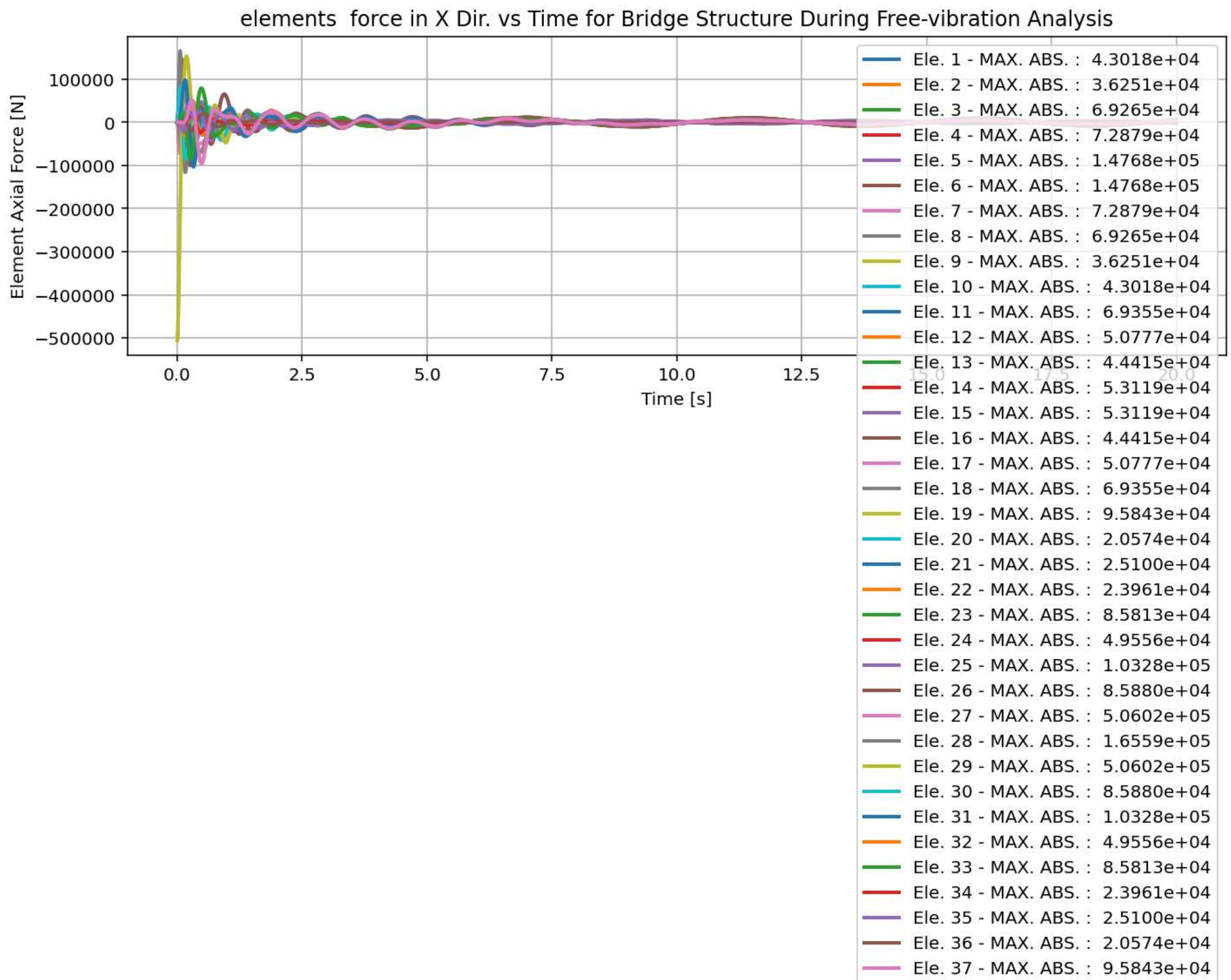

Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
4000
3500
3000
2500
2000
1500
1000
500
0
10000
20000
30000
40000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

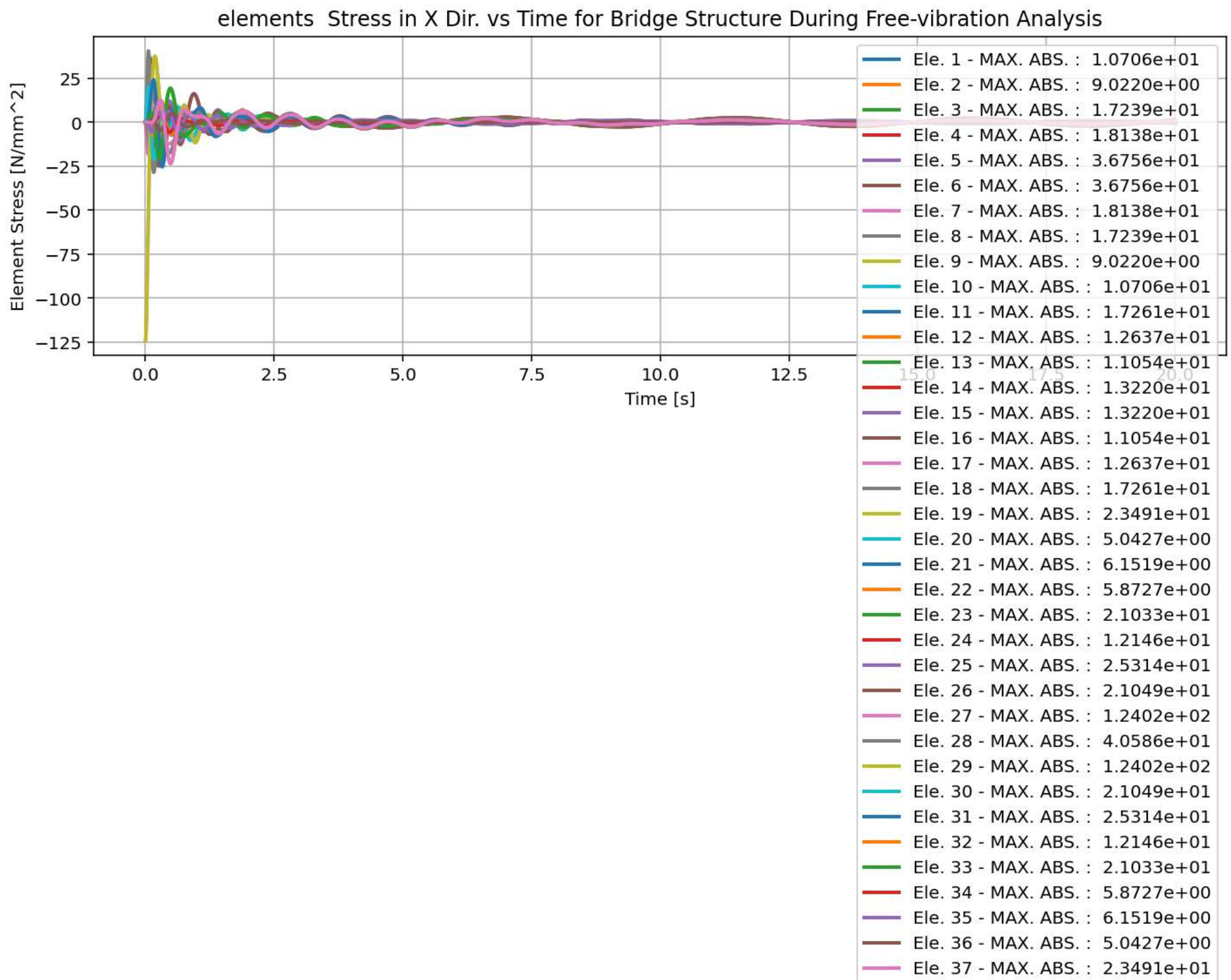
FREE-VIBRATION ANALYSIS


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_WARREN_TRUSS
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_WARREN_TRUSS\BRIDGE_WARREN_TRUSS.py
BRIDGE_WARREN_TRUSS.py
FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITY
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'FREE-VIBRATION'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(time_FV, reaction_FV, disp_mid_FV,
ele_force_FV, ele_stress_FV, ele_strain_FV,
node_displacementsX_FV, node_displacementsY_FV,
dispX_FV, dispY_FV,
veloX_FV, veloY_FV,
accX_FV, accY_FV,
stiffness_FV, PERIOD_FV, damping_ratio_FV,
PERIOD_MIN_FV, PERIOD_MAX_FV) = DATA
XDATA = disp_mid_FV
YDATA = reaction_FV
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Dispalcement of Structure During Free-vibration Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_FV, reaction_FV, disp_mid_FV,
                  dispX_FV, dispY_FV,
                  veloX_FV, veloY_FV,
                  accX_FV, accY_FV)
# PLOT ELEMENTS AXIAL FORCE
YLABEL = 'Element Axial Force [N]'
TITLE = "elements force in X Dir. vs Time for Bridge Structure During Free-vibration Anal
PLOT_FORCES(time_FV, ele_force_FV, YLABEL, TITLE) # BRIDGE STRUCTURE - ELEMENTS AXIAL FORC
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 13, Col 57 UTF-8 CRLF RW Mem 60%

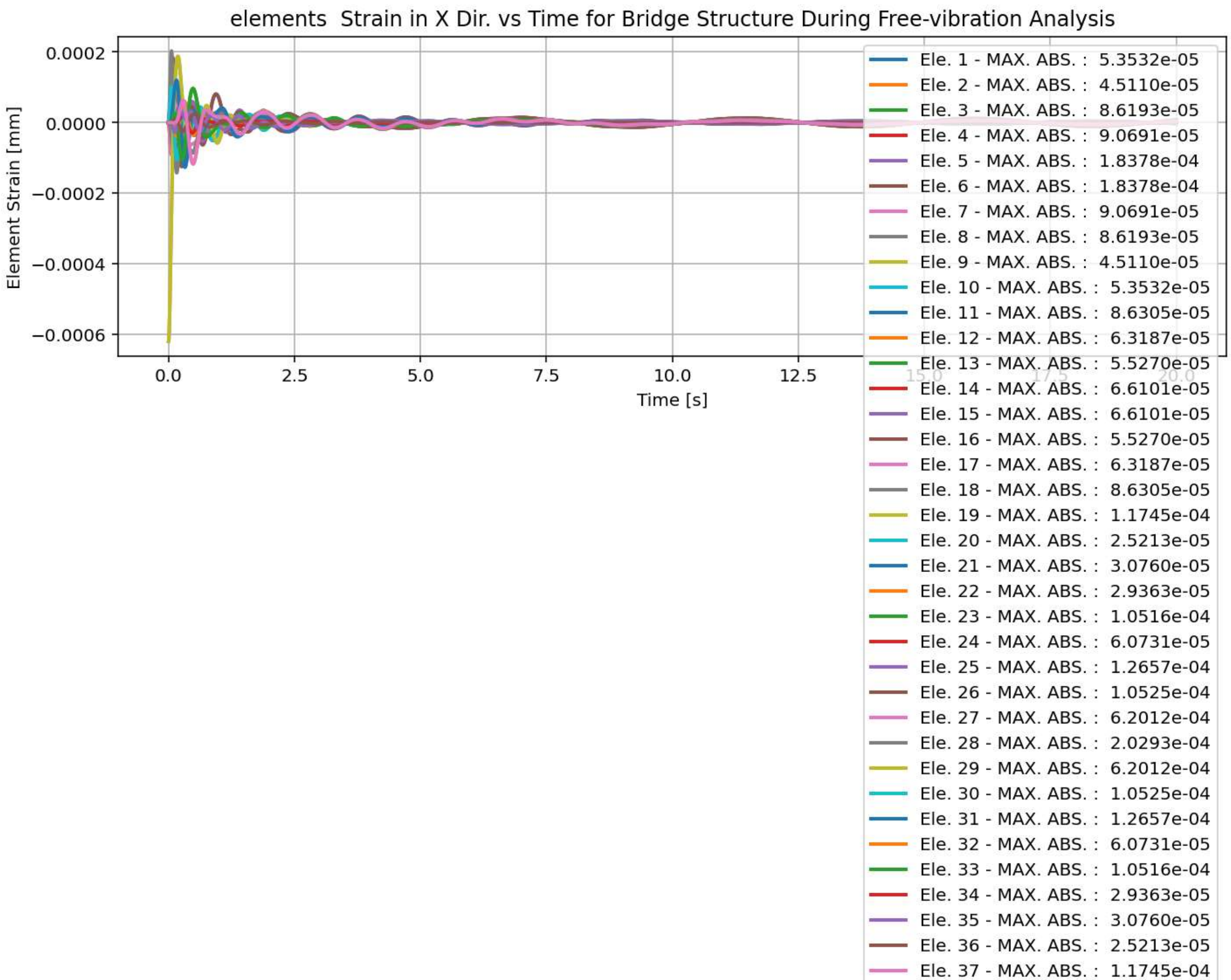

Base Reaction and Dispalcement of Structure During Free-vibration Analysis
125000
100000
75000
50000
25000
0
−25000
−50000
−75000
Base Reaction [N]
−5
−4
−3
−2
−1
0
1
Displacement in Middle Span [mm]

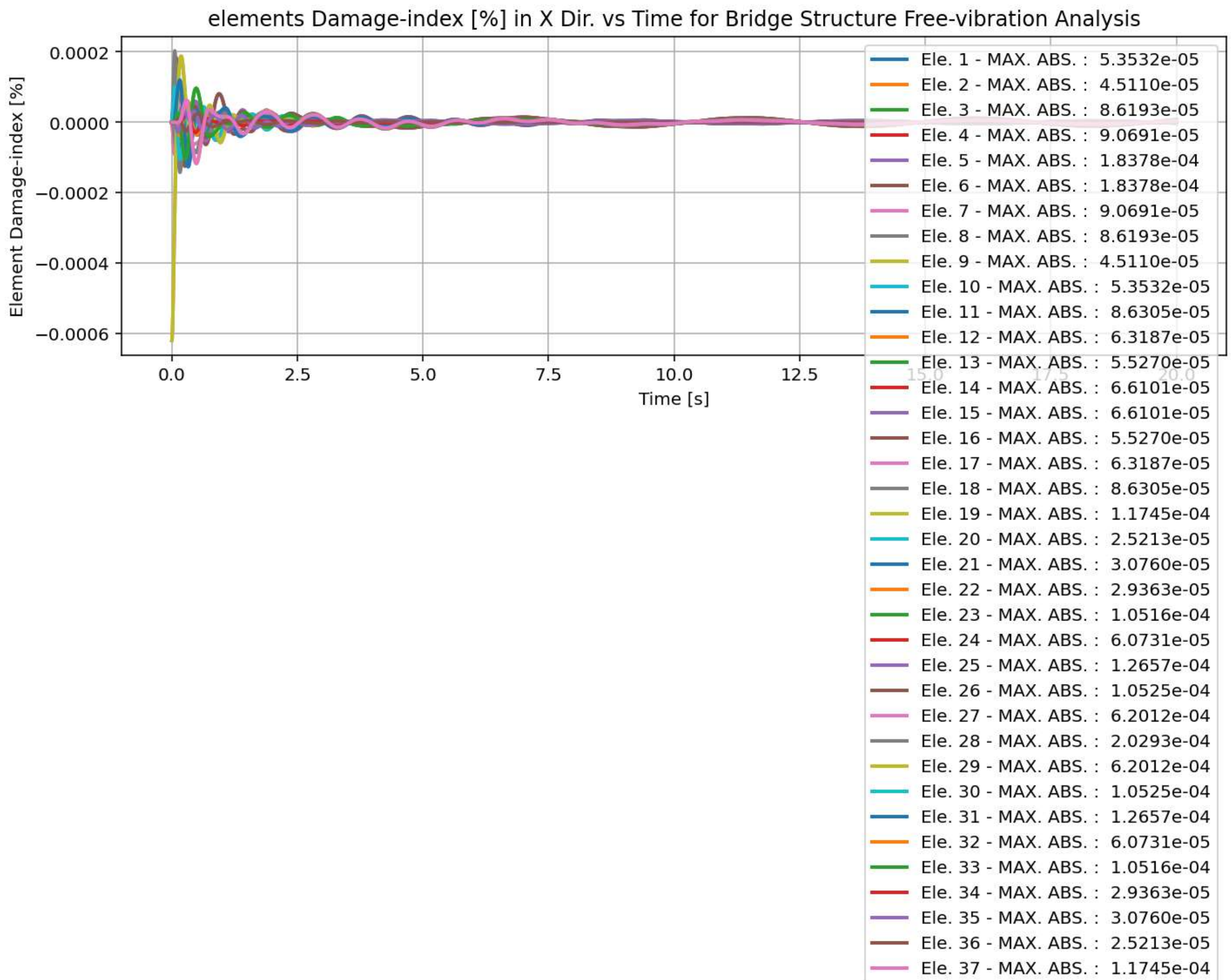

Reaction [N]
Mid Displacement [mm]
Disp X [mm]
Disp Y [mm]
Velocity X [mm/s]
Velocity Y [mm/s]
Acceleration X [mm/s²]
Acceleration Y [mm/s²]
Time (s)
1e−13
1e−12
1e−9


elements force in X Dir. vs Time for Bridge Structure During Free-vibration Analysis
Element Axial Force [N]
Time [s]
100000
0
−100000
−200000
−300000
−400000
−500000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 4.3018e+04
Ele. 2 - MAX. ABS. : 3.6251e+04
Ele. 3 - MAX. ABS. : 6.9265e+04
Ele. 4 - MAX. ABS. : 7.2879e+04
Ele. 5 - MAX. ABS. : 1.4768e+05
Ele. 6 - MAX. ABS. : 1.4768e+05
Ele. 7 - MAX. ABS. : 7.2879e+04
Ele. 8 - MAX. ABS. : 6.9265e+04
Ele. 9 - MAX. ABS. : 3.6251e+04
Ele. 10 - MAX. ABS. : 4.3018e+04
Ele. 11 - MAX. ABS. : 6.9355e+04
Ele. 12 - MAX. ABS. : 5.0777e+04
Ele. 13 - MAX. ABS. : 4.4415e+04
Ele. 14 - MAX. ABS. : 5.3119e+04
Ele. 15 - MAX. ABS. : 5.3119e+04
Ele. 16 - MAX. ABS. : 4.4415e+04
Ele. 17 - MAX. ABS. : 5.0777e+04
Ele. 18 - MAX. ABS. : 6.9355e+04
Ele. 19 - MAX. ABS. : 9.5843e+04
Ele. 20 - MAX. ABS. : 2.0574e+04
Ele. 21 - MAX. ABS. : 2.5100e+04
Ele. 22 - MAX. ABS. : 2.3961e+04
Ele. 23 - MAX. ABS. : 8.5813e+04
Ele. 24 - MAX. ABS. : 4.9556e+04
Ele. 25 - MAX. ABS. : 1.0328e+05
Ele. 26 - MAX. ABS. : 8.5880e+04
Ele. 27 - MAX. ABS. : 5.0602e+05
Ele. 28 - MAX. ABS. : 1.6559e+05
Ele. 29 - MAX. ABS. : 5.0602e+05
Ele. 30 - MAX. ABS. : 8.5880e+04
Ele. 31 - MAX. ABS. : 1.0328e+05
Ele. 32 - MAX. ABS. : 4.9556e+04
Ele. 33 - MAX. ABS. : 8.5813e+04
Ele. 34 - MAX. ABS. : 2.3961e+04
Ele. 35 - MAX. ABS. : 2.5100e+04
Ele. 36 - MAX. ABS. : 2.0574e+04
Ele. 37 - MAX. ABS. : 9.5843e+04

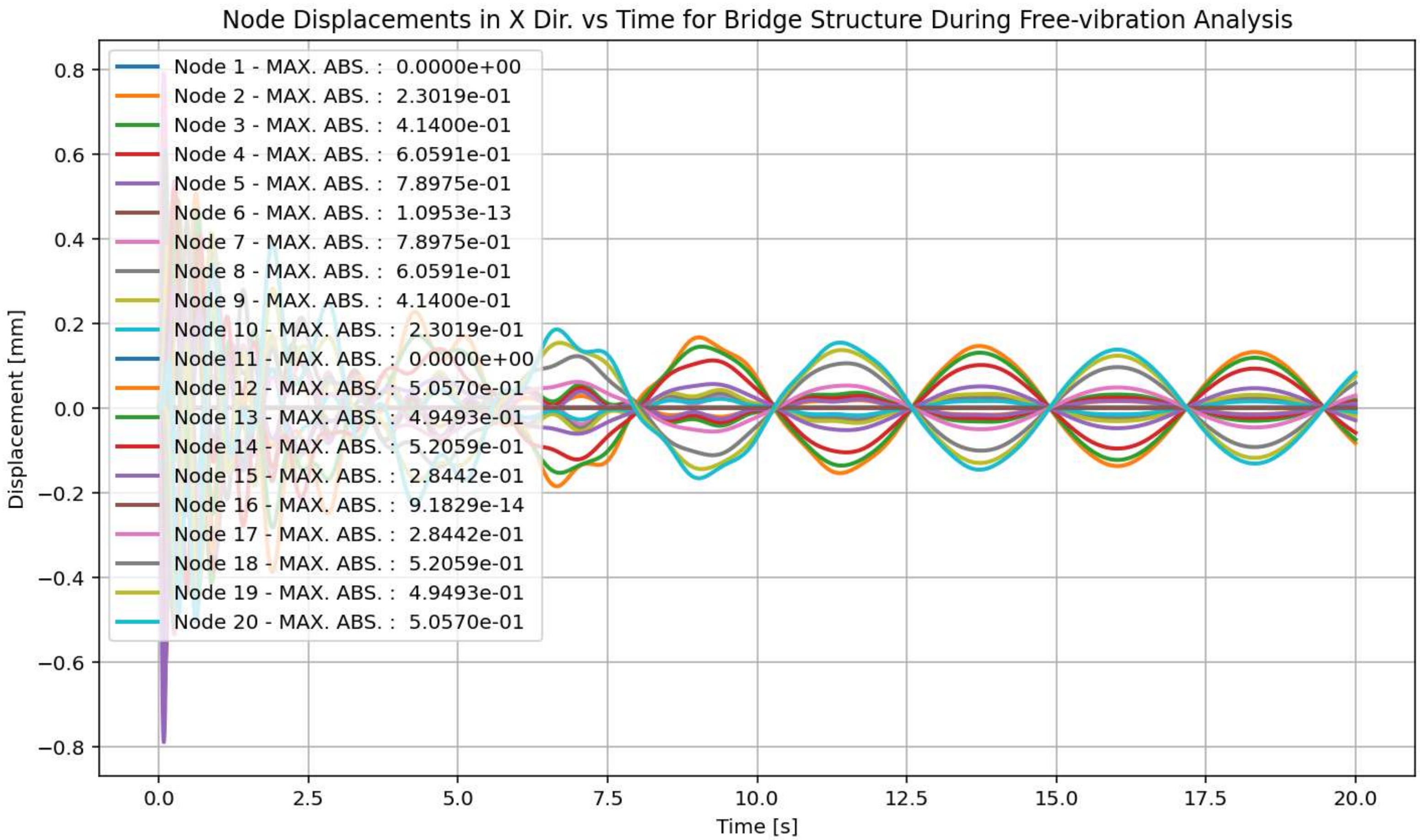


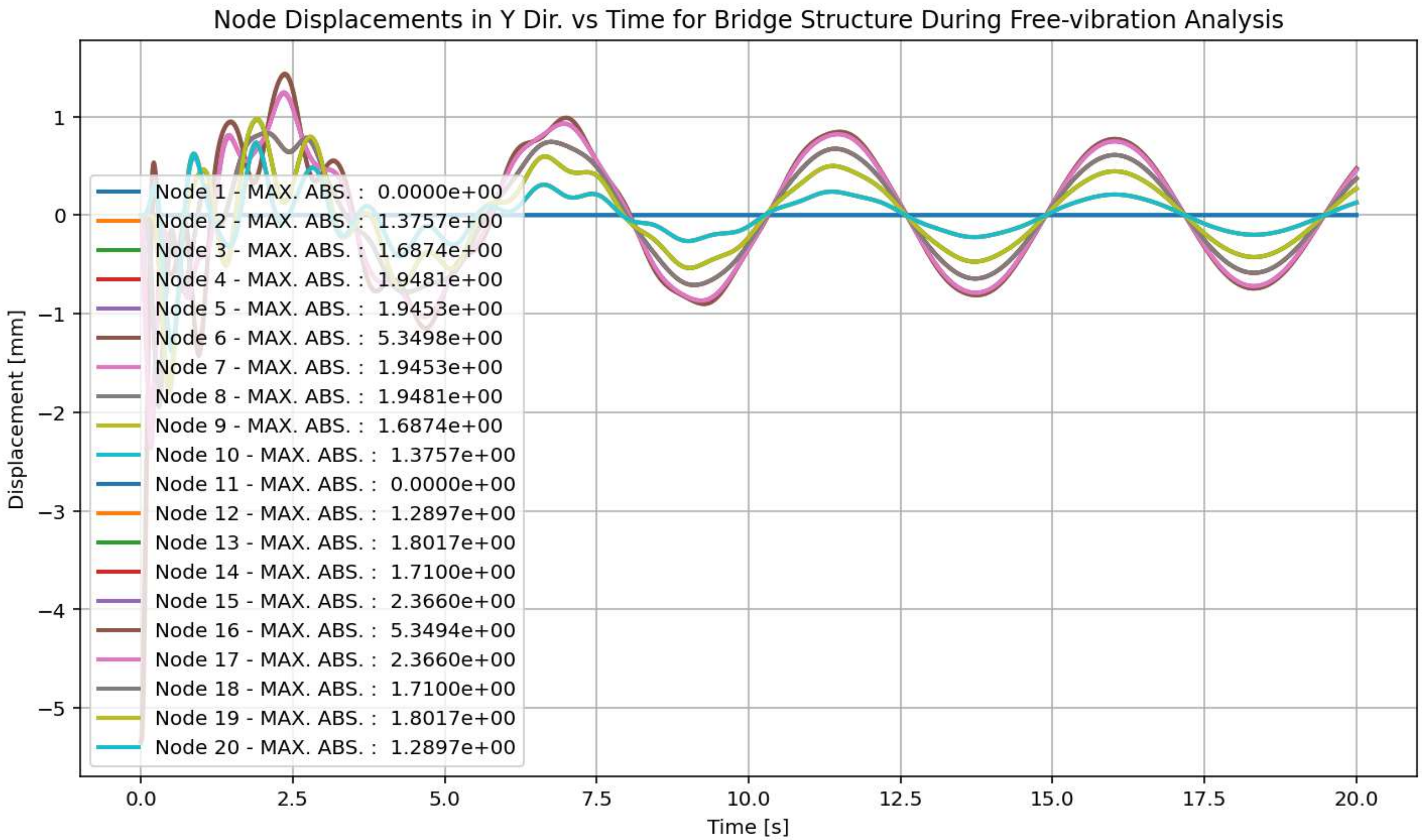
elements Stress in X Dir. vs Time for Bridge Structure During Free-vibration Analysis
Element Stress [N/mm^2]
Time [s]
Ele. 1 - MAX. ABS. : 1.0706e+01
Ele. 2 - MAX. ABS. : 9.0220e+00
Ele. 3 - MAX. ABS. : 1.7239e+01
Ele. 4 - MAX. ABS. : 1.8138e+01
Ele. 5 - MAX. ABS. : 3.6756e+01
Ele. 6 - MAX. ABS. : 3.6756e+01
Ele. 7 - MAX. ABS. : 1.8138e+01
Ele. 8 - MAX. ABS. : 1.7239e+01
Ele. 9 - MAX. ABS. : 9.0220e+00
Ele. 10 - MAX. ABS. : 1.0706e+01
Ele. 11 - MAX. ABS. : 1.7261e+01
Ele. 12 - MAX. ABS. : 1.2637e+01
Ele. 13 - MAX. ABS. : 1.1054e+01
Ele. 14 - MAX. ABS. : 1.3220e+01
Ele. 15 - MAX. ABS. : 1.3220e+01
Ele. 16 - MAX. ABS. : 1.1054e+01
Ele. 17 - MAX. ABS. : 1.2637e+01
Ele. 18 - MAX. ABS. : 1.7261e+01
Ele. 19 - MAX. ABS. : 2.3491e+01
Ele. 20 - MAX. ABS. : 5.0427e+00
Ele. 21 - MAX. ABS. : 6.1519e+00
Ele. 22 - MAX. ABS. : 5.8727e+00
Ele. 23 - MAX. ABS. : 2.1033e+01
Ele. 24 - MAX. ABS. : 1.2146e+01
Ele. 25 - MAX. ABS. : 2.5314e+01
Ele. 26 - MAX. ABS. : 2.1049e+01
Ele. 27 - MAX. ABS. : 1.2402e+02
Ele. 28 - MAX. ABS. : 4.0586e+01
Ele. 29 - MAX. ABS. : 1.2402e+02
Ele. 30 - MAX. ABS. : 2.1049e+01
Ele. 31 - MAX. ABS. : 2.5314e+01
Ele. 32 - MAX. ABS. : 1.2146e+01
Ele. 33 - MAX. ABS. : 2.1033e+01
Ele. 34 - MAX. ABS. : 5.8727e+00
Ele. 35 - MAX. ABS. : 6.1519e+00
Ele. 36 - MAX. ABS. : 5.0427e+00
Ele. 37 - MAX. ABS. : 2.3491e+01

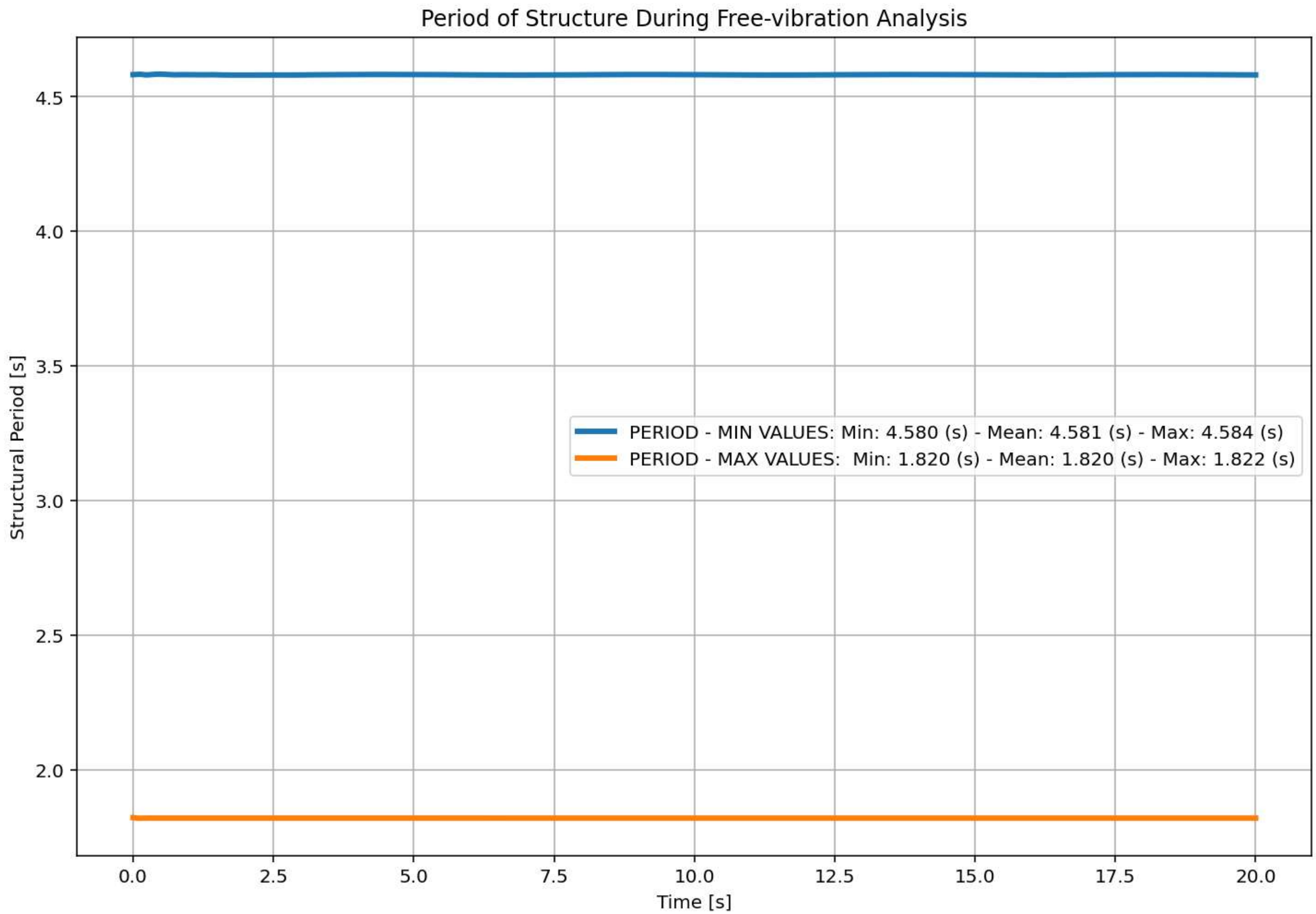

elements Strain in X Dir. vs Time for Bridge Structure During Free-vibration Analysis
Element Strain [mm]
0.0002
0.0000
−0.0002
−0.0004
−0.0006
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
Ele. 1 - MAX. ABS. : 5.3532e-05
Ele. 2 - MAX. ABS. : 4.5110e-05
Ele. 3 - MAX. ABS. : 8.6193e-05
Ele. 4 - MAX. ABS. : 9.0691e-05
Ele. 5 - MAX. ABS. : 1.8378e-04
Ele. 6 - MAX. ABS. : 1.8378e-04
Ele. 7 - MAX. ABS. : 9.0691e-05
Ele. 8 - MAX. ABS. : 8.6193e-05
Ele. 9 - MAX. ABS. : 4.5110e-05
Ele. 10 - MAX. ABS. : 5.3532e-05
Ele. 11 - MAX. ABS. : 8.6305e-05
Ele. 12 - MAX. ABS. : 6.3187e-05
Ele. 13 - MAX. ABS. : 5.5270e-05
Ele. 14 - MAX. ABS. : 6.6101e-05
Ele. 15 - MAX. ABS. : 6.6101e-05
Ele. 16 - MAX. ABS. : 5.5270e-05
Ele. 17 - MAX. ABS. : 6.3187e-05
Ele. 18 - MAX. ABS. : 8.6305e-05
Ele. 19 - MAX. ABS. : 1.1745e-04
Ele. 20 - MAX. ABS. : 2.5213e-05
Ele. 21 - MAX. ABS. : 3.0760e-05
Ele. 22 - MAX. ABS. : 2.9363e-05
Ele. 23 - MAX. ABS. : 1.0516e-04
Ele. 24 - MAX. ABS. : 6.0731e-05
Ele. 25 - MAX. ABS. : 1.2657e-04
Ele. 26 - MAX. ABS. : 1.0525e-04
Ele. 27 - MAX. ABS. : 6.2012e-04
Ele. 28 - MAX. ABS. : 2.0293e-04
Ele. 29 - MAX. ABS. : 6.2012e-04
Ele. 30 - MAX. ABS. : 1.0525e-04
Ele. 31 - MAX. ABS. : 1.2657e-04
Ele. 32 - MAX. ABS. : 6.0731e-05
Ele. 33 - MAX. ABS. : 1.0516e-04
Ele. 34 - MAX. ABS. : 2.9363e-05
Ele. 35 - MAX. ABS. : 3.0760e-05
Ele. 36 - MAX. ABS. : 2.5213e-05
Ele. 37 - MAX. ABS. : 1.1745e-04



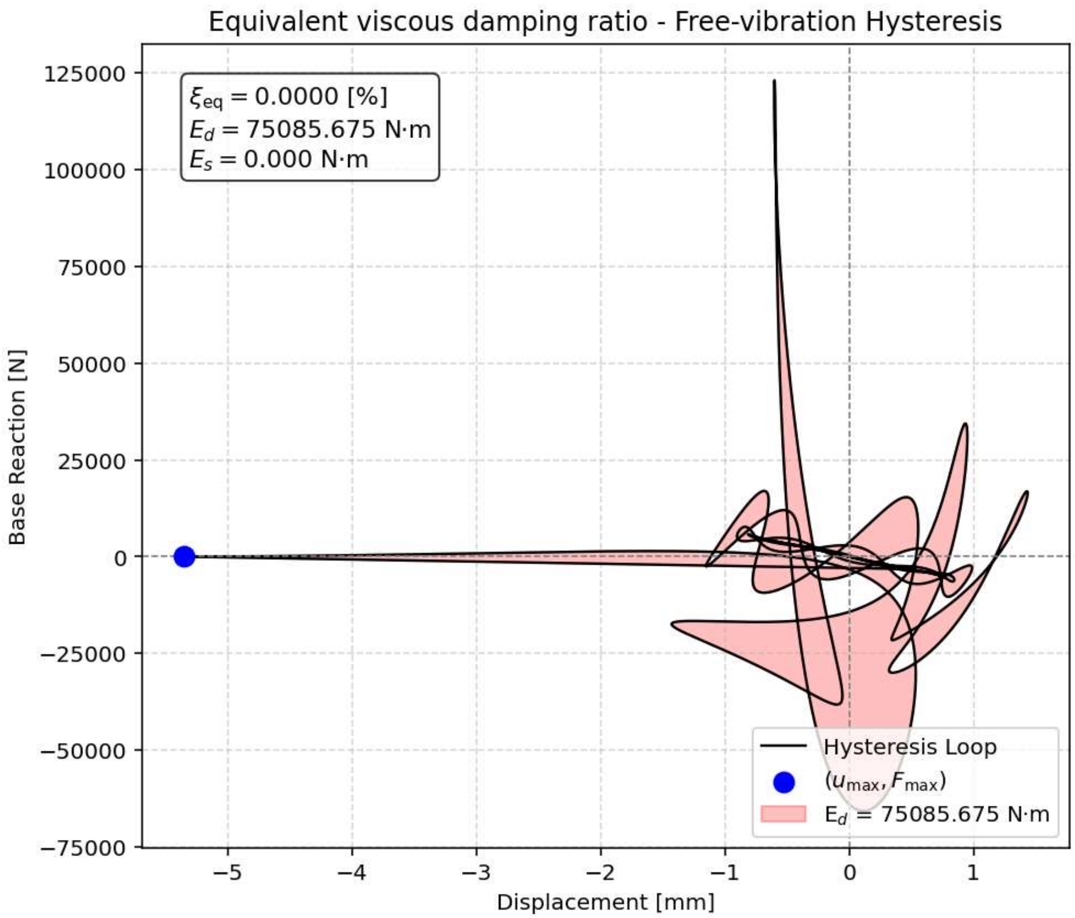
elements Damage-index [%] in X Dir. vs Time for Bridge Structure Free-vibration Analysis
Element Damage-index [%]
Time [s]
0.0002
0.0000
−0.0002
−0.0004
−0.0006
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 5.3532e-05
Ele. 2 - MAX. ABS. : 4.5110e-05
Ele. 3 - MAX. ABS. : 8.6193e-05
Ele. 4 - MAX. ABS. : 9.0691e-05
Ele. 5 - MAX. ABS. : 1.8378e-04
Ele. 6 - MAX. ABS. : 1.8378e-04
Ele. 7 - MAX. ABS. : 9.0691e-05
Ele. 8 - MAX. ABS. : 8.6193e-05
Ele. 9 - MAX. ABS. : 4.5110e-05
Ele. 10 - MAX. ABS. : 5.3532e-05
Ele. 11 - MAX. ABS. : 8.6305e-05
Ele. 12 - MAX. ABS. : 6.3187e-05
Ele. 13 - MAX. ABS. : 5.5270e-05
Ele. 14 - MAX. ABS. : 6.6101e-05
Ele. 15 - MAX. ABS. : 6.6101e-05
Ele. 16 - MAX. ABS. : 5.5270e-05
Ele. 17 - MAX. ABS. : 6.3187e-05
Ele. 18 - MAX. ABS. : 8.6305e-05
Ele. 19 - MAX. ABS. : 1.1745e-04
Ele. 20 - MAX. ABS. : 2.5213e-05
Ele. 21 - MAX. ABS. : 3.0760e-05
Ele. 22 - MAX. ABS. : 2.9363e-05
Ele. 23 - MAX. ABS. : 1.0516e-04
Ele. 24 - MAX. ABS. : 6.0731e-05
Ele. 25 - MAX. ABS. : 1.2657e-04
Ele. 26 - MAX. ABS. : 1.0525e-04
Ele. 27 - MAX. ABS. : 6.2012e-04
Ele. 28 - MAX. ABS. : 2.0293e-04
Ele. 29 - MAX. ABS. : 6.2012e-04
Ele. 30 - MAX. ABS. : 1.0525e-04
Ele. 31 - MAX. ABS. : 1.2657e-04
Ele. 32 - MAX. ABS. : 6.0731e-05
Ele. 33 - MAX. ABS. : 1.0516e-04
Ele. 34 - MAX. ABS. : 2.9363e-05
Ele. 35 - MAX. ABS. : 3.0760e-05
Ele. 36 - MAX. ABS. : 2.5213e-05
Ele. 37 - MAX. ABS. : 1.1745e-04


Node Displacements in X Dir. vs Time for Bridge Structure During Free-vibration Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 2.3019e-01
Node 3 - MAX. ABS. : 4.1400e-01
Node 4 - MAX. ABS. : 6.0591e-01
Node 5 - MAX. ABS. : 7.8975e-01
Node 6 - MAX. ABS. : 1.0953e-13
Node 7 - MAX. ABS. : 7.8975e-01
Node 8 - MAX. ABS. : 6.0591e-01
Node 9 - MAX. ABS. : 4.1400e-01
Node 10 - MAX. ABS. : 2.3019e-01
Node 11 - MAX. ABS. : 0.0000e+00
Node 12 - MAX. ABS. : 5.0570e-01
Node 13 - MAX. ABS. : 4.9493e-01
Node 14 - MAX. ABS. : 5.2059e-01
Node 15 - MAX. ABS. : 2.8442e-01
Node 16 - MAX. ABS. : 9.1829e-14
Node 17 - MAX. ABS. : 2.8442e-01
Node 18 - MAX. ABS. : 5.2059e-01
Node 19 - MAX. ABS. : 4.9493e-01
Node 20 - MAX. ABS. : 5.0570e-01
Displacement [mm]
Time [s]

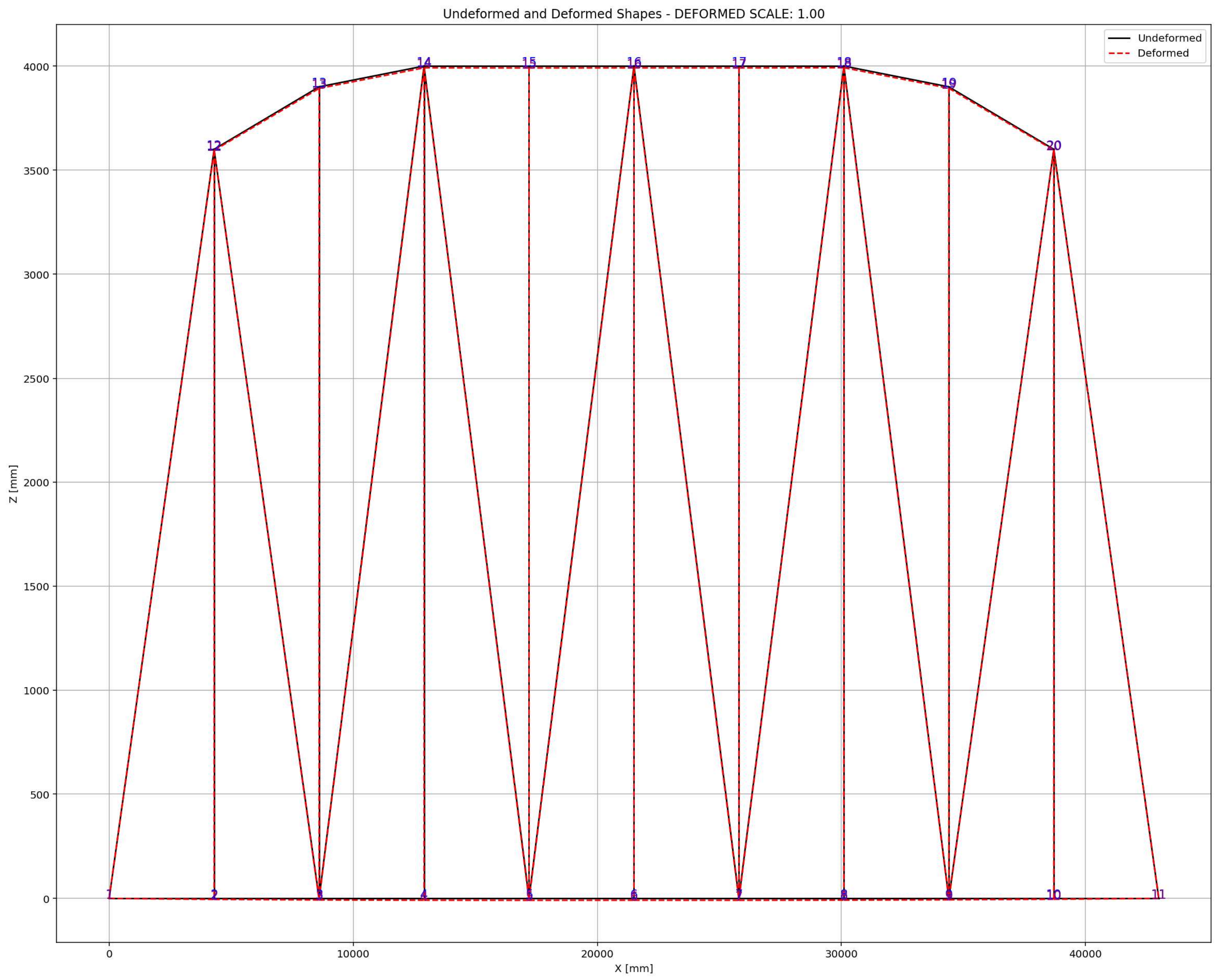

Node Displacements in Y Dir. vs Time for Bridge Structure During Free-vibration Analysis
Displacement [mm]
Time [s]
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 1.3757e+00
Node 3 - MAX. ABS. : 1.6874e+00
Node 4 - MAX. ABS. : 1.9481e+00
Node 5 - MAX. ABS. : 1.9453e+00
Node 6 - MAX. ABS. : 5.3498e+00
Node 7 - MAX. ABS. : 1.9453e+00
Node 8 - MAX. ABS. : 1.9481e+00
Node 9 - MAX. ABS. : 1.6874e+00
Node 10 - MAX. ABS. : 1.3757e+00
Node 11 - MAX. ABS. : 0.0000e+00
Node 12 - MAX. ABS. : 1.2897e+00
Node 13 - MAX. ABS. : 1.8017e+00
Node 14 - MAX. ABS. : 1.7100e+00
Node 15 - MAX. ABS. : 2.3660e+00
Node 16 - MAX. ABS. : 5.3494e+00
Node 17 - MAX. ABS. : 2.3660e+00
Node 18 - MAX. ABS. : 1.7100e+00
Node 19 - MAX. ABS. : 1.8017e+00
Node 20 - MAX. ABS. : 1.2897e+00



Period of Structure During Free-vibration Analysis
Structural Period [s]
Time [s]
PERIOD - MIN VALUES: Min: 4.580 (s) - Mean: 4.581 (s) - Max: 4.584 (s)
PERIOD - MAX VALUES: Min: 1.820 (s) - Mean: 1.820 (s) - Max: 1.822 (s)
4.5
4.0
3.5
3.0
2.5
2.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

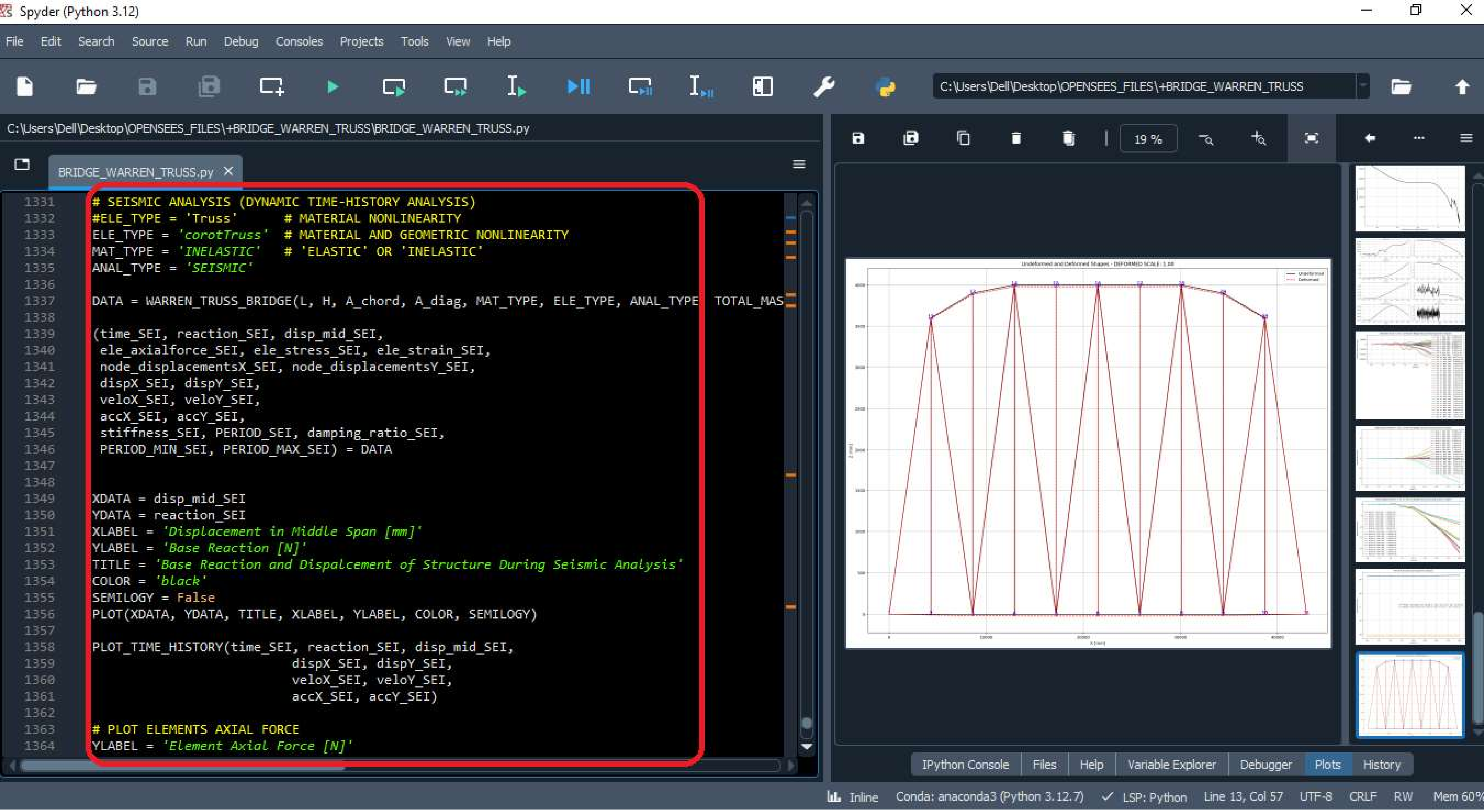


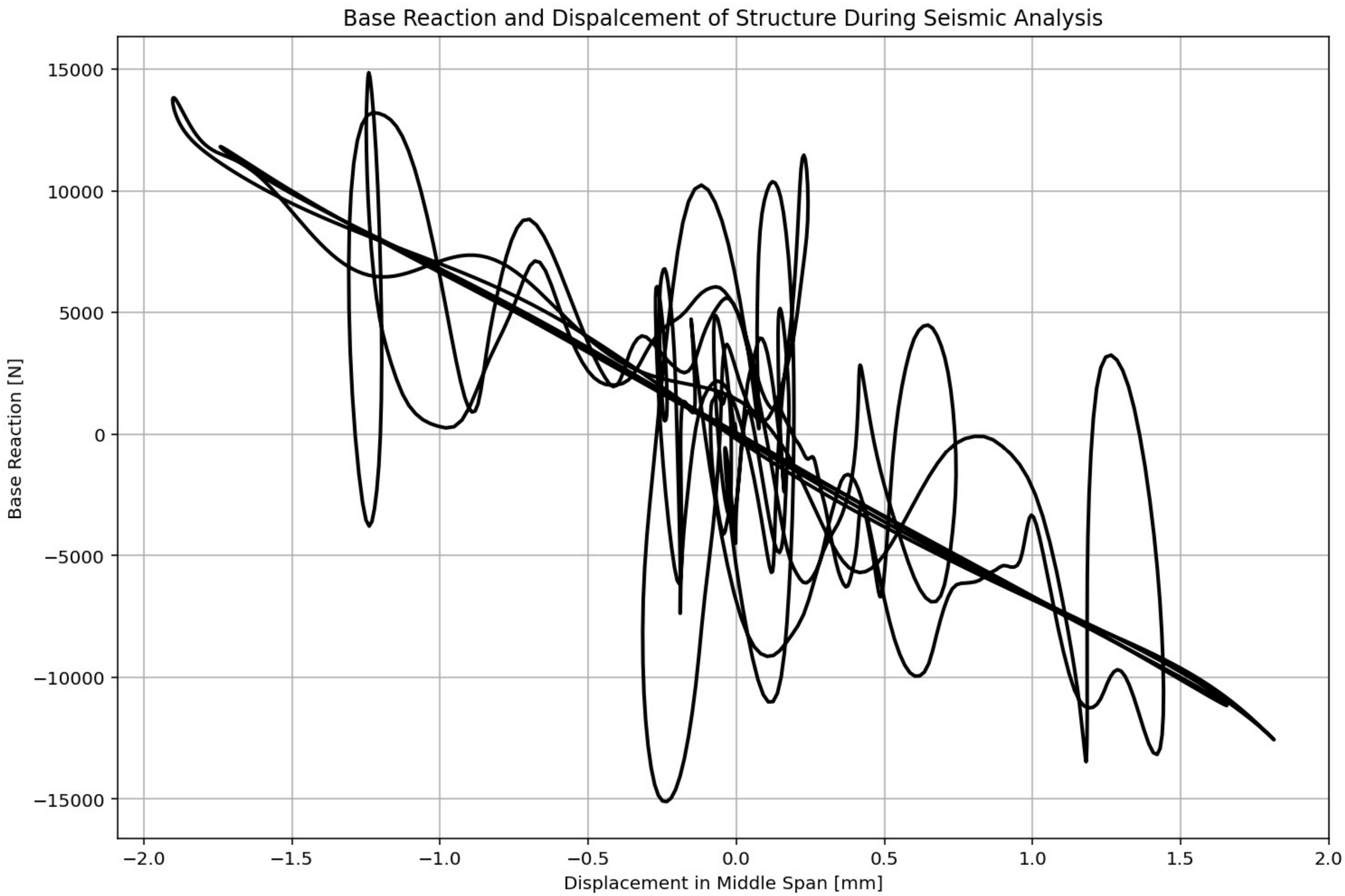
Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{\mathrm{eq}} = 0.0000$ [%]
$E_d = 75085.675$ N·m
$E_s = 0.000$ N·m
125000
100000
75000
50000
25000
0
−25000
−50000
−75000
Base Reaction [N]
−5
−4
−3
−2
−1
0
1
Displacement [mm]
Hysteresis Loop
$(u_{\mathrm{max}}, F_{\mathrm{max}})$
E_d = 75085.675 N·m

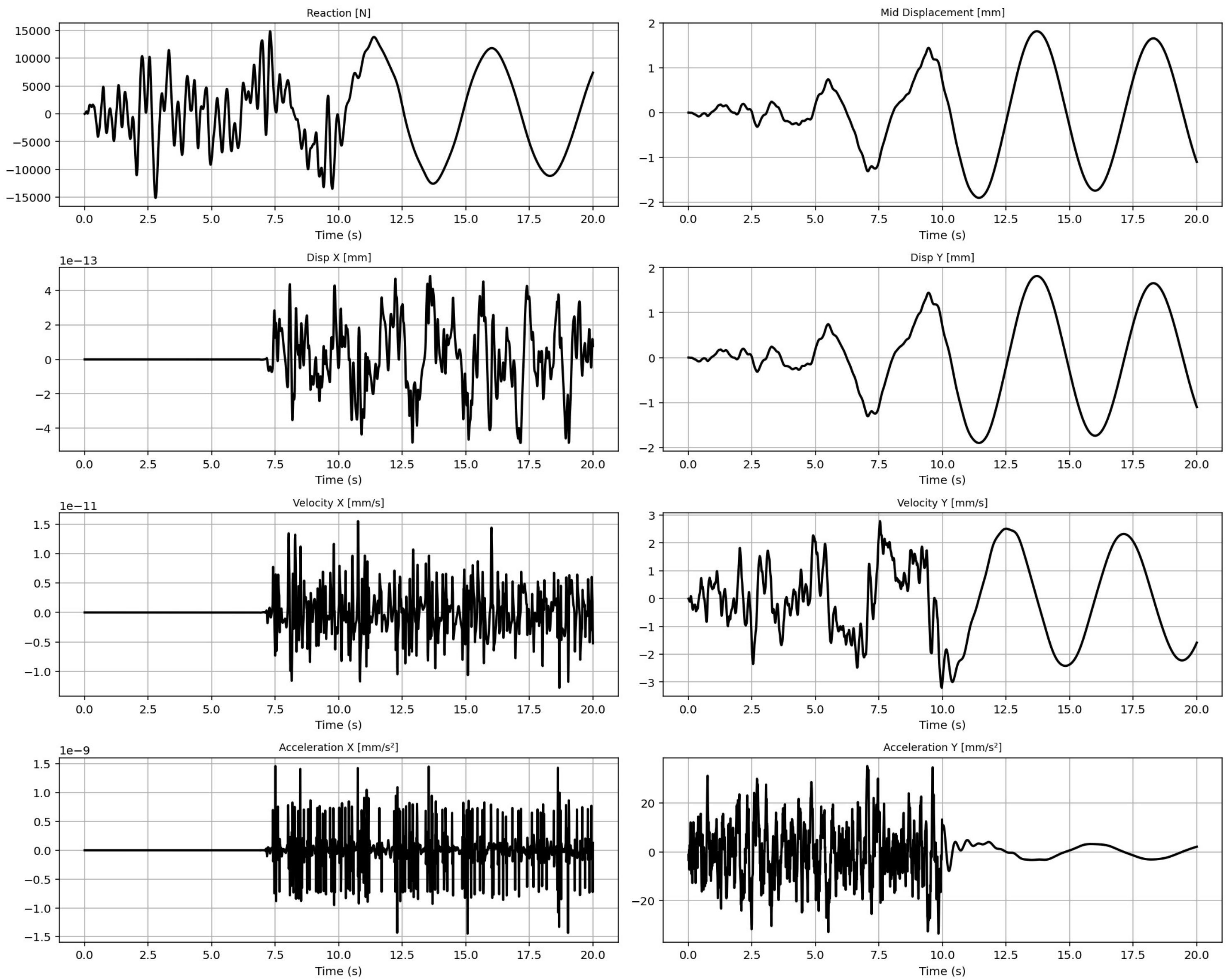


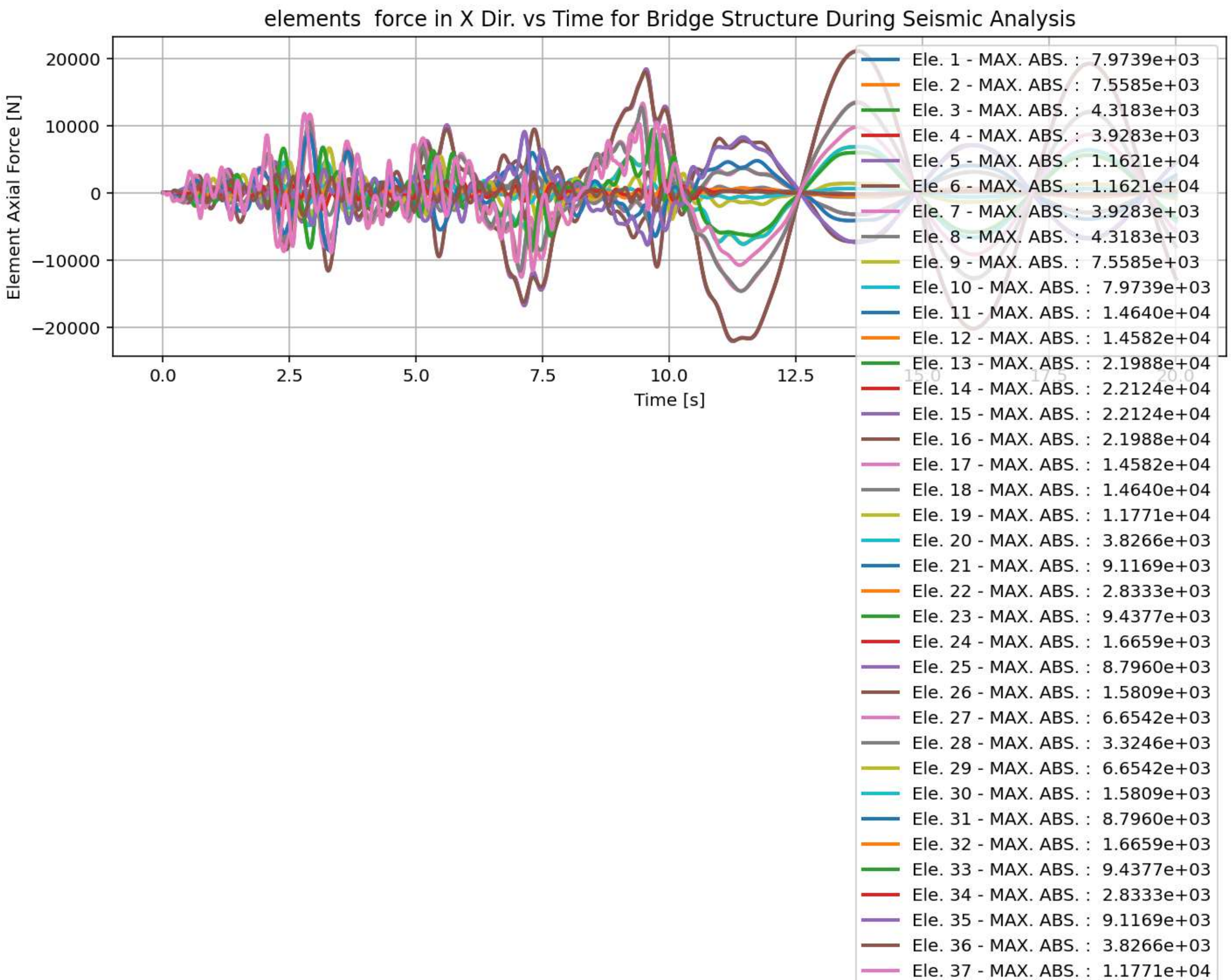
Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
500
1000
1500
2000
2500
3000
3500
4000
10000
20000
30000
40000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

SEISMIC ANALYSIS

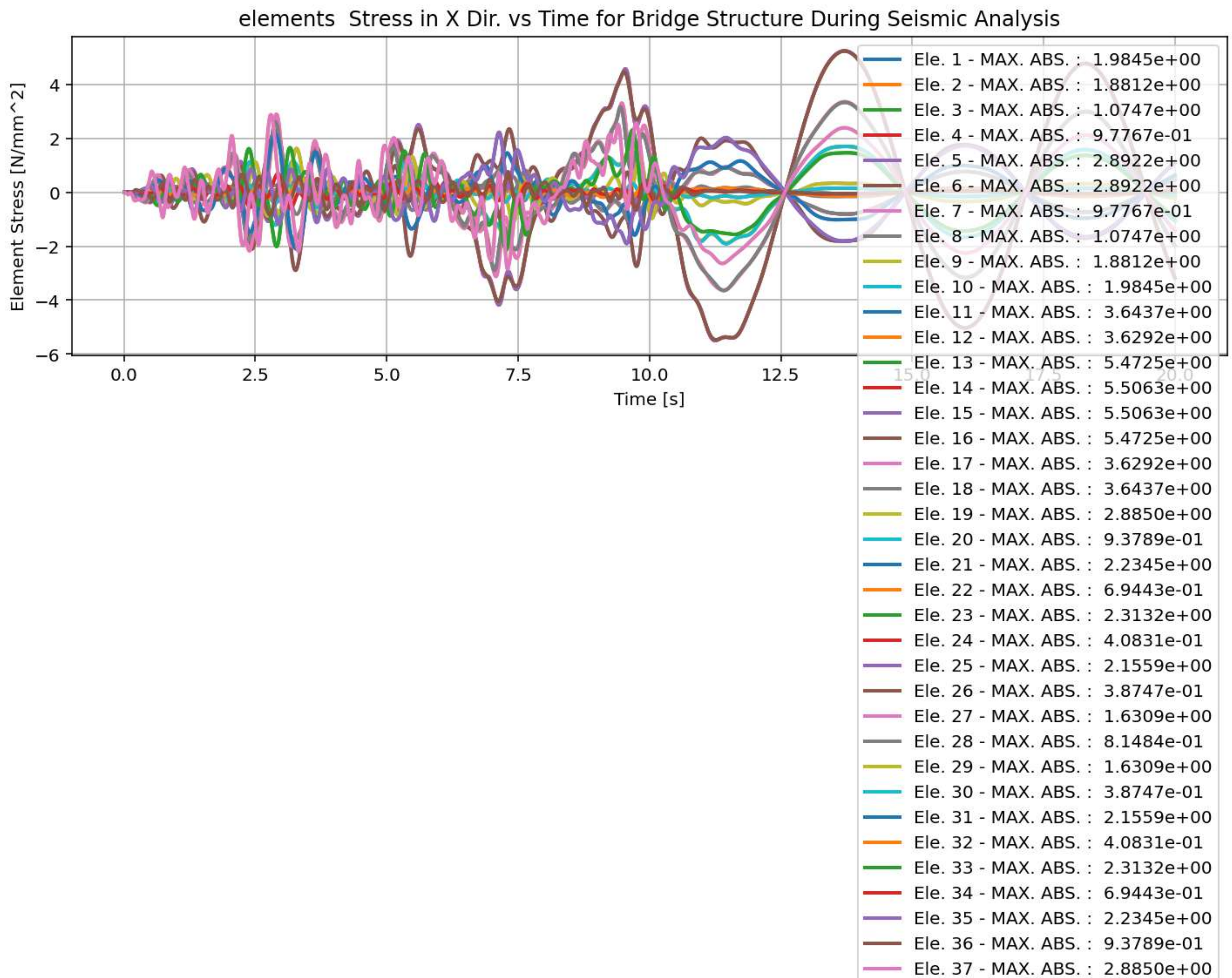

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_WARREN_TRUSS
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_WARREN_TRUSS\BRIDGE_WARREN_TRUSS.py
BRIDGE_WARREN_TRUSS.py
SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITY
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'SEISMIC'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(time_SEI, reaction_SEI, disp_mid_SEI,
 ele_axialforce_SEI, ele_stress_SEI, ele_strain_SEI,
 node_displacementsX_SEI, node_displacementsY_SEI,
 dispX_SEI, dispY_SEI,
 veloX_SEI, veloY_SEI,
 accX_SEI, accY_SEI,
 stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
 PERIOD_MIN_SEI, PERIOD_MAX_SEI) = DATA
XDATA = disp_mid_SEI
YDATA = reaction_SEI
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Dispalcement of Structure During Seismic Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_SEI, reaction_SEI, disp_mid_SEI,
                  dispX_SEI, dispY_SEI,
                  veloX_SEI, veloY_SEI,
                  accX_SEI, accY_SEI)
# PLOT ELEMENTS AXIAL FORCE
YLABEL = 'Element Axial Force [N]'
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 13, Col 57 UTF-8 CRLF RW Mem 60%

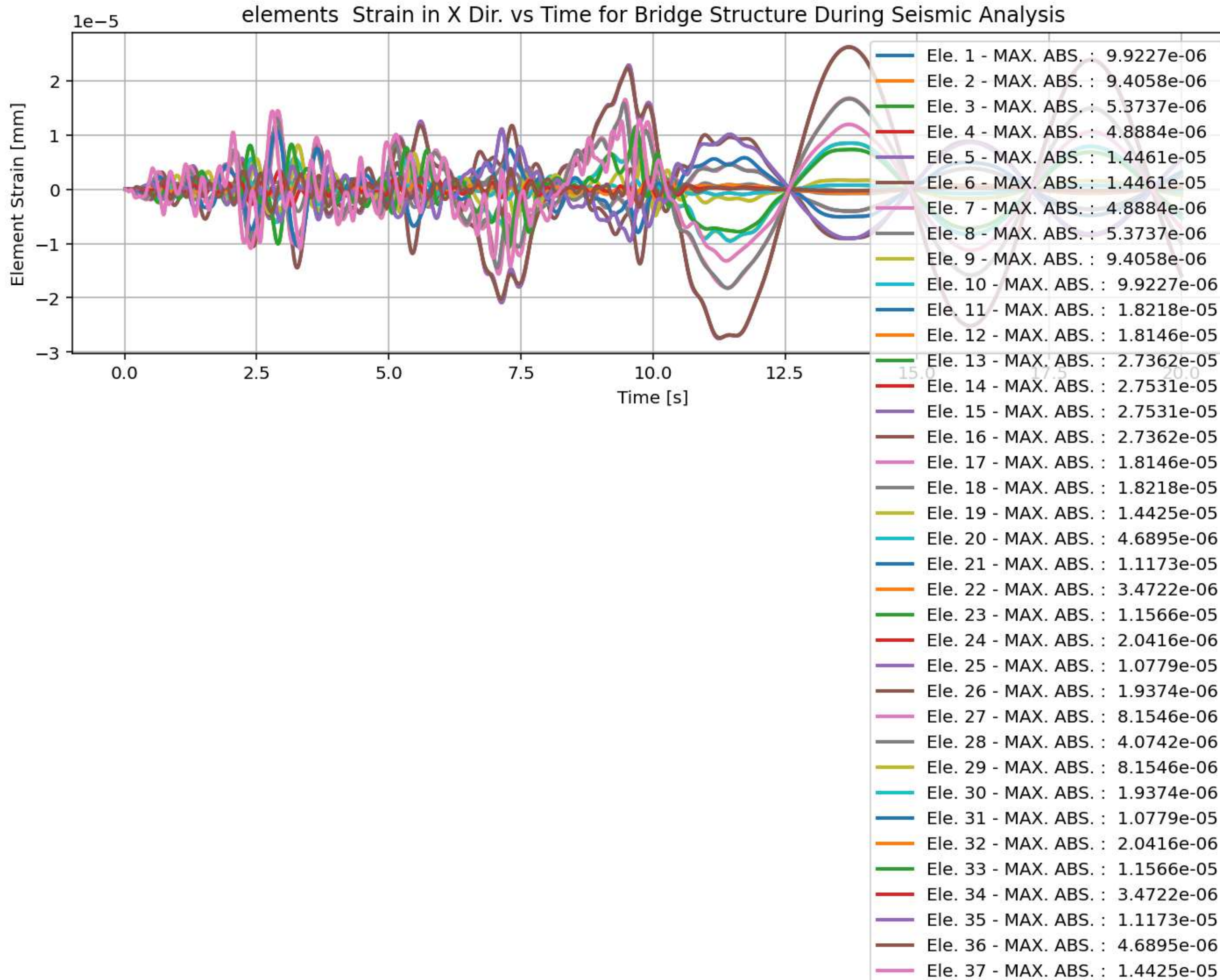

Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
15000
10000
5000
0
−5000
−10000
−15000
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Displacement in Middle Span [mm]

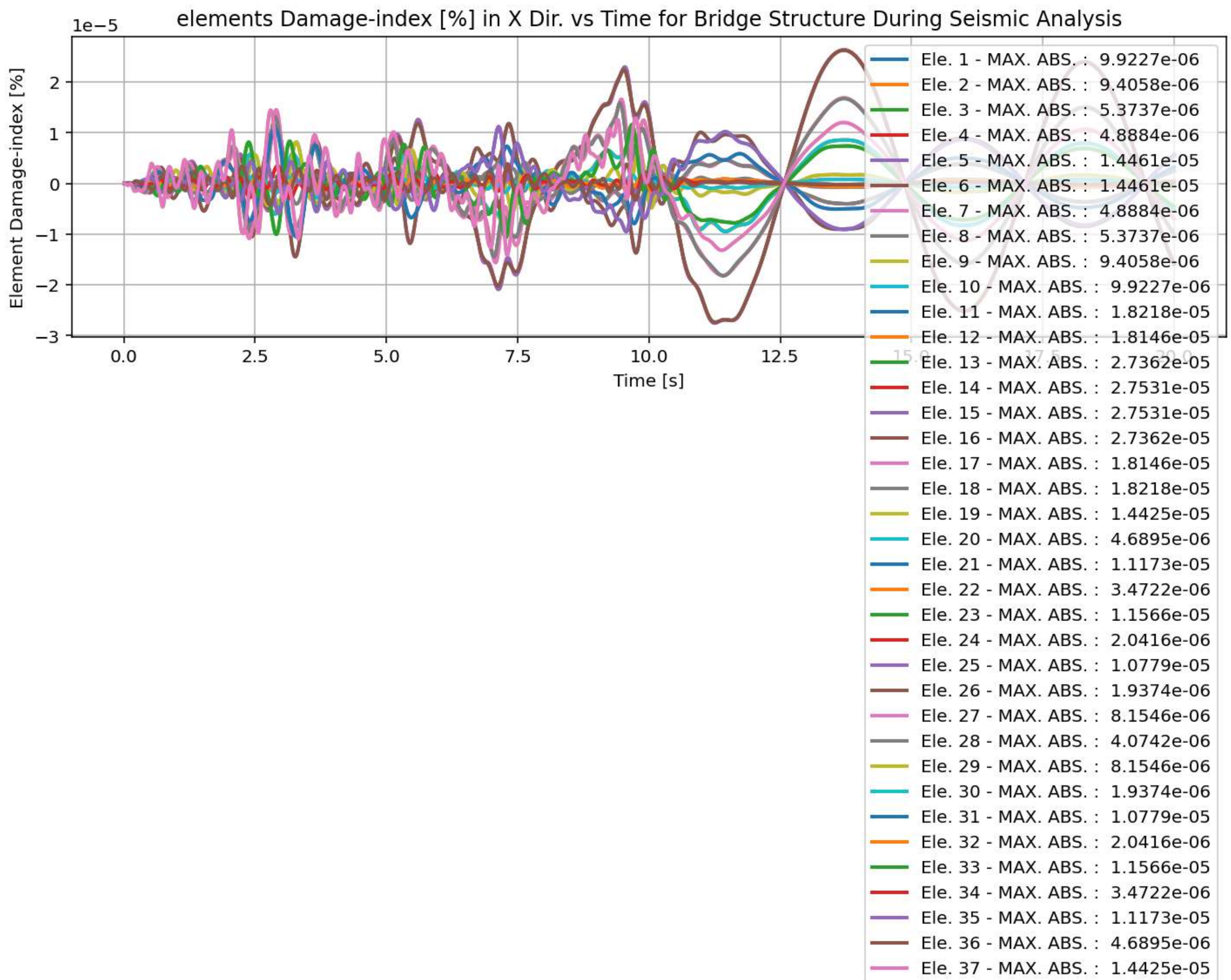

Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
1e−13
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
1e−11
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
1e−9
Time (s)
Acceleration Y [mm/s²]
Time (s)

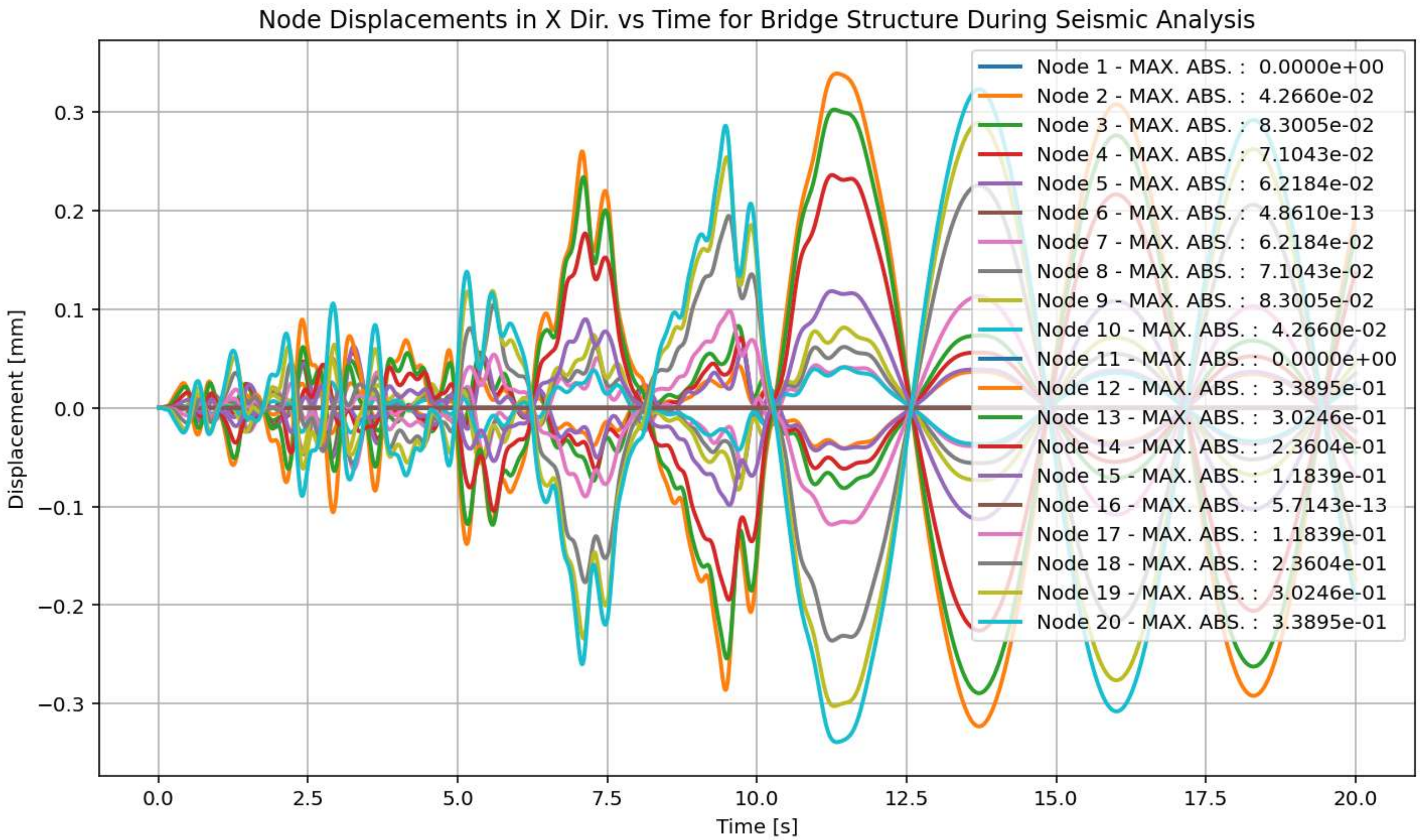


elements force in X Dir. vs Time for Bridge Structure During Seismic Analysis
Element Axial Force [N]
Time [s]
20000
10000
0
−10000
−20000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 7.9739e+03
Ele. 2 - MAX. ABS. : 7.5585e+03
Ele. 3 - MAX. ABS. : 4.3183e+03
Ele. 4 - MAX. ABS. : 3.9283e+03
Ele. 5 - MAX. ABS. : 1.1621e+04
Ele. 6 - MAX. ABS. : 1.1621e+04
Ele. 7 - MAX. ABS. : 3.9283e+03
Ele. 8 - MAX. ABS. : 4.3183e+03
Ele. 9 - MAX. ABS. : 7.5585e+03
Ele. 10 - MAX. ABS. : 7.9739e+03
Ele. 11 - MAX. ABS. : 1.4640e+04
Ele. 12 - MAX. ABS. : 1.4582e+04
Ele. 13 - MAX. ABS. : 2.1988e+04
Ele. 14 - MAX. ABS. : 2.2124e+04
Ele. 15 - MAX. ABS. : 2.2124e+04
Ele. 16 - MAX. ABS. : 2.1988e+04
Ele. 17 - MAX. ABS. : 1.4582e+04
Ele. 18 - MAX. ABS. : 1.4640e+04
Ele. 19 - MAX. ABS. : 1.1771e+04
Ele. 20 - MAX. ABS. : 3.8266e+03
Ele. 21 - MAX. ABS. : 9.1169e+03
Ele. 22 - MAX. ABS. : 2.8333e+03
Ele. 23 - MAX. ABS. : 9.4377e+03
Ele. 24 - MAX. ABS. : 1.6659e+03
Ele. 25 - MAX. ABS. : 8.7960e+03
Ele. 26 - MAX. ABS. : 1.5809e+03
Ele. 27 - MAX. ABS. : 6.6542e+03
Ele. 28 - MAX. ABS. : 3.3246e+03
Ele. 29 - MAX. ABS. : 6.6542e+03
Ele. 30 - MAX. ABS. : 1.5809e+03
Ele. 31 - MAX. ABS. : 8.7960e+03
Ele. 32 - MAX. ABS. : 1.6659e+03
Ele. 33 - MAX. ABS. : 9.4377e+03
Ele. 34 - MAX. ABS. : 2.8333e+03
Ele. 35 - MAX. ABS. : 9.1169e+03
Ele. 36 - MAX. ABS. : 3.8266e+03
Ele. 37 - MAX. ABS. : 1.1771e+04

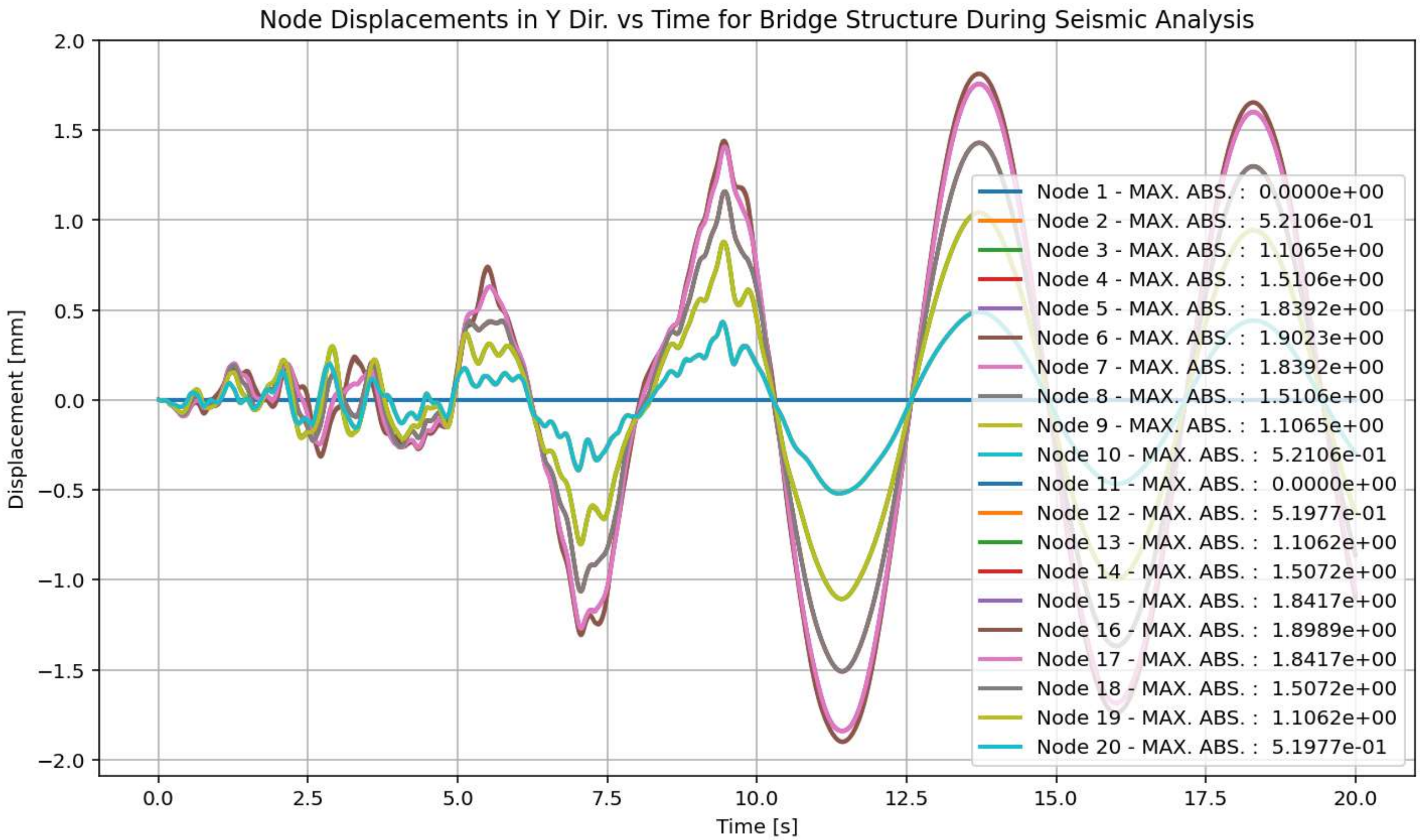

elements Stress in X Dir. vs Time for Bridge Structure During Seismic Analysis
Element Stress [N/mm^2]
Time [s]
Ele. 1 - MAX. ABS. : 1.9845e+00
Ele. 2 - MAX. ABS. : 1.8812e+00
Ele. 3 - MAX. ABS. : 1.0747e+00
Ele. 4 - MAX. ABS. : 9.7767e-01
Ele. 5 - MAX. ABS. : 2.8922e+00
Ele. 6 - MAX. ABS. : 2.8922e+00
Ele. 7 - MAX. ABS. : 9.7767e-01
Ele. 8 - MAX. ABS. : 1.0747e+00
Ele. 9 - MAX. ABS. : 1.8812e+00
Ele. 10 - MAX. ABS. : 1.9845e+00
Ele. 11 - MAX. ABS. : 3.6437e+00
Ele. 12 - MAX. ABS. : 3.6292e+00
Ele. 13 - MAX. ABS. : 5.4725e+00
Ele. 14 - MAX. ABS. : 5.5063e+00
Ele. 15 - MAX. ABS. : 5.5063e+00
Ele. 16 - MAX. ABS. : 5.4725e+00
Ele. 17 - MAX. ABS. : 3.6292e+00
Ele. 18 - MAX. ABS. : 3.6437e+00
Ele. 19 - MAX. ABS. : 2.8850e+00
Ele. 20 - MAX. ABS. : 9.3789e-01
Ele. 21 - MAX. ABS. : 2.2345e+00
Ele. 22 - MAX. ABS. : 6.9443e-01
Ele. 23 - MAX. ABS. : 2.3132e+00
Ele. 24 - MAX. ABS. : 4.0831e-01
Ele. 25 - MAX. ABS. : 2.1559e+00
Ele. 26 - MAX. ABS. : 3.8747e-01
Ele. 27 - MAX. ABS. : 1.6309e+00
Ele. 28 - MAX. ABS. : 8.1484e-01
Ele. 29 - MAX. ABS. : 1.6309e+00
Ele. 30 - MAX. ABS. : 3.8747e-01
Ele. 31 - MAX. ABS. : 2.1559e+00
Ele. 32 - MAX. ABS. : 4.0831e-01
Ele. 33 - MAX. ABS. : 2.3132e+00
Ele. 34 - MAX. ABS. : 6.9443e-01
Ele. 35 - MAX. ABS. : 2.2345e+00
Ele. 36 - MAX. ABS. : 9.3789e-01
Ele. 37 - MAX. ABS. : 2.8850e+00


elements Strain in X Dir. vs Time for Bridge Structure During Seismic Analysis
1e−5
Element Strain [mm]
Time [s]
2
1
0
−1
−2
−3
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Ele. 1 - MAX. ABS. : 9.9227e-06
Ele. 2 - MAX. ABS. : 9.4058e-06
Ele. 3 - MAX. ABS. : 5.3737e-06
Ele. 4 - MAX. ABS. : 4.8884e-06
Ele. 5 - MAX. ABS. : 1.4461e-05
Ele. 6 - MAX. ABS. : 1.4461e-05
Ele. 7 - MAX. ABS. : 4.8884e-06
Ele. 8 - MAX. ABS. : 5.3737e-06
Ele. 9 - MAX. ABS. : 9.4058e-06
Ele. 10 - MAX. ABS. : 9.9227e-06
Ele. 11 - MAX. ABS. : 1.8218e-05
Ele. 12 - MAX. ABS. : 1.8146e-05
Ele. 13 - MAX. ABS. : 2.7362e-05
Ele. 14 - MAX. ABS. : 2.7531e-05
Ele. 15 - MAX. ABS. : 2.7531e-05
Ele. 16 - MAX. ABS. : 2.7362e-05
Ele. 17 - MAX. ABS. : 1.8146e-05
Ele. 18 - MAX. ABS. : 1.8218e-05
Ele. 19 - MAX. ABS. : 1.4425e-05
Ele. 20 - MAX. ABS. : 4.6895e-06
Ele. 21 - MAX. ABS. : 1.1173e-05
Ele. 22 - MAX. ABS. : 3.4722e-06
Ele. 23 - MAX. ABS. : 1.1566e-05
Ele. 24 - MAX. ABS. : 2.0416e-06
Ele. 25 - MAX. ABS. : 1.0779e-05
Ele. 26 - MAX. ABS. : 1.9374e-06
Ele. 27 - MAX. ABS. : 8.1546e-06
Ele. 28 - MAX. ABS. : 4.0742e-06
Ele. 29 - MAX. ABS. : 8.1546e-06
Ele. 30 - MAX. ABS. : 1.9374e-06
Ele. 31 - MAX. ABS. : 1.0779e-05
Ele. 32 - MAX. ABS. : 2.0416e-06
Ele. 33 - MAX. ABS. : 1.1566e-05
Ele. 34 - MAX. ABS. : 3.4722e-06
Ele. 35 - MAX. ABS. : 1.1173e-05
Ele. 36 - MAX. ABS. : 4.6895e-06
Ele. 37 - MAX. ABS. : 1.4425e-05

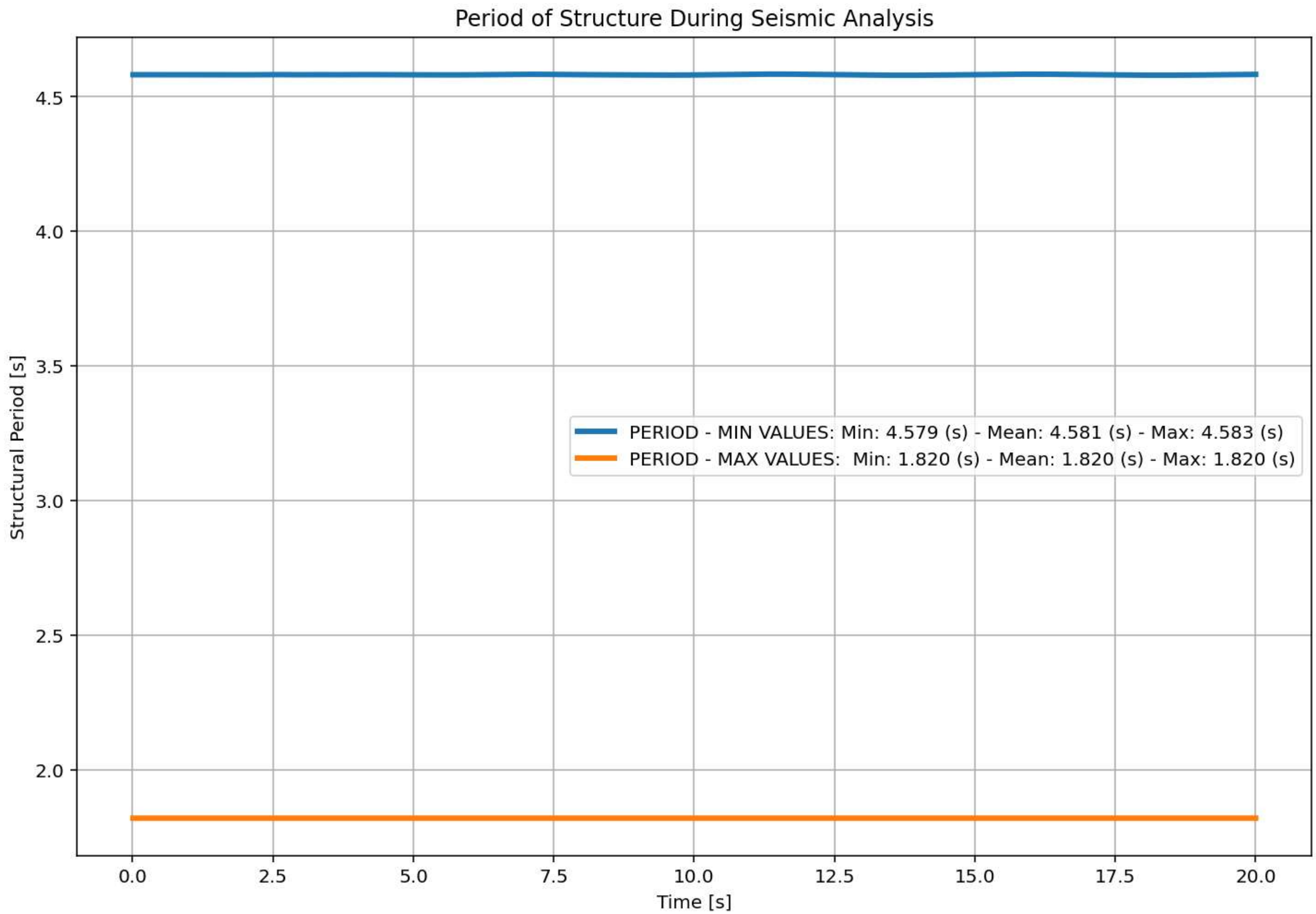

elements Damage-index [%] in X Dir. vs Time for Bridge Structure During Seismic Analysis
1e−5
Element Damage-index [%]
Time [s]
Ele. 1 - MAX. ABS. : 9.9227e-06
Ele. 2 - MAX. ABS. : 9.4058e-06
Ele. 3 - MAX. ABS. : 5.3737e-06
Ele. 4 - MAX. ABS. : 4.8884e-06
Ele. 5 - MAX. ABS. : 1.4461e-05
Ele. 6 - MAX. ABS. : 1.4461e-05
Ele. 7 - MAX. ABS. : 4.8884e-06
Ele. 8 - MAX. ABS. : 5.3737e-06
Ele. 9 - MAX. ABS. : 9.4058e-06
Ele. 10 - MAX. ABS. : 9.9227e-06
Ele. 11 - MAX. ABS. : 1.8218e-05
Ele. 12 - MAX. ABS. : 1.8146e-05
Ele. 13 - MAX. ABS. : 2.7362e-05
Ele. 14 - MAX. ABS. : 2.7531e-05
Ele. 15 - MAX. ABS. : 2.7531e-05
Ele. 16 - MAX. ABS. : 2.7362e-05
Ele. 17 - MAX. ABS. : 1.8146e-05
Ele. 18 - MAX. ABS. : 1.8218e-05
Ele. 19 - MAX. ABS. : 1.4425e-05
Ele. 20 - MAX. ABS. : 4.6895e-06
Ele. 21 - MAX. ABS. : 1.1173e-05
Ele. 22 - MAX. ABS. : 3.4722e-06
Ele. 23 - MAX. ABS. : 1.1566e-05
Ele. 24 - MAX. ABS. : 2.0416e-06
Ele. 25 - MAX. ABS. : 1.0779e-05
Ele. 26 - MAX. ABS. : 1.9374e-06
Ele. 27 - MAX. ABS. : 8.1546e-06
Ele. 28 - MAX. ABS. : 4.0742e-06
Ele. 29 - MAX. ABS. : 8.1546e-06
Ele. 30 - MAX. ABS. : 1.9374e-06
Ele. 31 - MAX. ABS. : 1.0779e-05
Ele. 32 - MAX. ABS. : 2.0416e-06
Ele. 33 - MAX. ABS. : 1.1566e-05
Ele. 34 - MAX. ABS. : 3.4722e-06
Ele. 35 - MAX. ABS. : 1.1173e-05
Ele. 36 - MAX. ABS. : 4.6895e-06
Ele. 37 - MAX. ABS. : 1.4425e-05

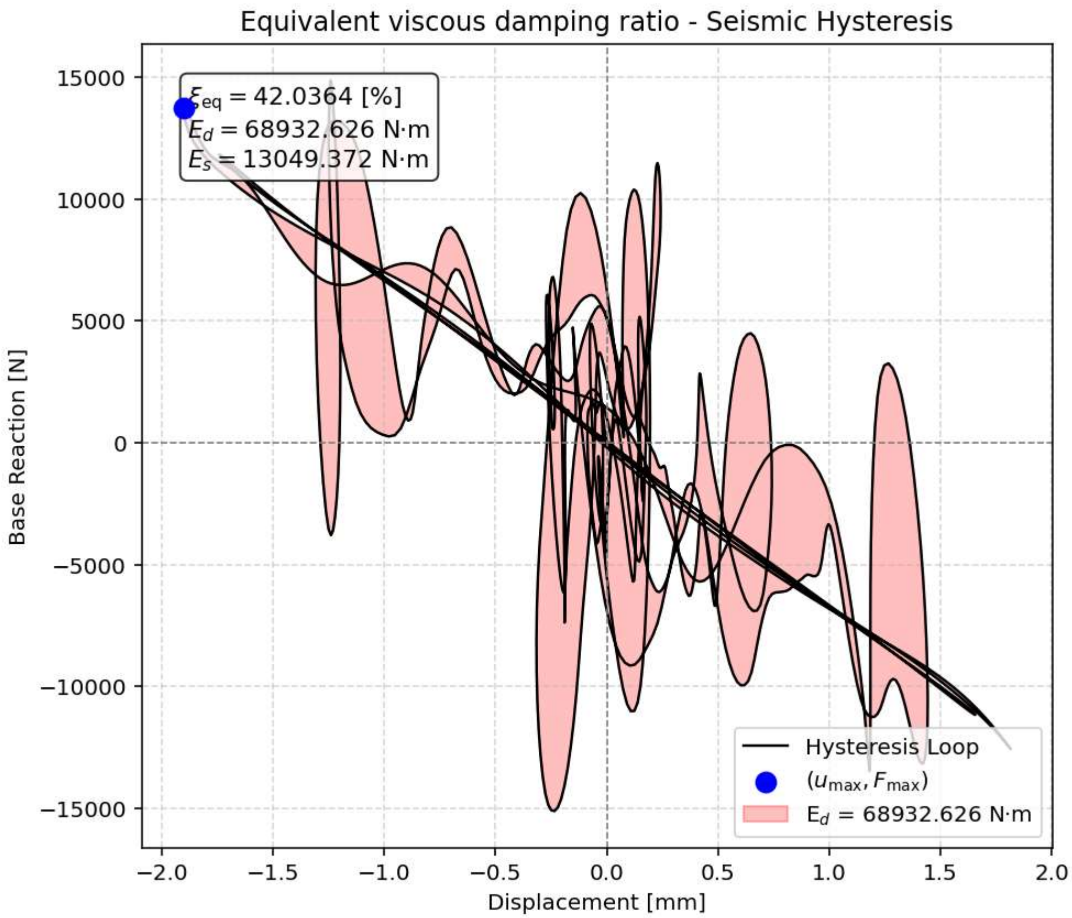


Node Displacements in X Dir. vs Time for Bridge Structure During Seismic Analysis
Displacement [mm]
Time [s]
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 4.2660e-02
Node 3 - MAX. ABS. : 8.3005e-02
Node 4 - MAX. ABS. : 7.1043e-02
Node 5 - MAX. ABS. : 6.2184e-02
Node 6 - MAX. ABS. : 4.8610e-13
Node 7 - MAX. ABS. : 6.2184e-02
Node 8 - MAX. ABS. : 7.1043e-02
Node 9 - MAX. ABS. : 8.3005e-02
Node 10 - MAX. ABS. : 4.2660e-02
Node 11 - MAX. ABS. : 0.0000e+00
Node 12 - MAX. ABS. : 3.3895e-01
Node 13 - MAX. ABS. : 3.0246e-01
Node 14 - MAX. ABS. : 2.3604e-01
Node 15 - MAX. ABS. : 1.1839e-01
Node 16 - MAX. ABS. : 5.7143e-13
Node 17 - MAX. ABS. : 1.1839e-01
Node 18 - MAX. ABS. : 2.3604e-01
Node 19 - MAX. ABS. : 3.0246e-01
Node 20 - MAX. ABS. : 3.3895e-01

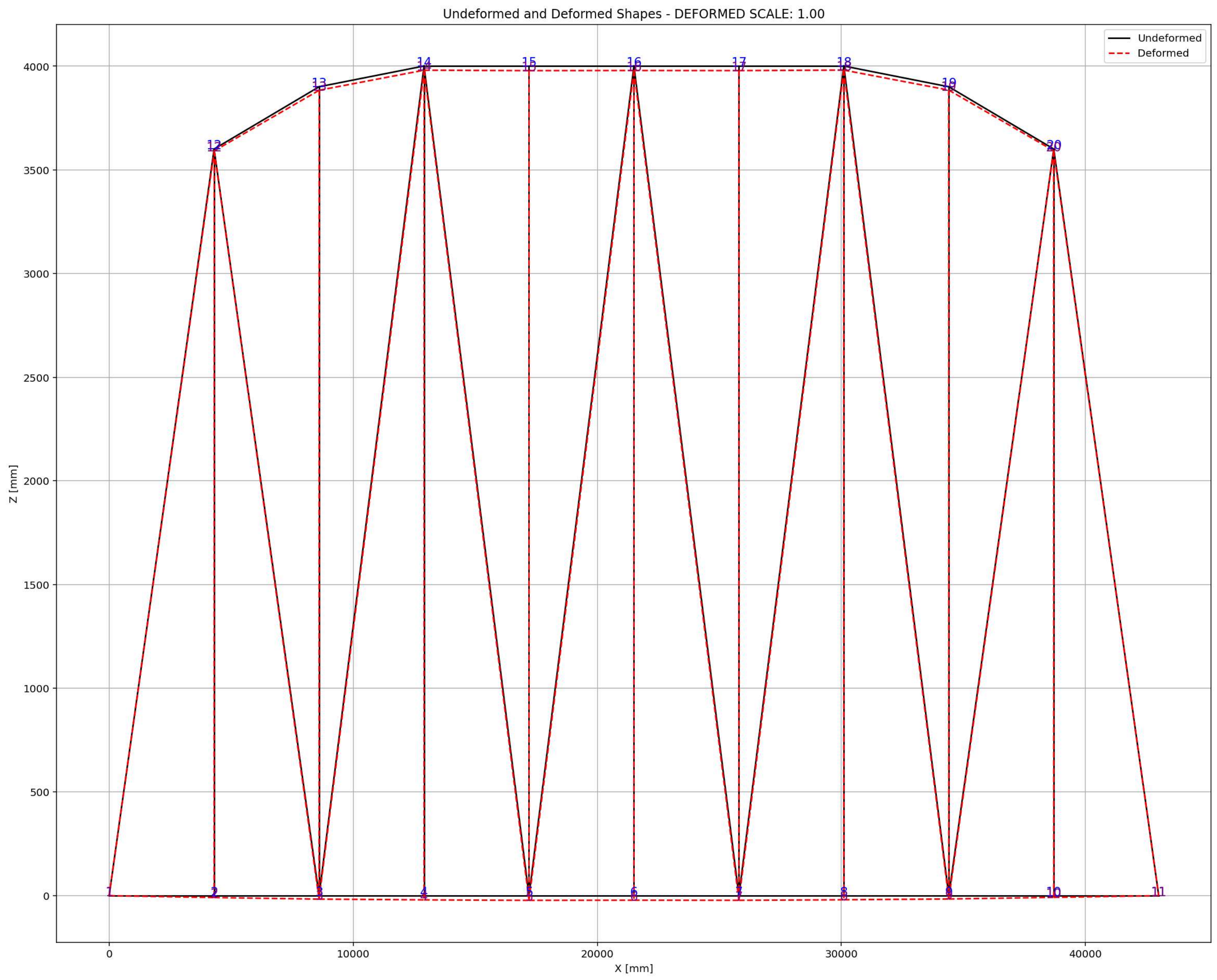


Node Displacements in Y Dir. vs Time for Bridge Structure During Seismic Analysis
Displacement [mm]
Time [s]
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 5.2106e-01
Node 3 - MAX. ABS. : 1.1065e+00
Node 4 - MAX. ABS. : 1.5106e+00
Node 5 - MAX. ABS. : 1.8392e+00
Node 6 - MAX. ABS. : 1.9023e+00
Node 7 - MAX. ABS. : 1.8392e+00
Node 8 - MAX. ABS. : 1.5106e+00
Node 9 - MAX. ABS. : 1.1065e+00
Node 10 - MAX. ABS. : 5.2106e-01
Node 11 - MAX. ABS. : 0.0000e+00
Node 12 - MAX. ABS. : 5.1977e-01
Node 13 - MAX. ABS. : 1.1062e+00
Node 14 - MAX. ABS. : 1.5072e+00
Node 15 - MAX. ABS. : 1.8417e+00
Node 16 - MAX. ABS. : 1.8989e+00
Node 17 - MAX. ABS. : 1.8417e+00
Node 18 - MAX. ABS. : 1.5072e+00
Node 19 - MAX. ABS. : 1.1062e+00
Node 20 - MAX. ABS. : 5.1977e-01

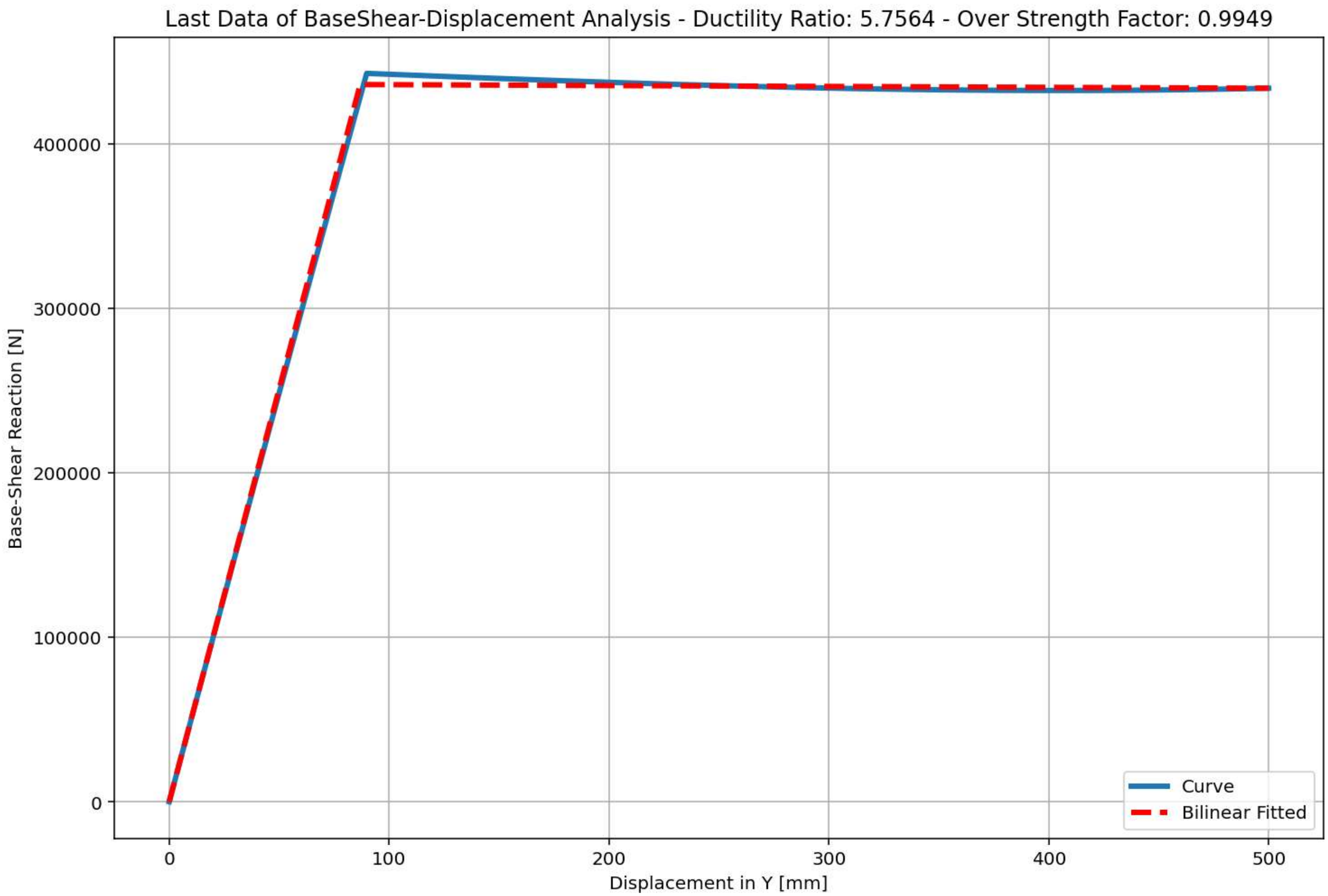

Period of Structure During Seismic Analysis
Structural Period [s]
4.5
4.0
3.5
3.0
2.5
2.0
PERIOD - MIN VALUES: Min: 4.579 (s) - Mean: 4.581 (s) - Max: 4.583 (s)
PERIOD - MAX VALUES: Min: 1.820 (s) - Mean: 1.820 (s) - Max: 1.820 (s)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]



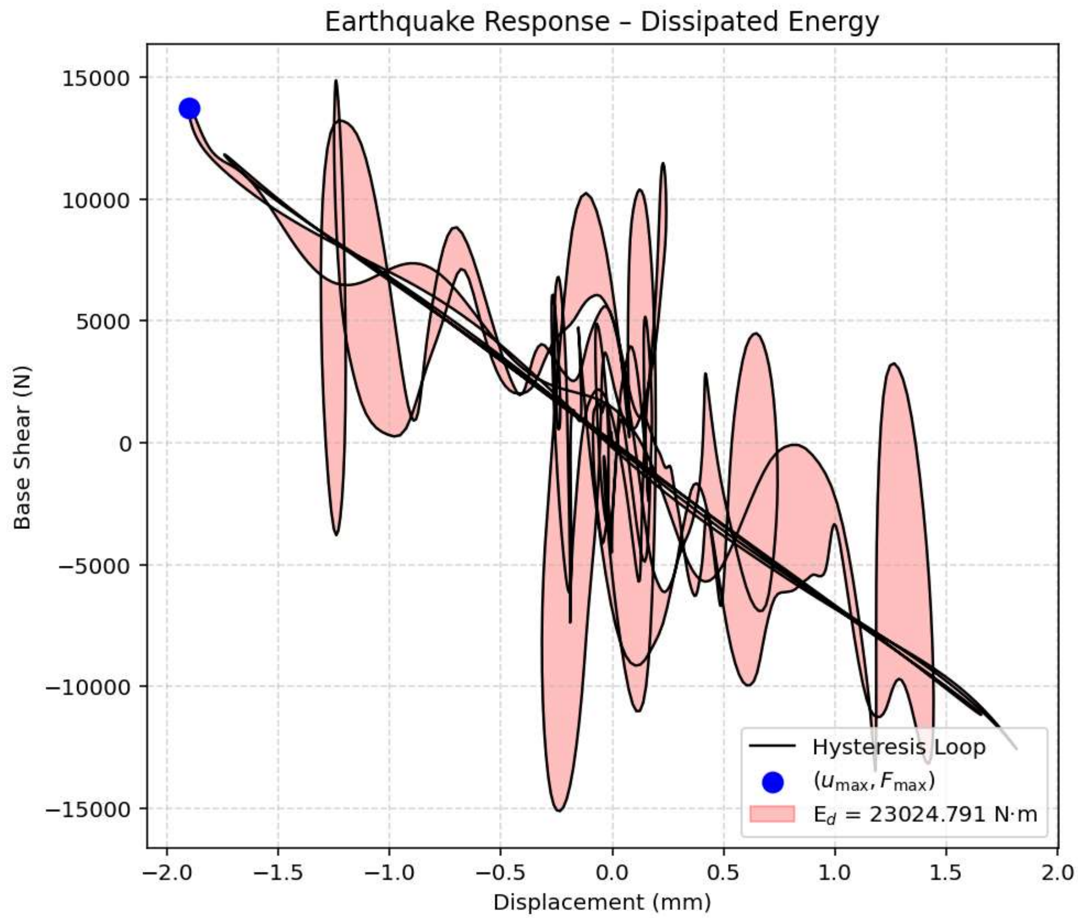
Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 42.0364$ [%]
$E_d = 68932.626$ N·m
$E_s = 13049.372$ N·m
Base Reaction [N]
Displacement [mm]
Hysteresis Loop
(u_{max}, F_{max})
E$_d$ = 68932.626 N·m



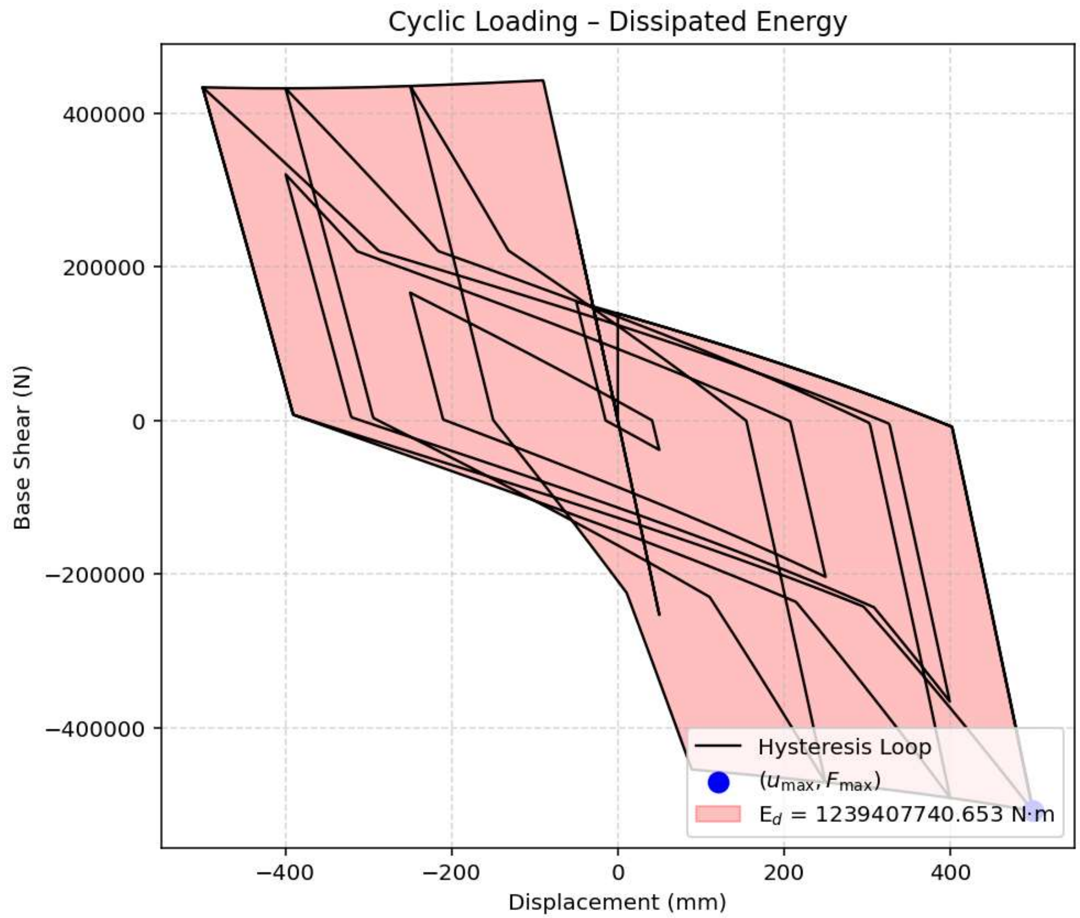
Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
10000
20000
30000
40000
500
1000
1500
2000
2500
3000
3500
4000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20


Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 5.7564 - Over Strength Factor: 0.9949
Base-Shear Reaction [N]
400000
300000
200000
100000
0
0
100
200
300
400
500
Displacement in Y [mm]
Curve
Bilinear Fitted

```
+===========================+
=   Analysis curve fitted =
    Disp       Base Shear
---------------------------
[[0.00000000e+00 0.00000000e+00]
 [8.68592608e+01 4.35874099e+05]
 [5.00000000e+02 4.33655994e+05]]
+===========================+
+--------------------------------------------------+
 Structure Elastic Stiffness :     5018.16
 Structure Plastic Stiffness :     867.31
 Structure Tangent Stiffness :     -5.37
 Structure Ductility Ratio :      5.76
 Structure Over Strength Factor:   0.99
+--------------------------------------------------+
Over Strength Coefficient (Ω0):     0.9949
Displacement Ductility Ratio (μ):    5.7564
Ductility Coefficient (Rμ):        5.7564
Structural Behavior Coefficient (R): 5.7271
Structural Ductility Damage Index in Y Direction: -20.5637 (%)
```



Earthquake Response – Dissipated Energy
Base Shear (N)
Displacement (mm)
15000
10000
5000
0
−5000
−10000
−15000
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Hysteresis Loop
$(u_{\max}, F_{\max})$
E_d = 23024.791 N·m



Cyclic Loading – Dissipated Energy
Base Shear (N)
400000
200000
0
−200000
−400000
−400
−200
0
200
400
Displacement (mm)
Hysteresis Loop
$(u_{\max}, F_{\max})$
E_d = 1239407740.653 N·m

Dissipated Energy from Earthquake= 23024.79 N·mm
Dissipated Energy from Cyclic Displacement= 1239407740.65 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 0.002 [%]
ZONE 1: VERY MINOR DAMAGE